

Table S1 *UCL* of EWMA TBFA schemes based on the statistic Z_i with both variables A and T following the gamma distribution when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

	Z_1					
	$\tau = 0.2$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	0.3183	0.2938	0.2450	0.1960	0.1451	0.0866
<i>UCL</i> -Frank	0.3251	0.2998	0.2498	0.1996	0.1476	0.0878
	$\tau = 0.5$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	0.2494	0.2291	0.1888	0.1481	0.1066	0.0603
<i>UCL</i> -Frank	0.2395	0.2204	0.1823	0.1442	0.1050	0.0609
	$\tau = 0.8$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	0.1649	0.1505	0.1220	0.0930	0.0635	0.0324
<i>UCL</i> -Frank	0.1399	0.1283	0.1046	0.0809	0.0566	0.0306
	Z_2					
	$\tau = 0.2$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	1.3598	1.3293	1.2688	1.2086	1.1469	1.0733
<i>UCL</i> -Frank	1.3835	1.3506	1.2854	1.2202	1.1538	1.0764
	$\tau = 0.5$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	1.2566	1.2340	1.1896	1.1453	1.0996	1.0476
<i>UCL</i> -Frank	1.2694	1.2459	1.1991	1.1527	1.1050	1.0505
	$\tau = 0.8$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	1.1540	1.1398	1.1116	1.0831	1.0546	1.0238
<i>UCL</i> -Frank	1.1495	1.1359	1.1090	1.0819	1.0543	1.0244
	Z_3					
	$\tau = 0.2$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	2.3391	2.3109	2.2555	2.2008	2.1440	2.0752
<i>UCL</i> -Frank	2.3621	2.3315	2.2716	2.2120	2.1505	2.0775
	$\tau = 0.5$					
λ	1	0.9	0.7	0.5	0.3	0.1
	2.2539	2.2321	2.1897	2.1474	2.1041	2.0542
<i>UCL</i> -Frank	2.2725	2.2495	2.2037	2.1579	2.1112	2.0574
	$\tau = 0.8$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	2.1738	2.1586	2.1285	2.0984	2.0678	2.0348
<i>UCL</i> -Frank	2.1847	2.1688	2.1367	2.1044	2.0717	2.0365

Table S2 *UCL* of EWMA TBFA schemes based on the statistic Z_i with variable A following the gamma distribution and T following the weibull distribution when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

	Z_1					
	$\tau = 0.2$					

λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	0.3322	0.3059	0.2543	0.2023	0.1488	0.0873
<i>UCL</i> -Frank	0.3613	0.3331	0.2761	0.2179	0.1584	0.0912
$\tau = 0.5$						
λ	1	0.9	0.7	0.5	0.3	0.1
	0.2402	0.2212	0.1826	0.1434	0.1033	0.0588
<i>UCL</i> -Frank	0.2732	0.2508	0.2060	0.1615	0.1154	0.0646
$\tau = 0.8$						
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	0.1688	0.1540	0.1250	0.0958	0.0659	0.0341
<i>UCL</i> -Frank	0.1914	0.1750	0.1414	0.1080	0.0740	0.0377
Z_2						
$\tau = 0.2$						
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	1.4106	1.3755	1.3044	1.2334	1.1609	1.0780
<i>UCL</i> -Frank	1.4906	1.4469	1.3594	1.2723	1.1836	1.0858
$\tau = 0.5$						
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	1.2556	1.2335	1.1892	1.1451	1.0997	1.0479
<i>UCL</i> -Frank	1.3665	1.3327	1.2667	1.2004	1.1329	1.0598
$\tau = 0.8$						
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	1.1743	1.1586	1.1267	1.0950	1.0624	1.0272
<i>UCL</i> -Frank	1.2551	1.2316	1.1832	1.1354	1.0867	1.0357
Z_3						
$\tau = 0.2$						
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	2.4136	2.3779	2.3071	2.2375	2.1652	2.0826
<i>UCL</i> -Frank	2.4781	2.4362	2.3524	2.2690	2.1833	2.0884
$\tau = 0.5$						
λ	1	0.9	0.7	0.5	0.3	0.1
	2.2962	2.2706	2.2203	2.1699	2.1186	2.0603
<i>UCL</i> -Frank	2.3950	2.3596	2.2888	2.2186	2.1468	2.0695
$\tau = 0.8$						
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	2.2567	2.2340	2.1880	2.1420	2.0959	2.0463
<i>UCL</i> -Frank	2.3236	2.2943	2.2351	2.1753	2.1154	2.0525

Table S3 *UCL* of EWMA TBEA schemes based on the statistic Z_i with both variables A and T following the weibull distribution when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

Z_1						
$\tau = 0.2$						
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	0.3027	0.2791	0.2328	0.1865	0.1383	0.0827
<i>UCL</i> -Frank	0.3459	0.3186	0.2639	0.2088	0.1520	0.0885
$\tau = 0.5$						
λ	1	0.9	0.7	0.5	0.3	0.1
	0.1945	0.1793	0.1491	0.1188	0.0874	0.0515

<i>UCL</i> -Frank	0.2649	0.2431	0.1999	0.1561	0.1113	0.0624
	$\tau = 0.8$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	0.1039	0.0952	0.0780	0.0605	0.0427	0.0234
<i>UCL</i> -Frank	0.1640	0.1494	0.1212	0.0923	0.0633	0.0327
	Z_2					
	$\tau = 0.2$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	1.3896	1.3555	1.2886	1.2213	1.1529	1.0746
<i>UCL</i> -Frank	1.4824	1.4399	1.3542	1.2686	1.1810	1.0848
	$\tau = 0.5$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	1.2125	1.1942	1.1580	1.1219	1.0850	1.0417
<i>UCL</i> -Frank	1.3687	1.3345	1.2677	1.2008	1.1325	1.0593
	$\tau = 0.8$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	1.0988	1.0901	1.0725	1.0550	1.0372	1.0172
<i>UCL</i> -Frank	1.2315	1.2097	1.1661	1.1225	1.0780	1.0320
	Z_3					
	$\tau = 0.2$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	2.3918	2.3581	2.2913	2.2250	2.1567	2.0790
<i>UCL</i> -Frank	2.4739	2.4322	2.3492	2.2657	2.1805	2.0868
	$\tau = 0.5$					
λ	1	0.9	0.7	0.5	0.3	0.1
	2.2351	2.2154	2.1766	2.1381	2.0987	2.0526
<i>UCL</i> -Frank	2.3933	2.3587	2.2881	2.2172	2.1450	2.0680
	$\tau = 0.8$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	2.1636	2.1493	2.1215	2.0934	2.0648	2.0341
<i>UCL</i> -Frank	2.3010	2.2739	2.2186	2.1628	2.1063	2.0481

Table S4 *UCL* of EWMA TBEA schemes based on the statistic Z_i with variable A following the weibull distribution and T following the gamma distribution when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

	Z_1					
	$\tau = 0.2$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	0.2753	0.2552	0.2147	0.1747	0.1326	0.0822
<i>UCL</i> -Frank	0.2928	0.2707	0.2271	0.1835	0.1383	0.0848
	$\tau = 0.5$					
λ	1	0.9	0.7	0.5	0.3	0.1
	0.2011	0.1856	0.1552	0.1244	0.0924	0.0554
<i>UCL</i> -Frank	0.2174	0.2007	0.1671	0.1336	0.0988	0.0592
	$\tau = 0.8$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	0.1117	0.1027	0.0849	0.0670	0.0485	0.0282
<i>UCL</i> -Frank	0.1114	0.1029	0.0853	0.0676	0.0494	0.0294

	Z_2					
	$\tau = 0.2$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	1.3241	1.2966	1.2436	1.1913	1.1371	1.0705
<i>UCL</i> -Frank	1.3597	1.3289	1.2688	1.2089	1.1476	1.0748
	$\tau = 0.5$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	1.2155	1.1974	1.1610	1.1253	1.0886	1.0443
<i>UCL</i> -Frank	1.2613	1.2384	1.1936	1.1489	1.1031	1.0506
	$\tau = 0.8$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	1.1085	1.0991	1.0805	1.0619	1.0429	1.0211
<i>UCL</i> -Frank	1.1309	1.1197	1.0967	1.0737	1.0503	1.0244
	Z_3					
	$\tau = 0.2$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	2.3057	2.2809	2.2321	2.1844	2.1345	2.0719
<i>UCL</i> -Frank	2.3413	2.3129	2.2571	2.2016	2.1446	2.0757
	$\tau = 0.5$					
λ	1	0.9	0.7	0.5	0.3	0.1
	2.2084	2.1914	2.1579	2.1250	2.0909	2.0496
<i>UCL</i> -Frank	2.2618	2.2394	2.1958	2.1520	2.1074	2.0562
	$\tau = 0.8$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	2.1149	2.1055	2.0869	2.0684	2.0495	2.0284
<i>UCL</i> -Frank	2.1583	2.1446	2.1177	2.0908	2.0635	2.0338

Table S5 *UCL* of EWMA TBEA schemes based on the statistic Z_i with variable A following the normal distribution and T following the gamma distribution when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

	Z_1					
	$\tau = 0.2$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	0.3072	0.2838	0.2369	0.1906	0.1420	0.0855
<i>UCL</i> -Frank	0.3156	0.2919	0.2436	0.1951	0.1452	0.0871
	$\tau = 0.5$					
λ	1	0.9	0.7	0.5	0.3	0.1
	0.2368	0.2178	0.1799	0.1419	0.1028	0.0590
<i>UCL</i> -Frank	0.2320	0.2135	0.1771	0.1406	0.1029	0.0602
	$\tau = 0.8$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	0.1509	0.1380	0.1120	0.0859	0.0593	0.0309
<i>UCL</i> -Frank	0.1299	0.1192	0.0976	0.0759	0.0537	0.0297
	Z_2					
	$\tau = 0.2$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	1.3508	1.3207	1.2623	1.2042	1.1441	1.0726
<i>UCL</i> -Frank	1.3774	1.3446	1.2807	1.2173	1.1521	1.0758

	$\tau = 0.5$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	1.2458	1.2243	1.1820	1.1400	1.0967	1.0468
<i>UCL</i> -Frank	1.2656	1.2425	1.1964	1.1509	1.1040	1.0503
	$\tau = 0.8$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	1.1418	1.1288	1.1031	1.0774	1.0511	1.0227
<i>UCL</i> -Frank	1.1420	1.1292	1.1040	1.0783	1.0522	1.0239
	Z_3					
	$\tau = 0.2$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	2.3299	2.3024	2.2489	2.1961	2.1415	2.0744
<i>UCL</i> -Frank	2.3555	2.3262	2.2674	2.2090	2.1489	2.0771
	$\tau = 0.5$					
λ	1	0.9	0.7	0.5	0.3	0.1
	2.2412	2.2212	2.1811	2.1415	2.1005	2.0531
<i>UCL</i> -Frank	2.2678	2.2445	2.1999	2.1555	2.1096	2.0569
	$\tau = 0.8$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	2.1579	2.1447	2.1175	2.0903	2.0627	2.0329
<i>UCL</i> -Frank	2.1747	2.1598	2.1295	2.0995	2.0686	2.0353

Table S6 *UCL* of EWMA TBEA schemes based on the statistic Z_i with variable A following the normal distribution and T following the weibull distribution when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

	Z_1					
	$\tau = 0.2$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	0.3234	0.2985	0.2481	0.1980	0.1459	0.0861
<i>UCL</i> -Frank	0.3572	0.3284	0.2721	0.2154	0.1568	0.0906
	$\tau = 0.5$					
λ	1	0.9	0.7	0.5	0.3	0.1
	0.2273	0.2093	0.1731	0.1365	0.0989	0.0568
<i>UCL</i> -Frank	0.2682	0.2461	0.2025	0.1587	0.1137	0.0638
	$\tau = 0.8$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	0.1513	0.1383	0.1123	0.0863	0.0596	0.0311
<i>UCL</i> -Frank	0.1811	0.1653	0.1336	0.1024	0.0702	0.0359
	Z_2					
	$\tau = 0.2$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	1.4052	1.3701	1.2995	1.2302	1.1586	1.0770
<i>UCL</i> -Frank	1.4885	1.4449	1.3578	1.2710	1.1828	1.0856
	$\tau = 0.5$					
λ	1	0.9	0.7	0.5	0.3	0.1
<i>UCL</i> -Clayton	1.2415	1.2207	1.1792	1.1378	1.0952	1.0461
<i>UCL</i> -Frank	1.3653	1.3319	1.2663	1.1994	1.1322	1.0594
	$\tau = 0.8$					

0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.8374	53.5107	17.2554	7.2565	3.9693	2.6341	1.9739	1.5895	1.3443	1.1862	1.0945
0.95	17.2888	7.2312	3.9684	2.6328	1.9704	1.5927	1.3431	1.1878	1.0948	1.0454	1.0200
0.9	3.9576	2.6333	1.9719	1.5888	1.3449	1.1892	1.0954	1.0442	1.0192	1.0082	1.0032
0.85	1.9706	1.5889	1.3431	1.1860	1.0931	1.0438	1.0186	1.0076	1.0027	1.0010	1.0005
0.8	1.3412	1.1886	1.0938	1.0436	1.0182	1.0076	1.0025	1.0008	1.0003	1.0002	1.0001
0.75	1.0926	1.0435	1.0178	1.0070	1.0023	1.0008	1.0002	1.0001	1	1	1
0.7	1.0179	1.0071	1.0025	1.0008	1.0003	1.0001	1	1	1	1	1
0.65	1.0022	1.0008	1.0002	1.0001	1	1	1	1	1	1	1
0.6	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.7280	63.3197	22.3069	9.3578	4.7423	2.8865	2.0162	1.5636	1.3069	1.1546	1.0774
0.95	22.4196	9.4039	4.7224	2.8928	2.0189	1.5595	1.3055	1.1602	1.0749	1.0349	1.0158
0.9	4.7527	2.8907	2.0269	1.5632	1.3035	1.1583	1.0749	1.0342	1.0153	1.0059	1.0023
0.85	2.0265	1.5703	1.3043	1.1577	1.0764	1.0336	1.0144	1.0058	1.0019	1.0007	1.0004
0.8	1.3049	1.1577	1.0755	1.0339	1.0135	1.0058	1.0018	1.0008	1.0002	1.0001	1
0.75	1.0750	1.0334	1.0138	1.0054	1.0020	1.0007	1.0002	1.0001	1	1	1
0.7	1.0138	1.0047	1.0016	1.0005	1.0002	1	1	1	1	1	1
0.65	1.0018	1.0005	1.0001	1	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.4435	70.4318	27.5038	12.0402	5.8946	3.3498	2.1749	1.6003	1.3024	1.1454	1.0703
0.95	27.3358	11.9625	5.9091	3.3554	2.1751	1.5990	1.3011	1.1448	1.0690	1.0307	1.0132
0.9	5.9034	3.3488	2.1765	1.5964	1.2999	1.1482	1.0683	1.0301	1.0126	1.0048	1.0020
0.85	2.1699	1.5971	1.3038	1.1457	1.0688	1.0290	1.0122	1.0049	1.0019	1.0005	1.0002
0.8	1.3024	1.1467	1.0681	1.0295	1.0120	1.0045	1.0016	1.0005	1.0002	1.0001	1
0.75	1.0688	1.0295	1.0114	1.0039	1.0014	1.0005	1.0001	1	1	1	1
0.7	1.0113	1.0043	1.0014	1.0004	1.0002	1.0001	1	1	1	1	1
0.65	1.0015	1.0004	1.0002	1	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.5199	73.2599	29.7962	13.3402	6.5734	3.6730	2.2895	1.6250	1.3034	1.1452	1.0663
0.95	29.7972	13.3531	6.5791	3.6485	2.2925	1.6357	1.3054	1.1444	1.0666	1.0289	1.0126
0.9	6.6111	3.6671	2.2909	1.6347	1.3084	1.1451	1.0658	1.0295	1.0118	1.0047	1.0018

0.85	2.3018	1.6330	1.3042	1.1458	1.0659	1.0279	1.0117	1.0047	1.0016	1.0005	1.0003
0.8	1.3068	1.1473	1.0666	1.0282	1.0114	1.0044	1.0014	1.0005	1.0001	1.0001	1
0.75	1.0664	1.0286	1.0110	1.0037	1.0016	1.0005	1.0001	1.0001	1	1	1
0.7	1.0110	1.0041	1.0014	1.0004	1.0002	1	1	1	1	1	1
0.65	1.0012	1.0005	1.0001	1	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.3763	19.6054	5.9533	3.4202	2.4500	1.9985	1.7082	1.3759	1.1629	1.0679	1.0287
0.95	5.9696	3.4184	2.4593	1.9979	1.7051	1.3773	1.1613	1.0673	1.0281	1.0112	1.0043
0.9	2.4522	1.9983	1.7052	1.3741	1.1606	1.0682	1.0271	1.0103	1.0040	1.0014	1.0004
0.85	1.7057	1.3778	1.1607	1.0660	1.0271	1.0099	1.0034	1.0012	1.0004	1.0001	1.0001
0.8	1.1603	1.0659	1.0259	1.0099	1.0035	1.0011	1.0003	1.0001	1	1	1
0.75	1.0250	1.0093	1.0036	1.0011	1.0002	1.0001	1	1	1	1	1
0.7	1.0029	1.0009	1.0004	1.0001	1	1	1	1	1	1	1
0.65	1.0004	1.0001	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.2650	32.2799	7.7594	3.5590	2.3344	1.8168	1.4239	1.1781	1.0749	1.0307	1.0127
0.95	7.7795	3.5558	2.3331	1.8090	1.4208	1.1783	1.0731	1.0304	1.0115	1.0048	1.0016
0.9	2.3315	1.8132	1.4214	1.1781	1.0739	1.0288	1.0109	1.0040	1.0014	1.0006	1.0001
0.85	1.4234	1.1790	1.0723	1.0295	1.0107	1.0037	1.0015	1.0004	1.0001	1	1
0.8	1.0718	1.0282	1.0103	1.0036	1.0012	1.0003	1.0001	1	1	1	1
0.75	1.0101	1.0035	1.0010	1.0004	1.0001	1	1	1	1	1	1
0.7	1.0010	1.0003	1.0001	1	1	1	1	1	1	1	1
0.65	1.0001	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.7933	45.2906	11.6669	4.3631	2.4318	1.7336	1.3214	1.1307	1.0532	1.0226	1.0087
0.95	11.7517	4.3704	2.4277	1.7323	1.3194	1.1289	1.0528	1.0212	1.0079	1.0031	1.0011
0.9	2.4252	1.7304	1.3203	1.1327	1.0522	1.0204	1.0080	1.0026	1.0010	1.0002	1.0001
0.85	1.3217	1.1305	1.0527	1.0196	1.0076	1.0026	1.0009	1.0003	1.0001	1	1
0.8	1.0524	1.0195	1.0073	1.0022	1.0007	1.0002	1.0001	1	1	1	1
0.75	1.0069	1.0021	1.0007	1.0002	1.0001	1	1	1	1	1	1
0.7	1.0006	1.0001	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1

0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.4347	56.9673	17.2197	6.1713	2.8127	1.7389	1.2857	1.1129	1.0474	1.0183	1.0074
0.95	17.3008	6.1413	2.8175	1.7390	1.2878	1.1136	1.0457	1.0176	1.0072	1.0025	1.0009
0.9	2.8216	1.7376	1.2881	1.1137	1.0449	1.0168	1.0062	1.0020	1.0006	1.0002	1.0001
0.85	1.2892	1.1118	1.0451	1.0165	1.0060	1.0020	1.0007	1.0001	1	1	1
0.8	1.0450	1.0163	1.0061	1.0018	1.0005	1.0002	1.0001	1	1	1	1
0.75	1.0057	1.0018	1.0005	1.0002	1	1	1	1	1	1	1
0.7	1.0006	1.0001	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.1878	64.8840	22.8372	8.7237	3.6890	1.8536	1.2821	1.1069	1.0423	1.0154	1.0068
0.95	22.9252	8.7789	3.6803	1.8609	1.2791	1.1071	1.0413	1.0162	1.0058	1.0022	1.0008
0.9	3.6757	1.8639	1.2823	1.1056	1.0417	1.0157	1.0058	1.0019	1.0007	1.0002	1.0001
0.85	1.2841	1.1061	1.0402	1.0150	1.0056	1.0016	1.0005	1.0001	1.0001	1	1
0.8	1.0392	1.0143	1.0051	1.0018	1.0006	1.0001	1	1	1	1	1
0.75	1.0048	1.0015	1.0005	1.0002	1	1	1	1	1	1	1
0.7	1.0004	1.0001	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.6411	68.3866	25.6351	10.1592	4.3118	1.9887	1.2928	1.1052	1.0419	1.0160	1.0060
0.95	25.7416	10.2504	4.3082	1.9894	1.2940	1.1061	1.0411	1.0152	1.0059	1.0020	1.0007
0.9	4.2920	1.9857	1.2972	1.1049	1.0392	1.0153	1.0052	1.0018	1.0006	1.0002	1.0001
0.85	1.2925	1.1052	1.0407	1.0141	1.0047	1.0021	1.0005	1.0002	1	1	1
0.8	1.0391	1.0145	1.0044	1.0015	1.0005	1.0001	1	1	1	1	1
0.75	1.0046	1.0014	1.0004	1.0001	1	1	1	1	1	1	1
0.7	1.0005	1.0001	1	1	1	1	1	1	1	1	1
0.65	1.0001	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.8$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.1540	7.4004	2.9272	2.0039	1.4907	1.0798	1.0235	1.0082	1.0029	1.0010	1.0003

0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.2617	56.4209	14.9848	2.7223	1.1393	1.0362	1.0115	1.0038	1.0012	1.0003	1.0002
0.95	15.0803	2.7260	1.1416	1.0353	1.0109	1.0038	1.0010	1.0002	1.0001	1.0001	1
0.9	1.1401	1.0352	1.0111	1.0032	1.0011	1.0003	1.0001	1	1	1	1
0.85	1.0100	1.0031	1.0010	1.0003	1.0001	1	1	1	1	1	1
0.8	1.0008	1.0002	1.0001	1	1	1	1	1	1	1	1
0.75	1.0001	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.2445	61.1984	18.3624	3.8077	1.1415	1.0350	1.0111	1.0039	1.0014	1.0005	1.0001
0.95	18.5712	3.8179	1.1418	1.0359	1.0108	1.0035	1.0010	1.0004	1.0001	1.0001	1
0.9	1.1437	1.0345	1.0109	1.0037	1.0011	1.0003	1.0001	1	1	1	1
0.85	1.0103	1.0033	1.0008	1.0002	1	1	1	1	1	1	1
0.8	1.0008	1.0002	1.0001	1	1	1	1	1	1	1	1
0.75	1.0001	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S8 ARL of the EWMA TBEA schemes based on the statistic Z_2 when with both variables A and T following the Gamma distribution assuming a Clayton copula when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.0348	23.2096	8.7623	5.1191	3.5585	2.7067	2.1712	1.7921	1.5140	1.3011	1.1609
0.95	10.3757	5.8654	4.0236	3.0408	2.4348	2.0247	1.7093	1.4566	1.2665	1.1384	1.0658
0.9	4.5221	3.3947	2.7006	2.2412	1.9109	1.6450	1.4116	1.2387	1.1206	1.0559	1.0235
0.85	2.9802	2.4609	2.1064	1.8311	1.5849	1.3717	1.2127	1.1055	1.0476	1.0194	1.0085
0.8	2.2885	2.0046	1.7640	1.5395	1.3348	1.1833	1.0904	1.0401	1.0167	1.0067	1.0022
0.75	1.9265	1.7108	1.4918	1.2988	1.1633	1.0777	1.0331	1.0140	1.0053	1.0017	1.0006
0.7	1.6641	1.4508	1.2701	1.1403	1.0658	1.0286	1.0101	1.0036	1.0014	1.0004	1.0002
0.65	1.4118	1.2390	1.1251	1.0541	1.0232	1.0086	1.0030	1.0009	1.0004	1.0001	1
0.6	1.2129	1.1043	1.0466	1.0189	1.0068	1.0025	1.0007	1.0002	1	1	1

0.55	1.0915	1.0393	1.0144	1.0054	1.0018	1.0005	1.0003	1.0001	1	1	1
0.5	1.0318	1.0120	1.0037	1.0012	1.0004	1.0001	1.0001	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	202.2646	31.6889	9.4903	4.6656	2.9836	2.1760	1.7069	1.4082	1.2193	1.1108	1.0511
0.95	12.8170	5.6958	3.4666	2.4801	1.9246	1.5614	1.3255	1.1744	1.0845	1.0385	1.0161
0.9	4.0625	2.8199	2.1617	1.7485	1.4573	1.2603	1.1335	1.0622	1.0270	1.0113	1.0042
0.85	2.4198	1.9448	1.6128	1.3730	1.2058	1.1020	1.0464	1.0190	1.0075	1.0026	1.0010
0.8	1.7920	1.5114	1.3046	1.1609	1.0775	1.0336	1.0129	1.0046	1.0018	1.0006	1.0002
0.75	1.4264	1.2449	1.1259	1.0579	1.0239	1.0091	1.0035	1.0011	1.0005	1.0002	1.0001
0.7	1.1943	1.0945	1.0428	1.0167	1.0061	1.0022	1.0007	1.0003	1.0001	1	1
0.65	1.0720	1.0309	1.0110	1.0043	1.0013	1.0004	1.0001	1	1	1	1
0.6	1.0219	1.0079	1.0025	1.0008	1.0002	1	1	1	1	1	1
0.55	1.0051	1.0017	1.0004	1.0001	1	1	1	1	1	1	1
0.5	1.0011	1.0003	1.0001	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.0486	40.4362	11.7714	5.1156	2.9562	2.0363	1.5663	1.2975	1.1491	1.0705	1.0312
0.95	17.8059	6.7588	3.6228	2.3771	1.7685	1.4265	1.2228	1.1097	1.0513	1.0213	1.0088
0.9	4.4829	2.8020	2.0226	1.5832	1.3191	1.1681	1.0792	1.0345	1.0148	1.0052	1.0020
0.85	2.3221	1.7825	1.4536	1.2470	1.1224	1.0566	1.0228	1.0096	1.0035	1.0012	1.0005
0.8	1.6109	1.3474	1.1864	1.0902	1.0395	1.0149	1.0060	1.0019	1.0008	1.0003	1.0001
0.75	1.2688	1.1371	1.0639	1.0268	1.0102	1.0038	1.0012	1.0005	1.0001	1	1
0.7	1.1002	1.0444	1.0179	1.0063	1.0020	1.0006	1.0001	1	1	1	1
0.65	1.0290	1.0112	1.0043	1.0013	1.0005	1.0001	1	1	1	1	1
0.6	1.0071	1.0025	1.0007	1.0002	1	1	1	1	1	1	1
0.55	1.0015	1.0003	1.0002	1	1	1	1	1	1	1	1
0.5	1.0002	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.8917	49.0518	14.9436	6.0067	3.1518	2.0355	1.5281	1.2607	1.1244	1.0566	1.0252
0.95	24.6416	8.6964	4.1218	2.4639	1.7419	1.3812	1.1888	1.0880	1.0377	1.0156	1.0071
0.9	5.5673	3.0544	2.0400	1.5362	1.2746	1.1347	1.0613	1.0247	1.0099	1.0037	1.0015
0.85	2.4182	1.7485	1.3945	1.1975	1.0944	1.0412	1.0165	1.0064	1.0022	1.0008	1.0003
0.8	1.5515	1.2905	1.1420	1.0645	1.0269	1.0099	1.0037	1.0013	1.0003	1.0002	1.0001
0.75	1.2114	1.1017	1.0444	1.0173	1.0063	1.0021	1.0009	1.0003	1.0001	1	1
0.7	1.0709	1.0298	1.0109	1.0037	1.0015	1.0003	1.0001	1	1	1	1
0.65	1.0194	1.0069	1.0023	1.0006	1.0002	1.0001	1	1	1	1	1
0.6	1.0041	1.0011	1.0004	1.0001	1	1	1	1	1	1	1
0.55	1.0007	1.0003	1	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.8621	55.4777	18.1676	7.1502	3.5061	2.1142	1.5232	1.2490	1.1147	1.0518	1.0220
0.95	31.2034	11.1286	4.8973	2.6730	1.7665	1.3686	1.1727	1.0778	1.0338	1.0135	1.0060

0.9	7.2365	3.5456	2.1509	1.5443	1.2609	1.1205	1.0519	1.0216	1.0086	1.0029	1.0013
0.85	2.7284	1.8005	1.3842	1.1795	1.0822	1.0345	1.0141	1.0052	1.0019	1.0005	1.0002
0.8	1.5635	1.2674	1.1264	1.0554	1.0220	1.0087	1.0027	1.0011	1.0004	1.0001	1
0.75	1.1870	1.0856	1.0365	1.0142	1.0048	1.0019	1.0006	1.0001	1.0001	1	1
0.7	1.0580	1.0231	1.0086	1.0031	1.0009	1.0003	1.0001	1	1	1	1
0.65	1.0142	1.0048	1.0018	1.0005	1.0002	1	1	1	1	1	1
0.6	1.0029	1.0008	1.0003	1	1	1	1	1	1	1	1
0.55	1.0005	1.0001	1.0001	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
Shewhart											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.7077	57.9214	19.5511	7.8194	3.7260	2.1739	1.5364	1.2479	1.1105	1.0488	1.0210
0.95	34.2817	12.4433	5.3597	2.8301	1.8132	1.3734	1.1706	1.0750	1.0320	1.0132	1.0048
0.9	8.1879	3.8683	2.2421	1.5570	1.2565	1.1163	1.0506	1.0208	1.0079	1.0029	1.0009
0.85	2.9106	1.8478	1.3926	1.1774	1.0772	1.0322	1.0132	1.0048	1.0017	1.0006	1.0003
0.8	1.5919	1.2701	1.1234	1.0519	1.0211	1.0077	1.0026	1.0008	1.0002	1.0001	1
0.75	1.1884	1.0818	1.0335	1.0129	1.0051	1.0016	1.0005	1.0002	1	1	1
0.7	1.0559	1.0213	1.0081	1.0025	1.0008	1.0003	1.0001	1	1	1	1
0.65	1.0133	1.0046	1.0014	1.0004	1.0001	1	1	1	1	1	1
0.6	1.0027	1.0009	1.0002	1.0001	1	1	1	1	1	1	1
0.55	1.0005	1.0001	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.2302	14.0735	5.4922	3.3618	2.4347	1.9335	1.5220	1.2550	1.1168	1.0498	1.0203
0.95	6.2777	3.7635	2.6891	2.1341	1.7484	1.4008	1.1900	1.0843	1.0345	1.0138	1.0055
0.9	2.9732	2.3137	1.9536	1.5928	1.3039	1.1408	1.0613	1.0240	1.0091	1.0032	1.0012
0.85	2.1001	1.8043	1.4665	1.2279	1.1007	1.0411	1.0165	1.0057	1.0022	1.0006	1.0002
0.8	1.6680	1.3615	1.1698	1.0724	1.0300	1.0108	1.0037	1.0012	1.0004	1.0001	1
0.75	1.2722	1.1246	1.0522	1.0201	1.0071	1.0022	1.0009	1.0002	1.0001	1	1
0.7	1.0892	1.0362	1.0133	1.0044	1.0012	1.0005	1.0001	1	1	1	1
0.65	1.0251	1.0090	1.0030	1.0008	1.0002	1.0001	1	1	1	1	1
0.6	1.0056	1.0016	1.0003	1.0001	1	1	1	1	1	1	1
0.55	1.0010	1.0003	1.0001	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.3915	20.9446	5.4969	2.8736	1.9635	1.4646	1.2061	1.0903	1.0382	1.0142	1.0054
0.95	6.8342	3.3115	2.2255	1.6573	1.3046	1.1364	1.0571	1.0221	1.0086	1.0030	1.0011
0.9	2.4822	1.8688	1.4439	1.2016	1.0871	1.0369	1.0135	1.0047	1.0018	1.0005	1.0002
0.85	1.6169	1.2948	1.1330	1.0555	1.0219	1.0078	1.0025	1.0010	1.0003	1.0001	1
0.8	1.1985	1.0856	1.0344	1.0124	1.0040	1.0013	1.0004	1.0002	1.0001	1	1
0.75	1.0545	1.0209	1.0066	1.0026	1.0005	1.0002	1.0001	1	1	1	1
0.7	1.0121	1.0040	1.0012	1.0003	1.0001	1	1	1	1	1	1
0.65	1.0020	1.0005	1.0001	1.0001	1	1	1	1	1	1	1

0.6	1.0003	1.0001	1	1	1	1	1	1	1	1	1
0.55	1.0001	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.0846	32.0260	7.2127	2.9380	1.8206	1.3467	1.1494	1.0629	1.0247	1.0101	1.0036
0.95	10.2313	3.6401	2.1221	1.4965	1.2156	1.0917	1.0382	1.0141	1.0051	1.0019	1.0007
0.9	2.4733	1.7004	1.3128	1.1340	1.0535	1.0216	1.0079	1.0029	1.0009	1.0004	1.0001
0.85	1.4410	1.1923	1.0830	1.0319	1.0118	1.0039	1.0014	1.0004	1.0002	1	1
0.8	1.1208	1.0493	1.0182	1.0066	1.0021	1.0006	1.0003	1.0001	1	1	1
0.75	1.0283	1.0109	1.0035	1.0009	1.0002	1.0002	1	1	1	1	1
0.7	1.0054	1.0018	1.0005	1.0001	1	1	1	1	1	1	1
0.65	1.0007	1.0002	1.0001	1	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.7365	42.2158	10.2505	3.3634	1.7953	1.3086	1.1263	1.0518	1.0200	1.0075	1.0032
0.95	16.1307	4.6484	2.1818	1.4510	1.1841	1.0734	1.0285	1.0117	1.0043	1.0015	1.0005
0.9	2.7094	1.6483	1.2644	1.1093	1.0421	1.0167	1.0063	1.0021	1.0007	1.0003	1.0001
0.85	1.3797	1.1569	1.0643	1.0238	1.0089	1.0030	1.0009	1.0003	1.0002	1	1
0.8	1.0908	1.0366	1.0131	1.0045	1.0014	1.0004	1.0001	1	1	1	1
0.75	1.0209	1.0074	1.0024	1.0007	1.0002	1.0001	1	1	1	1	1
0.7	1.0038	1.0012	1.0003	1.0001	1	1	1	1	1	1	1
0.65	1.0005	1.0001	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.9023	50.6382	14.2216	4.2174	1.8554	1.3020	1.1172	1.0459	1.0179	1.0070	1.0023
0.95	22.8558	6.6589	2.3846	1.4432	1.1696	1.0660	1.0263	1.0095	1.0036	1.0011	1.0004
0.9	3.3341	1.6859	1.2502	1.0979	1.0385	1.0143	1.0047	1.0016	1.0006	1.0001	1
0.85	1.3722	1.1405	1.0550	1.0223	1.0078	1.0024	1.0008	1.0001	1.0001	1	1
0.8	1.0811	1.0316	1.0116	1.0040	1.0010	1.0005	1.0001	1.0001	1	1	1
0.75	1.0162	1.0060	1.0020	1.0005	1.0002	1.0001	1	1	1	1	1
0.7	1.0027	1.0008	1.0003	1.0001	1	1	1	1	1	1	1
0.65	1.0004	1.0001	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.8241	54.4268	16.0113	4.8203	1.9177	1.3009	1.1130	1.0451	1.0173	1.0066	1.0022

0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.1407	32.2990	3.1859	1.3081	1.0731	1.0237	1.0083	1.0028	1.0008	1.0004	1.0001
0.95	4.8563	1.5052	1.1049	1.0316	1.0100	1.0033	1.0008	1.0004	1.0001	1	1
0.9	1.1494	1.0418	1.0136	1.0045	1.0012	1.0004	1.0002	1	1	1	1
0.85	1.0179	1.0054	1.0017	1.0004	1.0001	1.0001	1	1	1	1	1
0.8	1.0025	1.0008	1.0002	1	1	1	1	1	1	1	1
0.75	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.7027	44.8134	5.8472	1.3076	1.0690	1.0224	1.0071	1.0024	1.0008	1.0002	1.0001
0.95	11.6548	1.5263	1.0977	1.0282	1.0095	1.0030	1.0009	1.0003	1.0001	1	1
0.9	1.1412	1.0392	1.0125	1.0038	1.0013	1.0003	1.0001	1	1	1	1
0.85	1.0168	1.0051	1.0017	1.0004	1.0001	1	1	1	1	1	1
0.8	1.0020	1.0007	1.0001	1	1	1	1	1	1	1	1
0.75	1.0002	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.4896	50.2809	8.5383	1.3165	1.0689	1.0214	1.0076	1.0024	1.0007	1.0003	1.0001
0.95	15.9287	1.5747	1.0963	1.0280	1.0094	1.0029	1.0010	1.0004	1	1	1
0.9	1.1430	1.0383	1.0128	1.0040	1.0013	1.0004	1.0001	1	1	1	1
0.85	1.0157	1.0052	1.0016	1.0005	1.0001	1	1	1	1	1	1
0.8	1.0017	1.0005	1.0001	1	1	1	1	1	1	1	1
0.75	1.0003	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S9 ARL of the EWMA TBEA schemes based on the statistic Z_3 with both variables A and T following the Gamma distribution assuming a Clayton copula when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in$

{0.2,0.5,0.8}

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.2002	22.2476	8.3972	4.9284	3.4451	2.6259	2.1128	1.7490	1.4737	1.2731	1.1406
0.95	10.2200	5.9544	4.1376	3.1333	2.5150	2.0977	1.7782	1.5141	1.3073	1.1647	1.0798
0.9	4.4349	3.4540	2.8034	2.3602	2.0404	1.7765	1.5366	1.3288	1.1770	1.0869	1.0390
0.85	2.9183	2.4969	2.1915	1.9616	1.7505	1.5296	1.3273	1.1814	1.0895	1.0403	1.0170
0.8	2.2349	2.0409	1.8792	1.6989	1.4885	1.3042	1.1643	1.0805	1.0349	1.0142	1.0062
0.75	1.9279	1.7941	1.6166	1.4196	1.2513	1.1330	1.0622	1.0271	1.0108	1.0040	1.0016
0.7	1.6778	1.4984	1.3199	1.1808	1.0908	1.0421	1.0177	1.0063	1.0024	1.0007	1.0003
0.65	1.3419	1.2020	1.1083	1.0498	1.0225	1.0091	1.0030	1.0011	1.0003	1.0001	1.0001
0.6	1.0984	1.0476	1.0216	1.0080	1.0030	1.0012	1.0004	1.0001	1	1	1
0.55	1.0154	1.0057	1.0023	1.0007	1.0002	1.0001	1	1	1	1	1
0.5	1.0009	1.0003	1.0001	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.3674	28.9118	8.5706	4.2919	2.7775	2.0448	1.6124	1.3466	1.1777	1.0858	1.0380
0.95	12.0032	5.5862	3.4790	2.5067	1.9422	1.5819	1.3342	1.1786	1.0868	1.0383	1.0157
0.9	3.8387	2.8129	2.2044	1.8090	1.5153	1.3015	1.1605	1.0781	1.0347	1.0141	1.0055
0.85	2.3030	1.9437	1.6548	1.4197	1.2448	1.1258	1.0606	1.0249	1.0109	1.0037	1.0014
0.8	1.7114	1.4965	1.3120	1.1747	1.0853	1.0386	1.0158	1.0060	1.0024	1.0008	1.0003
0.75	1.3302	1.1915	1.0988	1.0471	1.0206	1.0083	1.0032	1.0010	1.0004	1.0001	1
0.7	1.0897	1.0436	1.0196	1.0077	1.0028	1.0011	1.0003	1.0001	1	1	1
0.65	1.0131	1.0051	1.0019	1.0006	1.0003	1.0001	1	1	1	1	1
0.6	1.0009	1.0002	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.1172	37.4741	10.2931	4.4944	2.6643	1.8776	1.4706	1.2408	1.1153	1.0526	1.0219
0.95	16.5624	6.5257	3.5639	2.3571	1.7626	1.4241	1.2209	1.1061	1.0474	1.0204	1.0079
0.9	4.1369	2.7388	2.0350	1.6157	1.3494	1.1828	1.0878	1.0388	1.0157	1.0064	1.0024
0.85	2.1564	1.7381	1.4488	1.2535	1.1312	1.0610	1.0264	1.0110	1.0040	1.0015	1.0005
0.8	1.4889	1.2895	1.1584	1.0782	1.0356	1.0145	1.0057	1.0020	1.0007	1.0002	1.0001
0.75	1.1559	1.0782	1.0366	1.0151	1.0059	1.0021	1.0007	1.0002	1.0001	1	1
0.7	1.0282	1.0121	1.0046	1.0016	1.0005	1.0002	1.0001	1	1	1	1
0.65	1.0024	1.0010	1.0002	1.0001	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.8537	45.7003	12.8603	5.0705	2.7398	1.8316	1.4116	1.2014	1.0929	1.0409	1.0165
0.95	22.8582	8.3117	3.9817	2.4080	1.7126	1.3658	1.1761	1.0836	1.0348	1.0143	1.0056

0.9	5.0021	2.9477	2.0201	1.5480	1.2847	1.1394	1.0642	1.0264	1.0107	1.0044	1.0015
0.85	2.1816	1.6607	1.3703	1.1927	1.0951	1.0427	1.0174	1.0066	1.0024	1.0009	1.0004
0.8	1.3914	1.2156	1.1092	1.0512	1.0213	1.0083	1.0032	1.0011	1.0003	1.0001	1
0.75	1.1003	1.0472	1.0213	1.0083	1.0029	1.0011	1.0003	1.0001	1.0001	1	1
0.7	1.0143	1.0057	1.0023	1.0007	1.0003	1.0001	1	1	1	1	1
0.65	1.0009	1.0004	1.0001	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.0281	53.5542	15.8954	5.9027	2.9426	1.8682	1.4017	1.1839	1.0830	1.0361	1.0148
0.95	29.8781	10.8558	4.7580	2.5775	1.7348	1.3454	1.1626	1.0717	1.0305	1.0121	1.0046
0.9	6.4964	3.4074	2.1258	1.5409	1.2626	1.1214	1.0544	1.0224	1.0086	1.0034	1.0011
0.85	2.3791	1.6900	1.3487	1.1701	1.0787	1.0341	1.0130	1.0051	1.0018	1.0007	1.0003
0.8	1.3652	1.1870	1.0879	1.0399	1.0169	1.0066	1.0022	1.0007	1.0003	1.0001	1
0.75	1.0784	1.0358	1.0148	1.0059	1.0020	1.0009	1.0003	1.0001	1	1	1
0.7	1.0100	1.0039	1.0015	1.0005	1.0001	1	1	1	1	1	1
0.65	1.0006	1.0002	1.0001	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.6005	56.3882	17.4291	6.4649	3.0840	1.8935	1.4034	1.1827	1.0814	1.0330	1.0135
0.95	32.8614	12.2406	5.1755	2.7270	1.7684	1.3461	1.1597	1.0688	1.0285	1.0110	1.0046
0.9	7.4477	3.7598	2.2297	1.5693	1.2613	1.1183	1.0514	1.0207	1.0079	1.0030	1.0011
0.85	2.5494	1.7418	1.3557	1.1655	1.0735	1.0321	1.0123	1.0046	1.0015	1.0007	1.0002
0.8	1.3812	1.1832	1.0845	1.0378	1.0147	1.0060	1.0022	1.0008	1.0003	1.0001	1
0.75	1.0740	1.0328	1.0141	1.0053	1.0016	1.0007	1.0001	1.0001	1	1	1
0.7	1.0092	1.0030	1.0013	1.0004	1.0002	1.0001	1	1	1	1	1
0.65	1.0006	1.0002	1.0001	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.5094	13.3713	5.3156	3.2806	2.3928	1.8864	1.5146	1.2633	1.1259	1.0539	1.0230
0.95	6.2715	3.8656	2.7837	2.2093	1.8308	1.4894	1.2520	1.1175	1.0514	1.0208	1.0080
0.9	2.9655	2.3652	2.0335	1.7418	1.4303	1.2131	1.0955	1.0410	1.0166	1.0061	1.0022
0.85	2.0701	1.8936	1.6171	1.3286	1.1521	1.0652	1.0270	1.0100	1.0039	1.0012	1.0004
0.8	1.7318	1.4255	1.2027	1.0865	1.0357	1.0138	1.0050	1.0016	1.0006	1.0002	1.0001
0.75	1.2115	1.0893	1.0360	1.0139	1.0049	1.0015	1.0005	1.0001	1.0001	1	1
0.7	1.0265	1.0095	1.0036	1.0012	1.0005	1.0001	1	1	1	1	1
0.65	1.0014	1.0004	1.0001	1.0001	1	1	1	1	1	1	1

0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.7521	18.9448	5.0119	2.7300	1.8798	1.4360	1.2067	1.0915	1.0369	1.0151	1.0059
0.95	6.6263	3.3466	2.2718	1.7001	1.3521	1.1637	1.0718	1.0282	1.0110	1.0038	1.0015
0.9	2.4234	1.9018	1.5048	1.2426	1.1083	1.0445	1.0172	1.0070	1.0022	1.0008	1.0003
0.85	1.6014	1.2955	1.1319	1.0553	1.0217	1.0082	1.0028	1.0009	1.0003	1.0001	1.0001
0.8	1.1215	1.0496	1.0196	1.0075	1.0026	1.0008	1.0004	1	1	1	1
0.75	1.0119	1.0043	1.0015	1.0005	1.0002	1.0001	1	1	1	1	1
0.7	1.0006	1.0002	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.8873	30.2661	6.0651	2.6856	1.7294	1.3286	1.1433	1.0598	1.0240	1.0095	1.0031
0.95	9.6050	3.6306	2.1519	1.5366	1.2413	1.1043	1.0432	1.0164	1.0066	1.0023	1.0008
0.9	2.3904	1.7156	1.3352	1.1456	1.0612	1.0242	1.0093	1.0034	1.0010	1.0004	1.0001
0.85	1.3776	1.1643	1.0680	1.0262	1.0099	1.0038	1.0014	1.0003	1.0001	1.0001	1
0.8	1.0560	1.0227	1.0077	1.0030	1.0010	1.0004	1.0001	1	1	1	1
0.75	1.0043	1.0012	1.0004	1.0001	1	1	1	1	1	1	1
0.7	1.0001	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.6816	43.0401	8.3719	2.8878	1.6952	1.2879	1.1225	1.0497	1.0196	1.0075	1.0028
0.95	15.1630	4.5990	2.2059	1.4869	1.2026	1.0853	1.0345	1.0126	1.0048	1.0016	1.0007
0.9	2.6085	1.6641	1.2750	1.1122	1.0473	1.0191	1.0067	1.0025	1.0007	1.0003	1.0001
0.85	1.2951	1.1211	1.0487	1.0185	1.0069	1.0023	1.0010	1.0003	1.0001	1	1
0.8	1.0374	1.0135	1.0049	1.0016	1.0005	1.0001	1.0001	1	1	1	1
0.75	1.0025	1.0008	1.0002	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.7920	54.0890	11.9897	3.3296	1.7259	1.2762	1.1131	1.0444	1.0182	1.0068	1.0025

0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.3984	18.8359	2.8383	1.5169	1.1321	1.0408	1.0144	1.0046	1.0014	1.0006	1.0003
0.95	3.8487	1.8637	1.2014	1.0515	1.0164	1.0054	1.0018	1.0007	1.0002	1	1
0.9	1.2305	1.0450	1.0130	1.0043	1.0014	1.0004	1.0001	1	1	1	1
0.85	1.0065	1.0020	1.0005	1.0002	1.0001	1	1	1	1	1	1
0.8	1.0001	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.3547	33.6401	3.6070	1.4833	1.1175	1.0356	1.0125	1.0045	1.0013	1.0005	1.0001
0.95	7.2091	1.8814	1.1662	1.0447	1.0147	1.0047	1.0012	1.0004	1.0001	1.0001	1
0.9	1.1748	1.0367	1.0108	1.0034	1.0011	1.0003	1.0001	1	1	1	1
0.85	1.0047	1.0016	1.0004	1.0001	1	1	1	1	1	1	1
0.8	1.0001	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.7480	48.2658	5.5643	1.5070	1.1113	1.0334	1.0114	1.0040	1.0012	1.0005	1.0001
0.95	13.8142	2.1569	1.1599	1.0408	1.0129	1.0042	1.0014	1.0005	1.0001	1.0001	1
0.9	1.1552	1.0317	1.0097	1.0030	1.0008	1.0003	1.0001	1	1	1	1
0.85	1.0042	1.0011	1.0004	1.0001	1	1	1	1	1	1	1
0.8	1	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2

1	200.5533	54.6854	7.0234	1.5451	1.1118	1.0342	1.0113	1.0037	1.0012	1.0004	1.0002
0.95	17.5716	2.5233	1.1596	1.0388	1.0125	1.0041	1.0014	1.0005	1.0001	1	1
0.9	1.1551	1.0320	1.0094	1.0031	1.0009	1.0003	1.0001	1	1	1	1
0.85	1.0044	1.0010	1.0003	1.0001	1.0001	1	1	1	1	1	1
0.8	1.0001	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S10 *ARL* of the EWMA TBEA schemes based on the statistic Z_1 with both variables A and T following the Gamma distribution assuming a Frank copula when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.3103	32.3707	10.2887	5.4241	3.6751	2.7950	2.2997	1.9745	1.7356	1.5301	1.3453
0.95	10.3153	5.4289	3.6714	2.8048	2.2973	1.9767	1.7339	1.5285	1.3431	1.2054	1.1083
0.9	3.6750	2.8019	2.2959	1.9754	1.7347	1.5318	1.3514	1.2034	1.1094	1.0547	1.0251
0.85	2.2975	1.9747	1.7375	1.5321	1.3481	1.2052	1.1089	1.0527	1.0236	1.0102	1.0042
0.8	1.7383	1.5327	1.3478	1.2033	1.1088	1.0536	1.0235	1.0096	1.0037	1.0015	1.0006
0.75	1.3489	1.2043	1.1089	1.0525	1.0240	1.0094	1.0042	1.0015	1.0005	1.0002	1.0001
0.7	1.1083	1.0521	1.0233	1.0092	1.0035	1.0014	1.0005	1.0001	1	1	1
0.65	1.0227	1.0098	1.0037	1.0014	1.0004	1.0002	1	1	1	1	1
0.6	1.0036	1.0013	1.0005	1.0002	1	1	1	1	1	1	1
0.55	1.0003	1.0001	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.7179	44.3157	13.4490	6.0786	3.6774	2.6316	2.0755	1.7216	1.4708	1.2833	1.1582
0.95	13.4477	6.0792	3.6765	2.6354	2.0702	1.7221	1.4675	1.2855	1.1571	1.0801	1.0373
0.9	3.6790	2.6288	2.0727	1.7189	1.4679	1.2855	1.1568	1.0798	1.0370	1.0164	1.0068
0.85	2.0726	1.7223	1.4718	1.2843	1.1561	1.0805	1.0368	1.0157	1.0068	1.0028	1.0010
0.8	1.4711	1.2854	1.1596	1.0797	1.0359	1.0156	1.0062	1.0026	1.0009	1.0004	1.0002
0.75	1.1572	1.0786	1.0368	1.0157	1.0062	1.0020	1.0007	1.0003	1.0001	1	1
0.7	1.0349	1.0155	1.0063	1.0026	1.0009	1.0003	1.0001	1	1	1	1
0.65	1.0059	1.0025	1.0007	1.0002	1.0001	1	1	1	1	1	1
0.6	1.0006	1.0002	1.0001	1	1	1	1	1	1	1	1
0.55	1.0001	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.4166	55.8876	18.1907	7.6290	4.1312	2.7158	2.0256	1.6256	1.3699	1.2066	1.1053
0.95	18.1721	7.6078	4.1244	2.7237	2.0245	1.6222	1.3703	1.2083	1.1051	1.0514	1.0238

0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	202.1420	19.0452	5.9146	3.3881	2.4399	1.9684	1.6524	1.3626	1.1560	1.0619	1.0228
0.95	5.9139	3.3936	2.4347	1.9654	1.6547	1.3604	1.1598	1.0603	1.0219	1.0075	1.0028
0.9	2.4432	1.9682	1.6535	1.3625	1.1580	1.0601	1.0221	1.0076	1.0027	1.0007	1.0003
0.85	1.6549	1.3643	1.1594	1.0615	1.0221	1.0075	1.0026	1.0008	1.0003	1.0001	1
0.8	1.1583	1.0604	1.0211	1.0071	1.0023	1.0007	1.0002	1.0001	1	1	1
0.75	1.0211	1.0070	1.0020	1.0008	1.0002	1.0001	1	1	1	1	1
0.7	1.0022	1.0007	1.0002	1.0001	1	1	1	1	1	1	1
0.65	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.1423	29.4488	7.2003	3.4273	2.2733	1.7368	1.3848	1.1671	1.0641	1.0226	1.0084
0.95	7.1974	3.4280	2.2737	1.7356	1.3881	1.1683	1.0651	1.0224	1.0080	1.0028	1.0009
0.9	2.2762	1.7400	1.3856	1.1706	1.0620	1.0229	1.0083	1.0026	1.0009	1.0004	1.0001
0.85	1.3889	1.1693	1.0629	1.0217	1.0075	1.0025	1.0007	1.0002	1.0001	1	1
0.8	1.0633	1.0221	1.0073	1.0024	1.0007	1.0002	1.0001	1	1	1	1
0.75	1.0073	1.0022	1.0009	1.0002	1.0001	1	1	1	1	1	1
0.7	1.0008	1.0001	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.8943	41.4872	10.1616	3.9705	2.3049	1.6504	1.2960	1.1176	1.0439	1.0159	1.0054
0.95	10.1169	3.9603	2.3119	1.6502	1.2969	1.1194	1.0440	1.0149	1.0051	1.0018	1.0007
0.9	2.3138	1.6480	1.3000	1.1175	1.0432	1.0149	1.0054	1.0017	1.0007	1.0002	1.0001
0.85	1.3001	1.1180	1.0426	1.0145	1.0047	1.0018	1.0005	1.0001	1.0001	1	1
0.8	1.0437	1.0145	1.0047	1.0016	1.0006	1.0002	1.0001	1	1	1	1
0.75	1.0044	1.0015	1.0005	1.0002	1	1	1	1	1	1	1
0.7	1.0004	1.0001	1	1	1	1	1	1	1	1	1
0.65	1.0001	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.7097	52.8270	14.4619	5.1069	2.5387	1.6453	1.2639	1.0976	1.0355	1.0129	1.0043

0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.5389	9.8082	2.5553	1.5907	1.0987	1.0200	1.0048	1.0014	1.0003	1.0001	1.0001
0.95	2.5587	1.5945	1.0994	1.0194	1.0048	1.0012	1.0003	1.0001	1	1	1
0.9	1.1002	1.0181	1.0045	1.0011	1.0004	1.0001	1	1	1	1	1
0.85	1.0047	1.0012	1.0002	1.0001	1	1	1	1	1	1	1
0.8	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.8390	19.0495	2.8892	1.5026	1.0769	1.0162	1.0039	1.0011	1.0003	1.0001	1.0001
0.95	2.8834	1.5028	1.0756	1.0160	1.0040	1.0008	1.0002	1.0001	1	1	1
0.9	1.0777	1.0159	1.0035	1.0009	1.0003	1	1	1	1	1	1
0.85	1.0037	1.0010	1.0002	1.0001	1	1	1	1	1	1	1
0.8	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.9172	30.9556	3.9466	1.4758	1.0678	1.0145	1.0036	1.0010	1.0003	1.0001	1
0.95	3.9410	1.4736	1.0671	1.0134	1.0037	1.0008	1.0002	1.0001	1	1	1
0.9	1.0672	1.0150	1.0034	1.0007	1.0002	1.0001	1	1	1	1	1
0.85	1.0031	1.0007	1.0002	1	1	1	1	1	1	1	1
0.8	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2

1	201.3153	41.5253	6.5321	1.5166	1.0631	1.0128	1.0033	1.0008	1.0003	1.0001	1
0.95	6.5469	1.5210	1.0632	1.0131	1.0031	1.0007	1.0002	1.0001	1	1	1
0.9	1.0634	1.0126	1.0030	1.0008	1.0002	1.0001	1	1	1	1	1
0.85	1.0028	1.0007	1.0002	1.0001	1	1	1	1	1	1	1
0.8	1.0002	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.2407	45.5871	8.3064	1.5730	1.0634	1.0124	1.0034	1.0008	1.0002	1.0001	1
0.95	8.3063	1.5760	1.0646	1.0122	1.0029	1.0008	1.0002	1.0001	1	1	1
0.9	1.0607	1.0126	1.0029	1.0008	1.0002	1.0001	1	1	1	1	1
0.85	1.0027	1.0008	1.0002	1.0001	1	1	1	1	1	1	1
0.8	1.0001	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S11 ARL of the EWMA TBEA schemes based on the statistic Z_2 with both variables A and T following the Gamma distribution assuming a Frank copula when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	202.3051	24.8793	9.2736	5.3601	3.7021	2.8095	2.2427	1.8648	1.5712	1.3400	1.1792
0.95	10.9693	6.1582	4.1886	3.1639	2.5186	2.0913	1.7758	1.5164	1.3056	1.1579	1.0753
0.9	4.7222	3.5269	2.8053	2.3129	1.9768	1.7086	1.4707	1.2731	1.1382	1.0636	1.0279
0.85	3.0995	2.5508	2.1685	1.8940	1.6549	1.4294	1.2449	1.1210	1.0570	1.0239	1.0096
0.8	2.3622	2.0583	1.8295	1.6067	1.3898	1.2153	1.1070	1.0478	1.0194	1.0078	1.0026
0.75	1.9775	1.7742	1.5636	1.3528	1.1947	1.0922	1.0397	1.0165	1.0064	1.0022	1.0008
0.7	1.7356	1.5279	1.3245	1.1722	1.0815	1.0346	1.0137	1.0051	1.0017	1.0005	1.0002
0.65	1.4872	1.2945	1.1509	1.0695	1.0282	1.0111	1.0037	1.0014	1.0005	1.0002	1
0.6	1.2641	1.1358	1.0583	1.0246	1.0092	1.0032	1.0010	1.0003	1.0001	1	1
0.55	1.1158	1.0507	1.0202	1.0071	1.0022	1.0005	1.0002	1.0001	1	1	1
0.5	1.0431	1.0164	1.0055	1.0022	1.0007	1.0002	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.7479	34.2308	10.3603	5.0208	3.1661	2.2864	1.7825	1.4586	1.2491	1.1245	1.0577
0.95	14.1366	6.1751	3.7156	2.6211	2.0249	1.6350	1.3714	1.1968	1.0955	1.0422	1.0174

0.9	4.3619	2.9946	2.2860	1.8368	1.5282	1.3025	1.1545	1.0739	1.0315	1.0132	1.0048
0.85	2.5500	2.0472	1.7001	1.4331	1.2422	1.1220	1.0541	1.0232	1.0090	1.0034	1.0013
0.8	1.8807	1.5892	1.3610	1.1940	1.0938	1.0403	1.0165	1.0065	1.0024	1.0007	1.0003
0.75	1.5012	1.2959	1.1543	1.0700	1.0295	1.0116	1.0041	1.0014	1.0004	1.0002	1.0001
0.7	1.2397	1.1199	1.0538	1.0218	1.0082	1.0028	1.0012	1.0003	1.0001	1.0001	1
0.65	1.0921	1.0396	1.0154	1.0052	1.0019	1.0006	1.0002	1.0001	1	1	1
0.6	1.0287	1.0116	1.0040	1.0011	1.0003	1.0001	1	1	1	1	1
0.55	1.0078	1.0025	1.0008	1.0003	1	1	1	1	1	1	1
0.5	1.0016	1.0005	1.0001	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.3074	44.4856	13.4413	5.7269	3.2389	2.1935	1.6540	1.3492	1.1758	1.0819	1.0364
0.95	20.6796	7.7024	4.0265	2.5864	1.8957	1.4983	1.2656	1.1274	1.0580	1.0260	1.0101
0.9	5.0625	3.0866	2.1774	1.6895	1.3822	1.1971	1.0951	1.0424	1.0180	1.0071	1.0029
0.85	2.5343	1.9159	1.5385	1.2960	1.1492	1.0704	1.0294	1.0115	1.0047	1.0016	1.0006
0.8	1.7217	1.4243	1.2264	1.1127	1.0505	1.0203	1.0078	1.0025	1.0010	1.0004	1.0001
0.75	1.3313	1.1708	1.0815	1.0354	1.0129	1.0053	1.0017	1.0006	1.0002	1.0001	1
0.7	1.1305	1.0584	1.0241	1.0088	1.0031	1.0011	1.0004	1.0001	1	1	1
0.65	1.0406	1.0162	1.0055	1.0021	1.0006	1.0002	1.0001	1	1	1	1
0.6	1.0105	1.0036	1.0013	1.0004	1.0001	1	1	1	1	1	1
0.55	1.0023	1.0008	1.0003	1.0001	1	1	1	1	1	1	1
0.5	1.0004	1.0001	1.0001	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.8984	53.8787	17.3337	7.0017	3.5977	2.2464	1.6250	1.3094	1.1466	1.0671	1.0290
0.95	28.5225	10.3041	4.7802	2.7784	1.8952	1.4605	1.2262	1.1050	1.0463	1.0192	1.0082
0.9	6.6557	3.5361	2.2647	1.6607	1.3406	1.1644	1.0744	1.0316	1.0129	1.0050	1.0019
0.85	2.7706	1.9247	1.4893	1.2489	1.1164	1.0530	1.0204	1.0080	1.0029	1.0011	1.0003
0.8	1.6908	1.3641	1.1833	1.0818	1.0359	1.0145	1.0052	1.0019	1.0006	1.0001	1.0001
0.75	1.2714	1.1316	1.0580	1.0240	1.0091	1.0032	1.0011	1.0004	1.0001	1	1
0.7	1.0935	1.0407	1.0161	1.0059	1.0024	1.0008	1.0002	1.0001	1	1	1
0.65	1.0285	1.0103	1.0035	1.0013	1.0003	1.0001	1.0001	1	1	1	1
0.6	1.0062	1.0023	1.0007	1.0002	1.0001	1	1	1	1	1	1
0.55	1.0014	1.0004	1.0001	1	1	1	1	1	1	1	1
0.5	1.0002	1.0001	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.2187	60.7992	21.1049	8.5518	4.1295	2.3920	1.6460	1.2979	1.1374	1.0606	1.0251
0.95	36.0984	13.4129	5.9036	3.1301	1.9804	1.4576	1.2107	1.0948	1.0400	1.0165	1.0063
0.9	8.8993	4.2959	2.4812	1.6986	1.3287	1.1480	1.0655	1.0275	1.0102	1.0039	1.0016
0.85	3.2605	2.0446	1.4959	1.2336	1.1037	1.0444	1.0180	1.0065	1.0026	1.0009	1.0004
0.8	1.7461	1.3506	1.1646	1.0712	1.0291	1.0115	1.0041	1.0017	1.0005	1.0001	1
0.75	1.2524	1.1149	1.0482	1.0188	1.0071	1.0025	1.0009	1.0002	1.0001	1	1
0.7	1.0783	1.0319	1.0122	1.0043	1.0014	1.0005	1.0001	1.0001	1	1	1
0.65	1.0210	1.0078	1.0027	1.0007	1.0003	1.0001	1	1	1	1	1
0.6	1.0044	1.0016	1.0003	1.0002	1.0001	1	1	1	1	1	1

0.55	1.0008	1.0003	1.0001	1	1	1	1	1	1	1	1
0.5	1.0001	1.0001	1	1	1	1	1	1	1	1	1
Shewhart											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.2626	63.5045	22.7706	9.2955	4.4231	2.4983	1.6702	1.3018	1.1348	1.0578	1.0244
0.95	39.3670	14.9445	6.5687	3.3539	2.0441	1.4728	1.2105	1.0933	1.0401	1.0156	1.0062
0.9	10.1121	4.7471	2.6460	1.7297	1.3324	1.1471	1.0630	1.0254	1.0098	1.0039	1.0015
0.85	3.5891	2.1416	1.5175	1.2333	1.1010	1.0427	1.0177	1.0063	1.0024	1.0007	1.0002
0.8	1.8018	1.3671	1.1610	1.0677	1.0265	1.0101	1.0038	1.0013	1.0004	1.0002	1.0001
0.75	1.2550	1.1102	1.0466	1.0183	1.0064	1.0023	1.0007	1.0003	1.0001	1	1
0.7	1.0758	1.0297	1.0115	1.0043	1.0014	1.0004	1.0002	1	1	1	1
0.65	1.0199	1.0071	1.0024	1.0009	1.0002	1.0001	1	1	1	1	1
0.6	1.0043	1.0012	1.0004	1.0001	1	1	1	1	1	1	1
0.55	1.0008	1.0003	1.0001	1	1	1	1	1	1	1	1
0.5	1.0001	1.0001	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.5529	15.1896	5.8971	3.5857	2.5532	2.0136	1.6462	1.3278	1.1330	1.0495	1.0192
0.95	6.7384	4.0210	2.8466	2.2184	1.8469	1.5212	1.2419	1.0944	1.0352	1.0129	1.0051
0.9	3.1481	2.4327	2.0132	1.7195	1.4050	1.1768	1.0660	1.0236	1.0084	1.0031	1.0010
0.85	2.1697	1.8842	1.6077	1.3121	1.1235	1.0473	1.0161	1.0058	1.0018	1.0006	1.0003
0.8	1.7829	1.4929	1.2283	1.0889	1.0309	1.0112	1.0037	1.0011	1.0006	1.0001	1.0001
0.75	1.3864	1.1631	1.0603	1.0214	1.0070	1.0021	1.0007	1.0002	1.0001	1	1
0.7	1.1146	1.0425	1.0141	1.0045	1.0016	1.0005	1.0001	1	1	1	1
0.65	1.0289	1.0093	1.0033	1.0009	1.0003	1	1	1	1	1	1
0.6	1.0065	1.0019	1.0005	1.0001	1	1	1	1	1	1	1
0.55	1.0012	1.0003	1.0001	1	1	1	1	1	1	1	1
0.5	1.0003	1.0001	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.1219	22.1606	6.0260	3.0959	2.0927	1.5813	1.2555	1.0974	1.0356	1.0128	1.0047
0.95	7.5084	3.5964	2.3567	1.7784	1.3957	1.1620	1.0577	1.0209	1.0077	1.0025	1.0010
0.9	2.6548	1.9756	1.5611	1.2548	1.0997	1.0348	1.0123	1.0043	1.0015	1.0005	1.0003
0.85	1.7366	1.3950	1.1637	1.0586	1.0205	1.0072	1.0025	1.0007	1.0002	1.0001	1
0.8	1.2614	1.1017	1.0365	1.0126	1.0037	1.0013	1.0004	1.0002	1	1	1
0.75	1.0604	1.0208	1.0067	1.0021	1.0007	1.0001	1.0001	1	1	1	1
0.7	1.0119	1.0039	1.0012	1.0003	1.0001	1	1	1	1	1	1
0.65	1.0023	1.0006	1.0002	1.0001	1	1	1	1	1	1	1
0.6	1.0003	1.0001	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.3161	32.4191	7.7518	3.2532	1.9953	1.4374	1.1705	1.0614	1.0224	1.0082	1.0029

0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.8$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.6860	6.3184	2.8409	1.9538	1.3737	1.0664	1.0165	1.0048	1.0017	1.0005	1.0002
0.95	3.1364	2.0793	1.6257	1.1251	1.0276	1.0073	1.0027	1.0007	1.0003	1.0001	1
0.9	1.8313	1.2508	1.0462	1.0120	1.0034	1.0010	1.0002	1.0001	1	1	1
0.85	1.0829	1.0195	1.0055	1.0015	1.0005	1.0001	1	1	1	1	1
0.8	1.0085	1.0022	1.0008	1.0002	1.0001	1	1	1	1	1	1
0.75	1.0011	1.0003	1.0001	1	1	1	1	1	1	1	1
0.7	1.0002	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.1243	7.8463	2.4577	1.5370	1.0890	1.0204	1.0055	1.0021	1.0006	1.0001	1.0001
0.95	2.7806	1.7602	1.1583	1.0306	1.0077	1.0023	1.0008	1.0002	1.0001	1	1
0.9	1.2731	1.0480	1.0113	1.0037	1.0010	1.0002	1	1	1	1	1
0.85	1.0169	1.0043	1.0012	1.0004	1.0001	1	1	1	1	1	1
0.8	1.0017	1.0004	1.0002	1	1	1	1	1	1	1	1
0.75	1.0002	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.2087	14.3869	2.5875	1.3824	1.0622	1.0144	1.0044	1.0011	1.0005	1.0001	1.0001
0.95	3.1031	1.6023	1.0981	1.0205	1.0056	1.0017	1.0005	1.0002	1.0001	1	1
0.9	1.1636	1.0306	1.0085	1.0016	1.0005	1.0002	1	1	1	1	1
0.85	1.0116	1.0034	1.0009	1.0002	1.0001	1	1	1	1	1	1
0.8	1.0010	1.0003	1.0001	1	1	1	1	1	1	1	1
0.75	1.0001	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2

1	199.4425	24.2370	3.1582	1.3299	1.0518	1.0124	1.0035	1.0010	1.0003	1.0001	1.0001
0.95	4.4028	1.5437	1.0808	1.0167	1.0047	1.0012	1.0004	1.0001	1.0001	1	1
0.9	1.1286	1.0243	1.0062	1.0018	1.0005	1.0001	1	1	1	1	1
0.85	1.0085	1.0024	1.0006	1.0002	1.0001	1	1	1	1	1	1
0.8	1.0008	1.0002	1.0001	1	1	1	1	1	1	1	1
0.75	1.0001	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.4850	33.5482	4.6488	1.3192	1.0464	1.0113	1.0035	1.0010	1.0003	1.0001	1.0001
0.95	7.4991	1.5790	1.0717	1.0155	1.0043	1.0011	1.0003	1.0001	1	1	1
0.9	1.1155	1.0219	1.0059	1.0015	1.0004	1.0002	1.0001	1	1	1	1
0.85	1.0076	1.0019	1.0005	1.0003	1.0001	1	1	1	1	1	1
0.8	1.0007	1.0001	1.0001	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.4643	37.6717	5.7218	1.3296	1.0457	1.0110	1.0034	1.0009	1.0004	1.0001	1.0001
0.95	9.5462	1.6547	1.0695	1.0157	1.0041	1.0011	1.0002	1.0001	1.0001	1	1
0.9	1.1137	1.0215	1.0049	1.0014	1.0004	1.0001	1	1	1	1	1
0.85	1.0074	1.0020	1.0006	1.0001	1	1	1	1	1	1	1
0.8	1.0006	1.0002	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S12 ARL of the EWMA TBEA schemes based on the statistic Z_3 with both variables A and T following the Gamma distribution assuming a Frank copula when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.5069	23.6354	8.8784	5.1598	3.5786	2.7146	2.1779	1.8071	1.5259	1.3015	1.1561
0.95	10.7514	6.2281	4.2994	3.2501	2.5997	2.1587	1.8438	1.5756	1.3483	1.1855	1.0883

0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.1162	57.2951	18.7029	7.3566	3.5446	2.0987	1.4874	1.2162	1.0959	1.0399	1.0162
0.95	36.2508	13.4462	5.9133	3.0969	1.9430	1.4339	1.1966	1.0850	1.0362	1.0148	1.0056
0.9	8.3050	4.2144	2.5061	1.7158	1.3375	1.1537	1.0664	1.0277	1.0107	1.0043	1.0016
0.85	2.8491	1.9359	1.4659	1.2229	1.1028	1.0442	1.0184	1.0070	1.0027	1.0010	1.0003
0.8	1.5035	1.2565	1.1249	1.0566	1.0244	1.0099	1.0037	1.0015	1.0005	1.0002	1.0001
0.75	1.1184	1.0575	1.0257	1.0106	1.0039	1.0016	1.0006	1.0002	1.0001	1.0001	1
0.7	1.0203	1.0087	1.0037	1.0013	1.0005	1.0001	1	1	1	1	1
0.65	1.0020	1.0008	1.0004	1.0001	1	1	1	1	1	1	1
0.6	1.0001	1.0001	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.0900	60.2838	20.4257	8.1072	3.7735	2.1741	1.5010	1.2184	1.0940	1.0389	1.0154
0.95	39.8739	15.2498	6.6108	3.3631	2.0136	1.4411	1.1970	1.0828	1.0342	1.0137	1.0051
0.9	9.6967	4.7661	2.6859	1.7569	1.3434	1.1498	1.0641	1.0271	1.0104	1.0040	1.0017
0.85	3.1502	2.0346	1.4820	1.2235	1.1001	1.0418	1.0171	1.0069	1.0022	1.0008	1.0004
0.8	1.5373	1.2634	1.1220	1.0543	1.0232	1.0085	1.0037	1.0013	1.0004	1.0001	1
0.75	1.1140	1.0545	1.0231	1.0092	1.0038	1.0015	1.0004	1.0002	1.0001	1	1
0.7	1.0178	1.0077	1.0029	1.0011	1.0003	1.0002	1	1	1	1	1
0.65	1.0020	1.0007	1.0003	1.0001	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.9556	15.0241	5.8310	3.5474	2.5350	1.9972	1.6380	1.3329	1.1462	1.0594	1.0238
0.95	6.8343	4.1631	2.9813	2.3154	1.9354	1.6337	1.3324	1.1428	1.0556	1.0216	1.0080
0.9	3.1682	2.5172	2.1063	1.8588	1.5848	1.2942	1.1196	1.0445	1.0173	1.0059	1.0024
0.85	2.1636	1.9559	1.7630	1.4867	1.2235	1.0848	1.0309	1.0117	1.0037	1.0016	1.0006
0.8	1.8350	1.6092	1.3261	1.1351	1.0473	1.0165	1.0055	1.0019	1.0007	1.0002	1.0001
0.75	1.3723	1.1628	1.0600	1.0204	1.0068	1.0024	1.0007	1.0003	1.0001	1	1
0.7	1.0564	1.0195	1.0065	1.0023	1.0006	1.0002	1.0001	1	1	1	1
0.65	1.0048	1.0015	1.0005	1.0001	1	1	1	1	1	1	1
0.6	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.8928	21.4816	5.8356	3.0369	2.0526	1.5561	1.2493	1.0963	1.0365	1.0141	1.0051

0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.0607	51.3589	14.9150	5.0243	2.1870	1.3601	1.1184	1.0416	1.0149	1.0055	1.0019
0.95	28.9168	9.4874	3.6056	1.7885	1.2404	1.0779	1.0273	1.0097	1.0031	1.0012	1.0004
0.9	4.9325	2.2790	1.3986	1.1253	1.0405	1.0133	1.0045	1.0017	1.0005	1.0001	1.0001
0.85	1.4662	1.1539	1.0517	1.0157	1.0048	1.0019	1.0006	1.0002	1.0001	1	1
0.8	1.0457	1.0155	1.0045	1.0017	1.0006	1.0001	1.0001	1	1	1	1
0.75	1.0035	1.0013	1.0003	1.0001	1	1	1	1	1	1	1
0.7	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.8$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.4331	7.2147	3.1199	2.0586	1.6081	1.1650	1.0404	1.0129	1.0040	1.0013	1.0004
0.95	3.5324	2.3035	1.8821	1.4082	1.0738	1.0174	1.0054	1.0017	1.0004	1.0002	1.0001
0.9	1.9510	1.6606	1.1228	1.0204	1.0059	1.0016	1.0005	1.0001	1.0001	1	1
0.85	1.1488	1.0192	1.0036	1.0008	1.0002	1.0001	1	1	1	1	1
0.8	1.0028	1.0003	1.0001	1	1	1	1	1	1	1	1
0.75	1.0001	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.2464	10.9324	2.8932	1.7886	1.2173	1.0454	1.0136	1.0042	1.0014	1.0005	1.0002
0.95	3.4072	2.0257	1.4725	1.0731	1.0158	1.0044	1.0013	1.0005	1.0001	1	1
0.9	1.6215	1.0984	1.0150	1.0034	1.0008	1.0002	1.0001	1	1	1	1
0.85	1.0129	1.0021	1.0005	1.0001	1	1	1	1	1	1	1
0.8	1.0002	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2

0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.2074	69.8076	26.0129	10.8831	5.4459	3.2478	2.2137	1.6756	1.3707	1.1911	1.0876
0.95	26.1138	10.9953	5.4707	3.2350	2.2168	1.6737	1.3744	1.1889	1.0852	1.0336	1.0117
0.9	5.4610	3.2450	2.2179	1.6827	1.3733	1.1868	1.0857	1.0318	1.0099	1.0028	1.0007
0.85	2.2134	1.6802	1.3734	1.1870	1.0830	1.0314	1.0099	1.0025	1.0005	1.0001	1
0.8	1.3702	1.1899	1.0822	1.0314	1.0090	1.0022	1.0006	1	1	1	1
0.75	1.0805	1.0304	1.0096	1.0022	1.0003	1.0001	1	1	1	1	1
0.7	1.0092	1.0021	1.0004	1.0001	1	1	1	1	1	1	1
0.65	1.0004	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.0581	77.3594	31.8377	14.0761	6.9695	3.9095	2.4544	1.7466	1.3756	1.1797	1.0780
0.95	31.8337	14.0948	7.0143	3.8941	2.4638	1.7527	1.3768	1.1798	1.0749	1.0281	1.0092
0.9	6.9682	3.8967	2.4739	1.7599	1.3798	1.1757	1.0751	1.0279	1.0084	1.0021	1.0005
0.85	2.4786	1.7508	1.3767	1.1757	1.0717	1.0267	1.0080	1.0018	1.0005	1.0001	1
0.8	1.3759	1.1762	1.0730	1.0252	1.0073	1.0018	1.0005	1.0001	1	1	1
0.75	1.0729	1.0242	1.0075	1.0018	1.0003	1.0001	1	1	1	1	1
0.7	1.0071	1.0016	1.0004	1	1	1	1	1	1	1	1
0.65	1.0003	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	202.0458	80.6521	34.4311	15.7840	7.9195	4.3497	2.6537	1.8186	1.3898	1.1799	1.0757
0.95	34.4404	15.8080	7.8990	4.3660	2.6602	1.8197	1.4022	1.1799	1.0744	1.0262	1.0085
0.9	7.9195	4.3653	2.6691	1.8278	1.3963	1.1788	1.0719	1.0263	1.0079	1.0020	1.0005
0.85	2.6735	1.8244	1.3963	1.1791	1.0712	1.0249	1.0072	1.0017	1.0003	1.0001	1
0.8	1.3960	1.1784	1.0701	1.0244	1.0070	1.0017	1.0003	1	1	1	1
0.75	1.0710	1.0236	1.0069	1.0017	1.0003	1	1	1	1	1	1
0.7	1.0060	1.0013	1.0003	1.0001	1	1	1	1	1	1	1
0.65	1.0003	1.0001	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.5931	18.8152	5.7743	3.3099	2.3832	1.9351	1.6255	1.3290	1.1349	1.0453	1.0127

0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.9141	64.2468	21.7192	7.7039	3.1804	1.7436	1.2568	1.0819	1.0234	1.0060	1.0013
0.95	21.6778	7.7218	3.1960	1.7438	1.2538	1.0808	1.0222	1.0051	1.0011	1.0001	1
0.9	3.1896	1.7457	1.2558	1.0780	1.0208	1.0049	1.0008	1.0001	1	1	1
0.85	1.2537	1.0784	1.0203	1.0046	1.0008	1.0001	1	1	1	1	1
0.8	1.0197	1.0043	1.0005	1.0001	1	1	1	1	1	1	1
0.75	1.0007	1.0001	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.6945	66.9567	24.0449	8.9668	3.6140	1.8274	1.2565	1.0801	1.0222	1.0054	1.0012
0.95	24.0886	8.9593	3.6297	1.8291	1.2607	1.0784	1.0213	1.0047	1.0009	1.0002	1
0.9	3.6243	1.8359	1.2573	1.0787	1.0207	1.0046	1.0008	1.0001	1	1	1
0.85	1.2569	1.0758	1.0195	1.0041	1.0007	1.0001	1	1	1	1	1
0.8	1.0182	1.0039	1.0006	1.0001	1	1	1	1	1	1	1
0.75	1.0006	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.8$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	197.7297	8.3869	3.0585	2.0155	1.6290	1.0979	1.0125	1.0019	1.0002	1	1
0.95	3.0628	2.0147	1.6298	1.0985	1.0119	1.0016	1.0002	1	1	1	1
0.9	1.6324	1.0972	1.0111	1.0013	1.0001	1	1	1	1	1	1
0.85	1.0102	1.0012	1.0002	1	1	1	1	1	1	1	1
0.8	1.0001	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2

0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.9580	59.3201	17.4404	4.1020	1.2089	1.0215	1.0029	1.0003	1	1	1
0.95	17.2235	4.1190	1.2126	1.0199	1.0028	1.0002	1	1	1	1	1
0.9	1.2117	1.0188	1.0021	1.0002	1	1	1	1	1	1	1
0.85	1.0018	1.0002	1	1	1	1	1	1	1	1	1
0.8	1	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S14 ARL of the EWMA TBFA schemes based on the statistic Z_2 when variable A following the gamma distribution and T following the weibull distribution assuming a Clayton when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	202.2921	26.3302	9.6881	5.5441	3.8089	2.8744	2.2936	1.9032	1.6134	1.3834	1.2020
0.95	11.5233	6.3757	4.3233	3.2468	2.5705	2.1291	1.8135	1.5648	1.3467	1.1792	1.0726
0.9	4.8796	3.6214	2.8620	2.3611	2.0102	1.7504	1.5233	1.3135	1.1544	1.0577	1.0178
0.85	3.1630	2.6025	2.2059	1.9193	1.6946	1.4840	1.2857	1.1317	1.0483	1.0131	1.0028
0.8	2.4055	2.0865	1.8580	1.6573	1.4499	1.2524	1.1126	1.0387	1.0096	1.0016	1.0002
0.75	1.9975	1.8097	1.6221	1.4167	1.2274	1.0947	1.0290	1.0064	1.0012	1.0002	1
0.7	1.7729	1.5931	1.3828	1.1968	1.0784	1.0224	1.0050	1.0006	1.0001	1	1
0.65	1.5562	1.3490	1.1735	1.0662	1.0172	1.0037	1.0004	1	1	1	1
0.6	1.3171	1.1496	1.0519	1.0129	1.0021	1.0003	1	1	1	1	1
0.55	1.1271	1.0403	1.0098	1.0017	1.0001	1	1	1	1	1	1
0.5	1.0333	1.0071	1.0009	1.0001	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.7699	39.0063	11.7819	5.5352	3.4195	2.4293	1.8769	1.5326	1.2976	1.1421	1.0568
0.95	16.3319	6.8752	4.0240	2.7917	2.1319	1.7260	1.4429	1.2372	1.1029	1.0376	1.0108
0.9	4.7787	3.2068	2.4104	1.9318	1.6074	1.3660	1.1878	1.0777	1.0240	1.0064	1.0013
0.85	2.7135	2.1506	1.7820	1.5165	1.2991	1.1435	1.0534	1.0151	1.0037	1.0004	1.0001
0.8	1.9718	1.6730	1.4380	1.2430	1.1076	1.0356	1.0088	1.0016	1.0002	1	1
0.75	1.5885	1.3715	1.1916	1.0773	1.0232	1.0051	1.0007	1.0001	1	1	1
0.7	1.3090	1.1490	1.0533	1.0150	1.0026	1.0004	1.0001	1	1	1	1
0.65	1.1100	1.0363	1.0088	1.0013	1.0002	1	1	1	1	1	1
0.6	1.0249	1.0049	1.0007	1	1	1	1	1	1	1	1

0.55	1.0025	1.0003	1	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.8758	51.4627	15.9829	6.6773	3.6611	2.4144	1.7888	1.4274	1.2106	1.0891	1.0312
0.95	25.1163	9.3030	4.6300	2.9003	2.0698	1.6112	1.3286	1.1542	1.0599	1.0193	1.0044
0.9	6.0077	3.4937	2.4070	1.8298	1.4801	1.2505	1.1109	1.0390	1.0106	1.0021	1.0003
0.85	2.8222	2.0868	1.6585	1.3762	1.1880	1.0747	1.0237	1.0057	1.0010	1.0001	1
0.8	1.8645	1.5296	1.2924	1.1374	1.0499	1.0135	1.0028	1.0004	1.0001	1	1
0.75	1.4263	1.2253	1.0954	1.0317	1.0075	1.0013	1.0002	1	1	1	1
0.7	1.1690	1.0664	1.0182	1.0042	1.0006	1.0001	1	1	1	1	1
0.65	1.0411	1.0105	1.0018	1.0003	1	1	1	1	1	1	1
0.6	1.0057	1.0007	1.0001	1	1	1	1	1	1	1	1
0.55	1.0003	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.1288	61.2933	20.8370	8.4793	4.2570	2.5690	1.7952	1.3936	1.1786	1.0689	1.0234
0.95	34.6721	12.8749	5.8640	3.2574	2.1477	1.5900	1.2886	1.1242	1.0443	1.0130	1.0030
0.9	8.3509	4.2478	2.6212	1.8456	1.4373	1.2064	1.0847	1.0264	1.0071	1.0013	1.0002
0.85	3.3023	2.1937	1.6356	1.3280	1.1442	1.0529	1.0153	1.0029	1.0004	1.0001	1
0.8	1.9038	1.4851	1.2378	1.0990	1.0315	1.0078	1.0015	1.0002	1	1	1
0.75	1.3674	1.1735	1.0653	1.0186	1.0042	1.0006	1.0001	1	1	1	1
0.7	1.1208	1.0409	1.0100	1.0018	1.0002	1	1	1	1	1	1
0.65	1.0234	1.0050	1.0010	1	1	1	1	1	1	1	1
0.6	1.0024	1.0003	1	1	1	1	1	1	1	1	1
0.55	1.0001	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.8754	67.5524	25.2735	10.5373	5.0399	2.8626	1.8651	1.3967	1.1694	1.0640	1.0195
0.95	42.9620	16.7721	7.4755	3.8633	2.3399	1.6297	1.2770	1.1119	1.0371	1.0099	1.0023
0.9	11.4875	5.4931	3.0399	1.9735	1.4507	1.1930	1.0728	1.0215	1.0049	1.0009	1.0001
0.85	4.1601	2.4807	1.7077	1.3170	1.1266	1.0430	1.0120	1.0023	1.0005	1	1
0.8	2.0826	1.5093	1.2209	1.0839	1.0254	1.0057	1.0010	1.0001	1	1	1
0.75	1.3666	1.1510	1.0525	1.0139	1.0028	1.0004	1	1	1	1	1
0.7	1.0995	1.0293	1.0075	1.0014	1.0001	1	1	1	1	1	1
0.65	1.0172	1.0035	1.0006	1	1	1	1	1	1	1	1
0.6	1.0016	1.0002	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.2992	69.8124	26.8279	11.4252	5.4720	3.0155	1.9132	1.4059	1.1667	1.0587	1.0184
0.95	46.0007	18.5165	8.3062	4.2221	2.4617	1.6595	1.2800	1.1074	1.0351	1.0090	1.0020

0.9	13.0326	6.1690	3.3185	2.0596	1.4699	1.1902	1.0674	1.0210	1.0044	1.0010	1.0001
0.85	4.6779	2.6885	1.7575	1.3281	1.1270	1.0416	1.0101	1.0024	1.0003	1.0001	1
0.8	2.2034	1.5327	1.2205	1.0798	1.0233	1.0054	1.0008	1.0001	1	1	1
0.75	1.3790	1.1468	1.0480	1.0125	1.0025	1.0003	1	1	1	1	1
0.7	1.0934	1.0274	1.0061	1.0011	1.0001	1	1	1	1	1	1
0.65	1.0152	1.0027	1.0003	1	1	1	1	1	1	1	1
0.6	1.0013	1.0001	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.5566	14.1322	5.5515	3.3979	2.4419	1.9256	1.5666	1.2814	1.1060	1.0328	1.0075
0.95	6.3222	3.7940	2.7059	2.1230	1.7601	1.4454	1.2002	1.0694	1.0186	1.0036	1.0005
0.9	2.9875	2.3187	1.9297	1.6297	1.3403	1.1356	1.0418	1.0092	1.0017	1.0002	1
0.85	2.0804	1.8028	1.5208	1.2518	1.0887	1.0222	1.0042	1.0007	1	1	1
0.8	1.6942	1.4070	1.1739	1.0540	1.0121	1.0018	1.0003	1	1	1	1
0.75	1.3057	1.1151	1.0303	1.0055	1.0008	1.0001	1	1	1	1	1
0.7	1.0721	1.0164	1.0026	1.0003	1	1	1	1	1	1	1
0.65	1.0080	1.0011	1.0001	1	1	1	1	1	1	1	1
0.6	1.0005	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.2677	20.2707	5.5452	2.9147	1.9842	1.5025	1.2223	1.0758	1.0208	1.0049	1.0006
0.95	6.8635	3.3717	2.2343	1.6832	1.3346	1.1304	1.0391	1.0087	1.0014	1.0002	1
0.9	2.5091	1.8765	1.4774	1.2113	1.0710	1.0180	1.0033	1.0005	1	1	1
0.85	1.6437	1.3267	1.1256	1.0353	1.0075	1.0011	1.0001	1	1	1	1
0.8	1.2063	1.0691	1.0162	1.0028	1.0004	1	1	1	1	1	1
0.75	1.0338	1.0064	1.0008	1.0001	1	1	1	1	1	1	1
0.7	1.0021	1.0002	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.9671	29.8846	6.9704	3.0207	1.8741	1.3827	1.1424	1.0436	1.0112	1.0021	1.0003
0.95	9.7584	3.7028	2.1682	1.5427	1.2235	1.0763	1.0196	1.0038	1.0005	1.0001	1
0.9	2.5153	1.7359	1.3349	1.1236	1.0365	1.0079	1.0014	1.0002	1	1	1
0.85	1.4753	1.1960	1.0639	1.0154	1.0029	1.0004	1	1	1	1	1
0.8	1.1060	1.0283	1.0060	1.0008	1	1	1	1	1	1	1
0.75	1.0120	1.0017	1.0002	1	1	1	1	1	1	1	1
0.7	1.0005	1.0001	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1

0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.5048	39.2812	9.3905	3.4376	1.8826	1.3362	1.1138	1.0330	1.0068	1.0015	1.0002
0.95	14.5057	4.6101	2.2690	1.5001	1.1832	1.0568	1.0135	1.0028	1.0004	1.0001	1
0.9	2.8027	1.7179	1.2799	1.0938	1.0242	1.0051	1.0007	1.0001	1	1	1
0.85	1.4164	1.1503	1.0444	1.0099	1.0016	1.0001	1	1	1	1	1
0.8	1.0749	1.0187	1.0031	1.0004	1	1	1	1	1	1	1
0.75	1.0069	1.0008	1	1	1	1	1	1	1	1	1
0.7	1.0002	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.8434	47.3380	12.3310	4.1207	1.9899	1.3321	1.1042	1.0295	1.0060	1.0012	1.0002
0.95	20.3732	6.1266	2.5398	1.5124	1.1678	1.0468	1.0114	1.0019	1.0002	1	1
0.9	3.4516	1.7981	1.2641	1.0815	1.0203	1.0041	1.0007	1	1	1	1
0.85	1.4166	1.1303	1.0357	1.0073	1.0012	1.0001	1	1	1	1	1
0.8	1.0625	1.0141	1.0022	1.0002	1	1	1	1	1	1	1
0.75	1.0049	1.0005	1.0001	1	1	1	1	1	1	1	1
0.7	1.0001	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.3158	50.0383	13.8261	4.5776	2.0648	1.3339	1.1019	1.0269	1.0059	1.0009	1.0001
0.95	22.9550	6.9643	2.7333	1.5345	1.1640	1.0462	1.0105	1.0016	1.0002	1	1
0.9	3.8518	1.8690	1.2677	1.0769	1.0185	1.0035	1.0004	1	1	1	1
0.85	1.4335	1.1291	1.0328	1.0066	1.0010	1.0001	1	1	1	1	1
0.8	1.0574	1.0124	1.0021	1.0002	1	1	1	1	1	1	1
0.75	1.0042	1.0005	1.0001	1	1	1	1	1	1	1	1
0.7	1.0001	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.8$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2

0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.3983	41.1946	8.2199	1.8508	1.0926	1.0128	1.0016	1.0002	1	1	1
0.95	12.9680	2.6143	1.1736	1.0220	1.0029	1.0002	1	1	1	1	1
0.9	1.3391	1.0372	1.0045	1.0005	1	1	1	1	1	1	1
0.85	1.0090	1.0008	1.0001	1	1	1	1	1	1	1	1
0.8	1.0001	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.0926	44.4321	9.8728	2.0044	1.0910	1.0120	1.0015	1.0001	1	1	1
0.95	15.5886	3.1259	1.1744	1.0210	1.0024	1.0002	1	1	1	1	1
0.9	1.3598	1.0370	1.0048	1.0005	1	1	1	1	1	1	1
0.85	1.0080	1.0008	1.0001	1	1	1	1	1	1	1	1
0.8	1.0001	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S15 ARL of the EWMA TBEA schemes based on the statistic Z_3 when variable A following the gamma distribution and T following the weibull distribution assuming a Clayton copula when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.7554	26.5167	9.7021	5.5519	3.8110	2.8705	2.2904	1.9080	1.6212	1.3907	1.2001
0.95	11.8375	6.7287	4.6011	3.4499	2.7442	2.2616	1.9241	1.6706	1.4488	1.2453	1.1078
0.9	4.9353	3.8050	3.0713	2.5558	2.1833	1.9159	1.6966	1.4831	1.2814	1.1270	1.0444
0.85	3.1916	2.7196	2.3549	2.0827	1.8787	1.6980	1.4983	1.2965	1.1361	1.0489	1.0126
0.8	2.4139	2.1640	1.9811	1.8385	1.6758	1.4812	1.2851	1.1317	1.0463	1.0126	1.0025
0.75	2.0124	1.9062	1.7866	1.6236	1.4314	1.2483	1.1111	1.0383	1.0098	1.0017	1.0003
0.7	1.8385	1.7055	1.5342	1.3495	1.1859	1.0790	1.0258	1.0062	1.0012	1.0001	1
0.65	1.5735	1.4022	1.2373	1.1160	1.0453	1.0134	1.0027	1.0005	1	1	1
0.6	1.2434	1.1280	1.0548	1.0186	1.0047	1.0010	1.0001	1	1	1	1

0.55	1.0476	1.0170	1.0051	1.0011	1.0001	1	1	1	1	1	1
0.5	1.0031	1.0009	1.0001	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.9386	39.7385	11.8738	5.5108	3.3922	2.4099	1.8751	1.5283	1.2829	1.1277	1.0436
0.95	17.6264	7.4723	4.3513	2.9961	2.2722	1.8322	1.5275	1.3026	1.1396	1.0492	1.0130
0.9	4.8460	3.3919	2.5879	2.0914	1.7529	1.4975	1.2859	1.1326	1.0475	1.0124	1.0024
0.85	2.7031	2.2363	1.9153	1.6570	1.4372	1.2464	1.1121	1.0390	1.0098	1.0017	1.0004
0.8	1.9650	1.7443	1.5426	1.3501	1.1839	1.0793	1.0266	1.0059	1.0011	1.0001	1
0.75	1.5805	1.4025	1.2356	1.1159	1.0452	1.0134	1.0029	1.0006	1	1	1
0.7	1.2446	1.1258	1.0545	1.0184	1.0048	1.0010	1.0001	1	1	1	1
0.65	1.0468	1.0175	1.0051	1.0012	1.0001	1	1	1	1	1	1
0.6	1.0032	1.0007	1.0001	1.0001	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.0770	52.7380	16.4662	6.7629	3.6641	2.4059	1.7702	1.4138	1.1949	1.0749	1.0216
0.95	29.3781	10.7029	5.2227	3.1687	2.2350	1.7179	1.4018	1.1962	1.0760	1.0226	1.0050
0.9	6.3485	3.8112	2.6450	2.0117	1.6176	1.3556	1.1736	1.0664	1.0191	1.0039	1.0008
0.85	2.8228	2.1871	1.7816	1.4930	1.2804	1.1331	1.0496	1.0142	1.0028	1.0004	1.0001
0.8	1.8343	1.5728	1.3571	1.1946	1.0831	1.0289	1.0074	1.0014	1.0001	1	1
0.75	1.3747	1.2200	1.1064	1.0418	1.0127	1.0030	1.0006	1.0001	1	1	1
0.7	1.1019	1.0411	1.0137	1.0038	1.0008	1.0002	1	1	1	1	1
0.65	1.0105	1.0031	1.0007	1.0001	1	1	1	1	1	1	1
0.6	1.0004	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.2393	62.1767	21.1486	8.5828	4.2597	2.5446	1.7719	1.3668	1.1590	1.0549	1.0156
0.95	42.4441	15.5723	6.9024	3.7017	2.3657	1.7030	1.3519	1.1524	1.0534	1.0144	1.0031
0.9	9.4726	4.8959	2.9983	2.0663	1.5802	1.2939	1.1283	1.0449	1.0125	1.0021	1.0003
0.85	3.3458	2.3348	1.7692	1.4355	1.2158	1.0931	1.0308	1.0076	1.0014	1.0002	1
0.8	1.8345	1.5060	1.2802	1.1348	1.0523	1.0161	1.0035	1.0006	1.0001	1	1
0.75	1.2850	1.1502	1.0627	1.0234	1.0060	1.0014	1.0002	1	1	1	1
0.7	1.0576	1.0210	1.0066	1.0015	1.0003	1	1	1	1	1	1
0.65	1.0044	1.0009	1.0003	1	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.0202	69.2625	25.9793	10.8066	5.1536	2.8529	1.8494	1.3740	1.1449	1.0468	1.0119
0.95	55.0478	21.1581	9.2781	4.6232	2.6605	1.7789	1.3470	1.1383	1.0459	1.0111	1.0020

0.9	14.4005	6.8375	3.7265	2.3095	1.6275	1.2841	1.1136	1.0359	1.0091	1.0017	1.0002
0.85	4.5337	2.7625	1.8979	1.4421	1.1987	1.0763	1.0237	1.0056	1.0010	1.0001	1
0.8	2.0107	1.5324	1.2591	1.1121	1.0398	1.0128	1.0023	1.0005	1.0001	1	1
0.75	1.2633	1.1243	1.0499	1.0154	1.0038	1.0007	1.0002	1	1	1	1
0.7	1.0414	1.0146	1.0043	1.0010	1.0002	1	1	1	1	1	1
0.65	1.0025	1.0006	1.0001	1.0001	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.0407	71.6576	27.7992	11.8102	5.6189	3.0391	1.8997	1.3761	1.1427	1.0441	1.0117
0.95	59.5172	24.0021	10.4569	5.1549	2.8752	1.8416	1.3572	1.1382	1.0429	1.0098	1.0021
0.9	17.1020	8.0140	4.2277	2.4908	1.6786	1.2889	1.1075	1.0342	1.0079	1.0014	1.0001
0.85	5.3557	3.1154	2.0093	1.4651	1.1992	1.0731	1.0210	1.0048	1.0009	1.0001	1
0.8	2.1783	1.5766	1.2680	1.1087	1.0382	1.0108	1.0023	1.0003	1	1	1
0.75	1.2733	1.1177	1.0459	1.0142	1.0035	1.0007	1.0001	1	1	1	1
0.7	1.0389	1.0126	1.0032	1.0007	1.0001	1	1	1	1	1	1
0.65	1.0023	1.0006	1.0001	1	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.1707	16.1169	6.1366	3.6985	2.6418	2.0679	1.7083	1.4090	1.1880	1.0650	1.0164
0.95	7.2348	4.3586	3.1015	2.3999	1.9869	1.6957	1.4172	1.1889	1.0640	1.0164	1.0032
0.9	3.2927	2.6112	2.1613	1.8971	1.6570	1.3827	1.1654	1.0525	1.0118	1.0023	1.0003
0.85	2.2238	1.9845	1.8162	1.5703	1.3038	1.1193	1.0351	1.0076	1.0011	1.0001	1
0.8	1.8776	1.6950	1.4246	1.1922	1.0660	1.0171	1.0032	1.0003	1	1	1
0.75	1.4868	1.2335	1.0868	1.0233	1.0051	1.0008	1.0001	1	1	1	1
0.7	1.0759	1.0225	1.0050	1.0008	1.0002	1	1	1	1	1	1
0.65	1.0027	1.0004	1.0001	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.4446	25.4905	6.6175	3.3187	2.2091	1.6670	1.3355	1.1320	1.0399	1.0091	1.0017
0.95	8.9709	4.1321	2.6573	1.9837	1.5765	1.2819	1.1065	1.0309	1.0064	1.0009	1.0001
0.9	2.8514	2.1703	1.7656	1.4473	1.2045	1.0689	1.0171	1.0032	1.0004	1	1
0.85	1.8414	1.5558	1.2861	1.1112	1.0337	1.0067	1.0012	1.0001	1	1	1
0.8	1.3128	1.1260	1.0387	1.0093	1.0018	1.0002	1	1	1	1	1
0.75	1.0302	1.0078	1.0016	1.0002	1	1	1	1	1	1	1
0.7	1.0008	1.0001	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1

0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.1872	37.9990	9.0128	3.6243	2.1562	1.5458	1.2327	1.0803	1.0204	1.0037	1.0005
0.95	14.9259	5.1280	2.7454	1.8767	1.4306	1.1785	1.0571	1.0134	1.0022	1.0003	1.0001
0.9	3.0752	2.1037	1.6070	1.2888	1.1083	1.0309	1.0069	1.0009	1.0001	1	1
0.85	1.6813	1.3590	1.1451	1.0470	1.0120	1.0022	1.0003	1	1	1	1
0.8	1.1449	1.0476	1.0120	1.0025	1.0004	1	1	1	1	1	1
0.75	1.0081	1.0017	1.0003	1	1	1	1	1	1	1	1
0.7	1.0001	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.1653	49.4974	12.7608	4.4624	2.2786	1.5149	1.1962	1.0608	1.0155	1.0027	1.0003
0.95	24.6140	7.5096	3.2175	1.9146	1.3799	1.1384	1.0415	1.0095	1.0013	1.0002	1
0.9	3.9480	2.2477	1.5636	1.2303	1.0773	1.0204	1.0038	1.0005	1.0001	1	1
0.85	1.6348	1.2816	1.1008	1.0294	1.0067	1.0012	1.0001	1	1	1	1
0.8	1.0954	1.0281	1.0064	1.0013	1.0002	1	1	1	1	1	1
0.75	1.0037	1.0009	1.0001	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.5232	57.5003	16.9017	5.6963	2.5374	1.5398	1.1799	1.0526	1.0118	1.0021	1.0004
0.95	35.1123	11.1926	4.1666	2.0792	1.3784	1.1232	1.0333	1.0067	1.0013	1.0001	1
0.9	5.8275	2.6668	1.6062	1.2102	1.0637	1.0158	1.0031	1.0003	1.0001	1	1
0.85	1.7166	1.2605	1.0841	1.0224	1.0048	1.0007	1.0001	1	1	1	1
0.8	1.0757	1.0210	1.0047	1.0009	1.0002	1	1	1	1	1	1
0.75	1.0023	1.0005	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.7935	60.5184	18.9189	6.4243	2.7314	1.5631	1.1792	1.0509	1.0118	1.0018	1.0003

0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.8327	53.0169	13.4182	3.7586	1.7244	1.1649	1.0282	1.0041	1.0004	1.0001	1
0.95	26.4818	6.9336	2.3422	1.4006	1.0655	1.0104	1.0013	1.0001	1	1	1
0.9	2.7732	1.6264	1.1085	1.0141	1.0020	1.0002	1	1	1	1	1
0.85	1.1114	1.0111	1.0011	1.0001	1	1	1	1	1	1	1
0.8	1.0005	1.0001	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.3008	61.9842	18.9615	5.7632	1.9417	1.1520	1.0251	1.0036	1.0004	1	1
0.95	40.4250	12.4402	3.5970	1.4184	1.0563	1.0082	1.0009	1.0001	1	1	1
0.9	5.3557	1.7767	1.0946	1.0120	1.0014	1.0002	1	1	1	1	1
0.85	1.0895	1.0086	1.0010	1.0001	1	1	1	1	1	1	1
0.8	1.0003	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.2431	64.9863	21.3448	6.9097	2.1475	1.1537	1.0232	1.0033	1.0004	1	1
0.95	46.4082	15.3256	4.6083	1.4769	1.0563	1.0074	1.0012	1.0001	1	1	1
0.9	7.3360	2.0750	1.0900	1.0107	1.0012	1.0002	1	1	1	1	1
0.85	1.0840	1.0079	1.0009	1.0001	1	1	1	1	1	1	1
0.8	1.0003	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S16 ARL of the EWMA TBEA schemes based on the statistic Z_1 when variable A following the gamma distribution and T following the weibull distribution assuming a Frank copula

when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.2812	38.2323	11.8191	6.0460	4.0048	3.0227	2.4576	2.1024	1.8583	1.6655	1.4788
0.95	11.8400	6.0380	4.0072	3.0208	2.4563	2.0993	1.8581	1.6651	1.4815	1.3079	1.1709
0.9	4.0053	3.0202	2.4549	2.1000	1.8567	1.6653	1.4801	1.3074	1.1703	1.0824	1.0352
0.85	2.4546	2.1011	1.8600	1.6684	1.4822	1.3082	1.1717	1.0803	1.0346	1.0124	1.0045
0.8	1.8599	1.6694	1.4831	1.3089	1.1691	1.0814	1.0337	1.0121	1.0038	1.0011	1.0002
0.75	1.4820	1.3080	1.1698	1.0825	1.0332	1.0121	1.0038	1.0010	1.0002	1	1
0.7	1.1716	1.0795	1.0332	1.0117	1.0039	1.0011	1.0003	1.0001	1	1	1
0.65	1.0325	1.0118	1.0033	1.0011	1.0002	1.0001	1	1	1	1	1
0.6	1.0036	1.0009	1.0003	1	1	1	1	1	1	1	1
0.55	1.0003	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.3833	55.8667	17.3795	7.4854	4.3107	2.9846	2.3150	1.9099	1.6290	1.4185	1.2495
0.95	17.3897	7.4944	4.3142	2.9830	2.3115	1.9107	1.6325	1.4208	1.2473	1.1308	1.0574
0.9	4.3105	2.9941	2.3149	1.9100	1.6323	1.4186	1.2499	1.1288	1.0579	1.0226	1.0076
0.85	2.3164	1.9083	1.6379	1.4151	1.2469	1.1275	1.0575	1.0222	1.0075	1.0024	1.0007
0.8	1.6343	1.4201	1.2473	1.1285	1.0553	1.0217	1.0079	1.0021	1.0006	1.0001	1
0.75	1.2442	1.1277	1.0566	1.0222	1.0077	1.0024	1.0005	1.0001	1	1	1
0.7	1.0565	1.0216	1.0071	1.0020	1.0006	1.0002	1	1	1	1	1
0.65	1.0068	1.0020	1.0005	1.0001	1	1	1	1	1	1	1
0.6	1.0005	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.0907	70.4421	25.3085	10.4163	5.3349	3.3190	2.3795	1.8617	1.5422	1.3226	1.1720
0.95	25.2985	10.4458	5.3510	3.3182	2.3734	1.8637	1.5421	1.3243	1.1727	1.0823	1.0340
0.9	5.3446	3.3148	2.3831	1.8657	1.5453	1.3235	1.1738	1.0818	1.0328	1.0118	1.0042
0.85	2.3811	1.8657	1.5416	1.3256	1.1735	1.0800	1.0335	1.0122	1.0038	1.0011	1.0003
0.8	1.5472	1.3231	1.1716	1.0810	1.0320	1.0112	1.0039	1.0007	1.0002	1.0001	1
0.75	1.1710	1.0796	1.0333	1.0115	1.0032	1.0009	1.0002	1	1	1	1
0.7	1.0332	1.0112	1.0036	1.0011	1.0002	1	1	1	1	1	1
0.65	1.0034	1.0009	1.0001	1.0001	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	202.3957	82.8510	34.0540	14.8436	7.2165	4.0755	2.6540	1.9332	1.5347	1.2886	1.1448
0.95	33.8345	14.7904	7.2025	4.0783	2.6596	1.9350	1.5341	1.2907	1.1457	1.0659	1.0264

0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.8808	39.8338	9.6325	4.1946	2.6329	1.9788	1.6079	1.3105	1.1238	1.0387	1.0122
0.95	9.6475	4.1731	2.6348	1.9822	1.6036	1.3108	1.1226	1.0380	1.0112	1.0028	1.0006
0.9	2.6343	1.9808	1.6081	1.3108	1.1202	1.0374	1.0108	1.0030	1.0007	1.0002	1
0.85	1.6112	1.3134	1.1216	1.0395	1.0107	1.0029	1.0009	1.0001	1	1	1
0.8	1.1216	1.0392	1.0110	1.0028	1.0007	1.0001	1	1	1	1	1
0.75	1.0111	1.0030	1.0009	1.0002	1	1	1	1	1	1	1
0.7	1.0009	1.0001	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.9379	56.3220	15.3036	5.5263	2.9140	1.9752	1.5193	1.2266	1.0803	1.0249	1.0068
0.95	15.3085	5.5127	2.9147	1.9823	1.5194	1.2293	1.0806	1.0244	1.0064	1.0017	1.0005
0.9	2.9166	1.9833	1.5180	1.2293	1.0812	1.0248	1.0069	1.0016	1.0004	1.0001	1
0.85	1.5232	1.2283	1.0808	1.0236	1.0069	1.0017	1.0005	1.0001	1	1	1
0.8	1.0803	1.0245	1.0064	1.0018	1.0006	1	1	1	1	1	1
0.75	1.0068	1.0016	1.0003	1.0001	1	1	1	1	1	1	1
0.7	1.0004	1.0001	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.1021	68.1535	22.4774	8.0835	3.6171	2.1201	1.4954	1.1948	1.0642	1.0193	1.0052
0.95	22.5418	8.0816	3.6295	2.1126	1.4960	1.1930	1.0637	1.0190	1.0053	1.0013	1.0003
0.9	3.6256	2.1129	1.4965	1.1948	1.0651	1.0183	1.0053	1.0014	1.0003	1.0001	1
0.85	1.4963	1.1925	1.0639	1.0200	1.0046	1.0011	1.0002	1.0001	1	1	1
0.8	1.0636	1.0190	1.0050	1.0011	1.0003	1.0001	1	1	1	1	1
0.75	1.0051	1.0014	1.0002	1	1	1	1	1	1	1	1
0.7	1.0003	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.2657	77.6591	30.0550	11.7989	4.9975	2.4811	1.5437	1.1875	1.0567	1.0170	1.0048

0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.9256	41.5221	7.0735	2.3891	1.5865	1.0793	1.0080	1.0011	1.0001	1	1
0.95	7.0745	2.3942	1.5871	1.0781	1.0077	1.0011	1.0001	1	1	1	1
0.9	1.5853	1.0808	1.0085	1.0010	1.0002	1	1	1	1	1	1
0.85	1.0087	1.0014	1.0002	1.0001	1	1	1	1	1	1	1
0.8	1.0003	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.6143	56.5818	13.4830	3.2205	1.5563	1.0614	1.0071	1.0010	1.0001	1	1
0.95	13.3580	3.2420	1.5553	1.0628	1.0066	1.0011	1.0001	1	1	1	1
0.9	1.5553	1.0625	1.0064	1.0009	1.0001	1	1	1	1	1	1
0.85	1.0068	1.0010	1.0001	1	1	1	1	1	1	1	1
0.8	1.0001	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.1143	68.1060	21.3052	5.8303	1.6672	1.0568	1.0062	1.0010	1.0001	1	1
0.95	21.1214	5.7910	1.6724	1.0580	1.0061	1.0011	1.0002	1	1	1	1
0.9	1.6698	1.0568	1.0065	1.0010	1.0002	1	1	1	1	1	1
0.85	1.0065	1.0008	1.0001	1	1	1	1	1	1	1	1
0.8	1.0002	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2

1	199.4813	72.0561	24.9348	7.6186	1.8551	1.0546	1.0058	1.0008	1	1	1
0.95	24.9293	7.6089	1.8678	1.0559	1.0057	1.0007	1.0001	1	1	1	1
0.9	1.8649	1.0563	1.0064	1.0009	1.0002	1	1	1	1	1	1
0.85	1.0059	1.0009	1.0001	1	1	1	1	1	1	1	1
0.8	1.0001	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S17 ARL of the EWMA TBEA schemes based on the statistic Z_2 when variable A following the gamma distribution and T following the weibull distribution assuming a Frank copula when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.6008	31.1998	11.0709	6.2087	4.2029	3.1408	2.4892	2.0510	1.7400	1.4930	1.2782
0.95	13.3617	7.1827	4.7971	3.5539	2.8032	2.2995	1.9496	1.6936	1.4596	1.2537	1.1124
0.9	5.4246	3.9862	3.1290	2.5632	2.1640	1.8800	1.6540	1.4343	1.2354	1.0985	1.0339
0.85	3.4780	2.8328	2.3850	2.0595	1.8275	1.6266	1.4055	1.2116	1.0864	1.0279	1.0072
0.8	2.6203	2.2492	1.9807	1.7900	1.5954	1.3819	1.1942	1.0746	1.0229	1.0053	1.0011
0.75	2.1395	1.9250	1.7579	1.5702	1.3555	1.1722	1.0651	1.0190	1.0043	1.0007	1.0001
0.7	1.8833	1.7356	1.5432	1.3304	1.1544	1.0553	1.0147	1.0035	1.0004	1.0001	1.0000
0.65	1.7132	1.5255	1.3059	1.1384	1.0466	1.0120	1.0023	1.0004	1.0001	1.0000	1.0000
0.6	1.4970	1.2819	1.1203	1.0401	1.0098	1.0018	1.0002	1.0000	1.0000	1.0000	1.0000
0.55	1.2579	1.1068	1.0339	1.0077	1.0015	1.0003	1.0000	1.0000	1.0000	1.0000	1.0000
0.5	1.0936	1.0273	1.0068	1.0010	1.0002	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.3087	52.4003	16.2303	7.0993	4.1244	2.8285	2.1443	1.7209	1.4321	1.2224	1.0959
0.95	23.5643	9.2256	5.0123	3.3171	2.4595	1.9543	1.6231	1.3661	1.1812	1.0738	1.0241
0.9	6.1352	3.8886	2.8133	2.2070	1.8214	1.5429	1.3130	1.1462	1.0547	1.0169	1.0040
0.85	3.2130	2.4841	2.0325	1.7225	1.4724	1.2613	1.1148	1.0392	1.0114	1.0026	1.0006
0.8	2.2522	1.9010	1.6411	1.4129	1.2125	1.0884	1.0281	1.0074	1.0015	1.0003	1
0.75	1.8069	1.5754	1.3594	1.1758	1.0665	1.0211	1.0047	1.0009	1.0002	1	1
0.7	1.5162	1.3055	1.1402	1.0502	1.0133	1.0031	1.0005	1.0001	1	1	1
0.65	1.2560	1.1089	1.0363	1.0097	1.0019	1.0002	1	1	1	1	1
0.6	1.0833	1.0256	1.0061	1.0012	1.0002	1.0000	1	1	1	1	1
0.55	1.0181	1.0038	1.0007	1.0000	1.0000	1.0000	1	1	1	1	1
0.5	1.0025	1.0004	1.0000	1.0000	1.0000	1.0000	1	1	1	1	1
$\lambda = 0.5$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.2814	68.8840	24.5187	9.9694	4.9919	3.0585	2.1392	1.6486	1.3429	1.1591	1.0628
0.95	39.0385	14.7186	6.7489	3.8385	2.5521	1.9116	1.5258	1.2749	1.1212	1.0446	1.0140

0.9	9.3991	4.8913	3.0875	2.2336	1.7451	1.4307	1.2147	1.0906	1.0301	1.0083	1.0019
0.85	3.8076	2.6209	2.0006	1.6175	1.3550	1.1684	1.0641	1.0201	1.0053	1.0011	1.0002
0.8	2.3023	1.8324	1.5168	1.2832	1.1282	1.0455	1.0131	1.0028	1.0004	1.0001	1
0.75	1.7046	1.4357	1.2302	1.0959	1.0315	1.0084	1.0019	1.0002	1.0001	1	1
0.7	1.3657	1.1797	1.0683	1.0206	1.0048	1.0009	1.0002	1	1	1	1
0.65	1.1380	1.0478	1.0134	1.0032	1.0005	1.0001	1	1	1	1	1
0.6	1.0333	1.0085	1.0015	1.0002	1	1	1	1	1	1	1
0.55	1.0053	1.0010	1.0001	1	1	1	1	1	1	1	1
0.5	1.0005	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.3123	79.5485	32.2803	13.7265	6.5216	3.6118	2.2920	1.6496	1.3165	1.1354	1.0513
0.95	52.8509	21.7556	9.7517	4.9444	2.9341	1.9937	1.5124	1.2409	1.0978	1.0335	1.0097
0.9	14.8936	7.0532	3.8733	2.4605	1.7738	1.3964	1.1797	1.0697	1.0217	1.0058	1.0013
0.85	5.2910	3.1416	2.1433	1.6143	1.3062	1.1363	1.0475	1.0138	1.0033	1.0008	1.0001
0.8	2.6244	1.8978	1.4853	1.2356	1.0963	1.0319	1.0088	1.0017	1.0004	1.0001	1
0.75	1.7189	1.3856	1.1792	1.0686	1.0208	1.0050	1.0009	1.0002	1	1	1
0.7	1.3008	1.1349	1.0473	1.0130	1.0031	1.0004	1	1	1	1	1
0.65	1.0971	1.0308	1.0083	1.0015	1.0002	1	1	1	1	1	1
0.6	1.0204	1.0048	1.0008	1.0001	1	1	1	1	1	1	1
0.55	1.0027	1.0004	1.0001	1	1	1	1	1	1	1	1
0.5	1.0002	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.8768	86.1644	38.3670	17.4945	8.4038	4.4208	2.5858	1.7244	1.3188	1.1257	1.0456
0.95	63.0573	28.1984	13.1778	6.5957	3.5712	2.2011	1.5482	1.2333	1.0880	1.0302	1.0086
0.9	21.1411	10.0740	5.2172	2.9535	1.9144	1.4151	1.1680	1.0608	1.0181	1.0049	1.0013
0.85	7.7934	4.1867	2.4969	1.6999	1.3050	1.1190	1.0412	1.0111	1.0027	1.0006	1.0001
0.8	3.4177	2.1417	1.5314	1.2235	1.0844	1.0262	1.0067	1.0014	1.0003	1.0001	1
0.75	1.8717	1.4008	1.1617	1.0574	1.0168	1.0043	1.0007	1.0001	1.0001	1	1
0.7	1.2969	1.1157	1.0381	1.0093	1.0024	1.0004	1.0001	1	1	1	1
0.65	1.0807	1.0242	1.0063	1.0012	1.0002	1	1	1	1	1	1
0.6	1.0152	1.0037	1.0007	1.0001	1	1	1	1	1	1	1
0.55	1.0022	1.0003	1	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.5876	89.0469	40.7928	19.1445	9.3742	4.8825	2.7711	1.7874	1.3277	1.1263	1.0436
0.95	66.6870	30.9972	14.8816	7.4768	3.9703	2.3609	1.5915	1.2377	1.0882	1.0291	1.0079
0.9	23.8465	11.6752	5.9847	3.3044	2.0224	1.4387	1.1695	1.0592	1.0176	1.0045	1.0011
0.85	9.2700	4.8278	2.7641	1.7754	1.3190	1.1193	1.0385	1.0101	1.0025	1.0005	1.0001
0.8	3.9932	2.3510	1.5845	1.2315	1.0793	1.0252	1.0062	1.0012	1.0002	1	1
0.75	2.0209	1.4268	1.1617	1.0555	1.0161	1.0037	1.0006	1.0001	1	1	1
0.7	1.3118	1.1112	1.0356	1.0096	1.0019	1.0004	1	1	1	1	1
0.65	1.0762	1.0229	1.0051	1.0010	1.0001	1	1	1	1	1	1
0.6	1.0141	1.0030	1.0005	1	1	1	1	1	1	1	1

0.55	1.0016	1.0002	1	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.1994	20.7504	7.3290	4.2666	2.9887	2.2919	1.8853	1.5923	1.2848	1.0933	1.0236
0.95	8.4957	4.8176	3.3451	2.5528	2.0690	1.7877	1.5014	1.2151	1.0637	1.0152	1.0030
0.9	3.7299	2.8296	2.2706	1.9350	1.7092	1.4173	1.1548	1.0415	1.0088	1.0015	1.0002
0.85	2.4972	2.0776	1.8567	1.6325	1.3279	1.1086	1.0263	1.0050	1.0009	1.0001	1.0001
0.8	1.9632	1.7958	1.5542	1.2530	1.0752	1.0158	1.0028	1.0005	1.0000	1	1
0.75	1.7403	1.4638	1.1855	1.0484	1.0096	1.0015	1.0003	1	1	1	1
0.7	1.3788	1.1303	1.0312	1.0052	1.0008	1.0001	1	1	1	1	1
0.65	1.0875	1.0196	1.0034	1.0005	1	1	1	1	1	1	1
0.6	1.0117	1.0019	1.0002	1	1	1	1	1	1	1	1
0.55	1.0010	1.0001	1	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.0406	41.4214	10.2282	4.4373	2.7280	1.9928	1.5759	1.2547	1.0803	1.0196	1.0040
0.95	13.9927	5.4209	3.1636	2.2524	1.7811	1.4316	1.1658	1.0464	1.0102	1.0019	1.0004
0.9	3.6643	2.5334	1.9753	1.6337	1.3118	1.1010	1.0244	1.0052	1.0011	1.0002	1
0.85	2.1773	1.8051	1.4991	1.2085	1.0591	1.0133	1.0024	1.0004	1.0001	1	1
0.8	1.6802	1.3728	1.1309	1.0329	1.0063	1.0010	1.0002	1	1	1	1
0.75	1.2592	1.0767	1.0174	1.0029	1.0005	1.0001	1	1	1	1	1
0.7	1.0437	1.0085	1.0017	1.0003	1	1	1	1	1	1	1
0.65	1.0045	1.0006	1.0001	1	1	1	1	1	1	1	1
0.6	1.0003	1.0001	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.1703	60.7409	17.5916	6.1293	3.0452	1.9779	1.4697	1.1722	1.0489	1.0120	1.0026
0.95	27.1060	8.6211	3.8316	2.3371	1.7009	1.3171	1.1041	1.0261	1.0062	1.0012	1.0002
0.9	4.9469	2.7717	1.9524	1.5063	1.2031	1.0574	1.0127	1.0024	1.0005	1	1
0.85	2.2361	1.7166	1.3531	1.1211	1.0294	1.0062	1.0011	1.0002	1	1	1
0.8	1.5451	1.2309	1.0665	1.0154	1.0028	1.0004	1.0001	1	1	1	1
0.75	1.1408	1.0354	1.0072	1.0015	1.0002	1.0001	1	1	1	1	1
0.7	1.0181	1.0029	1.0005	1.0001	1	1	1	1	1	1	1
0.65	1.0014	1.0003	1	1	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.7867	73.7142	26.0537	9.3235	3.8904	2.1244	1.4468	1.1448	1.0390	1.0091	1.0018

0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.9312	28.7170	5.0236	2.4607	1.7949	1.2254	1.0171	1.0017	1.0001	1	1
0.95	6.3379	2.8138	1.9325	1.5091	1.0524	1.0046	1.0003	1	1	1	1
0.9	2.0875	1.7458	1.1692	1.0130	1.0010	1.0001	1	1	1	1	1
0.85	1.4411	1.0366	1.0030	1.0003	1	1	1	1	1	1	1
0.8	1.0084	1.0007	1	1	1	1	1	1	1	1	1
0.75	1.0002	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.5919	52.0390	10.3482	2.8597	1.7430	1.1402	1.0116	1.0010	1.0001	1	1
0.95	15.6049	3.6775	1.9582	1.3672	1.0311	1.0029	1.0003	1	1	1	1
0.9	2.2369	1.6453	1.0925	1.0067	1.0005	1	1	1	1	1	1
0.85	1.2697	1.0187	1.0015	1.0001	1.0000	1	1	1	1	1	1
0.8	1.0042	1.0003	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.0514	66.8412	19.1457	4.6198	1.7852	1.1071	1.0084	1.0007	1.0001	1	1
0.95	28.9022	7.3855	2.2387	1.3001	1.0219	1.0020	1.0001	1	1	1	1
0.9	3.0188	1.6026	1.0661	1.0054	1.0004	1	1	1	1	1	1
0.85	1.1969	1.0141	1.0010	1.0001	1	1	1	1	1	1	1
0.8	1.0029	1.0002	1	1	1	1	1	1	1	1	1
0.75	1.0001	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2

1	201.4186	77.5135	27.7424	8.4981	2.1739	1.0958	1.0076	1.0007	1.0001	1	1
0.95	40.8085	13.6238	3.6134	1.2948	1.0198	1.0019	1.0001	1	1	1	1
0.9	6.2107	1.7542	1.0565	1.0041	1.0004	1	1	1	1	1	1
0.85	1.1760	1.0121	1.0010	1	1	1	1	1	1	1	1
0.8	1.0023	1.0002	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	202.4438	80.8913	30.9900	10.5065	2.6489	1.0955	1.0077	1.0007	1	1	1
0.95	45.4757	16.6763	4.8569	1.3191	1.0184	1.0017	1.0002	1	1	1	1
0.9	8.4150	2.0467	1.0549	1.0038	1.0003	1	1	1	1	1	1
0.85	1.1782	1.0104	1.0010	1.0001	1	1	1	1	1	1	1
0.8	1.0023	1.0001	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S18 ARL of the EWMA TBEA schemes based on the statistic Z_3 when variable A following the gamma distribution and T following the weibull distribution assuming a Frank copula when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.8190	30.8049	10.9489	6.1102	4.1365	3.0915	2.4463	2.0236	1.7239	1.4823	1.2652
0.95	13.4339	7.4504	5.0180	3.7385	2.9455	2.4136	2.0428	1.7764	1.5521	1.3298	1.1547
0.9	5.3963	4.1255	3.3128	2.7514	2.3296	2.0229	1.8056	1.6003	1.3801	1.1894	1.0728
0.85	3.4457	2.9287	2.5283	2.2052	1.9801	1.8073	1.6270	1.4147	1.2134	1.0842	1.0254
0.8	2.5946	2.3071	2.0810	1.9273	1.7974	1.6258	1.4137	1.2192	1.0876	1.0271	1.0067
0.75	2.1107	1.9811	1.8812	1.7632	1.5952	1.3858	1.2013	1.0812	1.0253	1.0063	1.0011
0.7	1.9222	1.8336	1.7051	1.5218	1.3250	1.1611	1.0625	1.0197	1.0045	1.0009	1.0002
0.65	1.7546	1.5989	1.4121	1.2362	1.1106	1.0425	1.0126	1.0030	1.0004	1	1
0.6	1.4349	1.2730	1.1408	1.0624	1.0222	1.0068	1.0013	1.0002	1	1	1
0.55	1.1400	1.0651	1.0281	1.0090	1.0024	1.0007	1.0001	1	1	1	1
0.5	1.0245	1.0088	1.0027	1.0008	1.0002	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.9733	51.2782	15.7105	6.8539	3.9743	2.7315	2.0827	1.6789	1.3979	1.1925	1.0730
0.95	24.6884	9.6901	5.2622	3.4706	2.5732	2.0396	1.6961	1.4285	1.2165	1.0851	1.0271

0.9	5.9953	3.9963	2.9649	2.3505	1.9589	1.6705	1.4270	1.2244	1.0892	1.0274	1.0066
0.85	3.1111	2.5254	2.1314	1.8474	1.6189	1.3958	1.2075	1.0843	1.0260	1.0064	1.0010
0.8	2.1860	1.9386	1.7384	1.5402	1.3319	1.1636	1.0662	1.0195	1.0048	1.0009	1.0001
0.75	1.7802	1.6088	1.4222	1.2426	1.1142	1.0428	1.0133	1.0029	1.0006	1.0001	1
0.7	1.4448	1.2763	1.1475	1.0631	1.0237	1.0069	1.0015	1.0003	1	1	1
0.65	1.1448	1.0694	1.0280	1.0097	1.0026	1.0006	1.0001	1	1	1	1
0.6	1.0244	1.0097	1.0025	1.0007	1.0002	1	1	1	1	1	1
0.55	1.0020	1.0008	1.0002	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.4160	67.7681	23.7289	9.6226	4.7885	2.9216	2.0587	1.5876	1.3001	1.1265	1.0417
0.95	44.7711	16.5182	7.3766	4.1154	2.7180	2.0058	1.5873	1.3098	1.1385	1.0452	1.0128
0.9	9.6515	5.1958	3.3343	2.4121	1.8911	1.5459	1.2970	1.1319	1.0469	1.0121	1.0026
0.85	3.6340	2.6611	2.1076	1.7446	1.4675	1.2561	1.1111	1.0392	1.0104	1.0024	1.0005
0.8	2.1831	1.8473	1.5864	1.3669	1.1908	1.0801	1.0269	1.0070	1.0013	1.0003	1
0.75	1.6263	1.4230	1.2465	1.1209	1.0473	1.0146	1.0042	1.0010	1.0001	1	1
0.7	1.2535	1.1350	1.0596	1.0224	1.0069	1.0020	1.0003	1.0001	1	1	1
0.65	1.0564	1.0244	1.0086	1.0025	1.0006	1.0001	1	1	1	1	1
0.6	1.0067	1.0022	1.0007	1.0002	1	1	1	1	1	1	1
0.55	1.0004	1.0001	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.4595	78.3883	31.2269	13.2451	6.2360	3.4223	2.1811	1.5842	1.2606	1.1032	1.0315
0.95	62.5127	25.5623	11.1979	5.5323	3.1832	2.1203	1.5699	1.2690	1.1058	1.0341	1.0076
0.9	16.8618	8.0454	4.3986	2.7692	1.9557	1.5180	1.2459	1.0969	1.0313	1.0080	1.0018
0.85	5.2119	3.2717	2.3011	1.7534	1.4158	1.1992	1.0790	1.0248	1.0068	1.0012	1.0002
0.8	2.4319	1.8909	1.5408	1.2980	1.1393	1.0538	1.0176	1.0038	1.0009	1.0001	1
0.75	1.5698	1.3419	1.1805	1.0783	1.0302	1.0086	1.0024	1.0004	1	1	1
0.7	1.1787	1.0884	1.0362	1.0131	1.0037	1.0010	1.0001	1	1	1	1
0.65	1.0316	1.0127	1.0039	1.0011	1.0004	1.0001	1	1	1	1	1
0.6	1.0030	1.0010	1.0002	1.0001	1	1	1	1	1	1	1
0.55	1.0002	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.2825	85.0765	36.9926	16.8128	8.0643	4.1917	2.4538	1.6353	1.2562	1.0891	1.0255
0.95	76.5959	33.8975	15.6065	7.6186	4.0550	2.4050	1.6306	1.2605	1.0934	1.0291	1.0063
0.9	26.2443	12.5042	6.3863	3.5344	2.2011	1.5484	1.2340	1.0870	1.0256	1.0062	1.0011
0.85	8.6177	4.6769	2.8283	1.9056	1.4280	1.1813	1.0656	1.0206	1.0050	1.0008	1.0001
0.8	3.1838	2.1563	1.5876	1.2842	1.1194	1.0418	1.0126	1.0032	1.0006	1.0001	1
0.75	1.6341	1.3325	1.1566	1.0649	1.0227	1.0065	1.0016	1.0003	1.0001	1	1
0.7	1.1558	1.0700	1.0284	1.0089	1.0026	1.0006	1.0001	1	1	1	1
0.65	1.0237	1.0089	1.0025	1.0008	1.0001	1	1	1	1	1	1
0.6	1.0021	1.0007	1.0001	1	1	1	1	1	1	1	1

0.55	1.0001	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.3610	86.9762	39.2361	18.3100	8.9327	4.6009	2.6087	1.6738	1.2617	1.0888	1.0252
0.95	81.8343	37.2933	17.8170	8.7453	4.5794	2.6061	1.6822	1.2663	1.0917	1.0266	1.0064
0.9	30.6504	14.9508	7.5790	4.0766	2.4048	1.5988	1.2395	1.0832	1.0240	1.0058	1.0012
0.85	10.6778	5.7160	3.2580	2.0474	1.4601	1.1831	1.0632	1.0181	1.0039	1.0008	1.0002
0.8	3.8371	2.4081	1.6715	1.2949	1.1173	1.0389	1.0122	1.0027	1.0004	1.0001	1
0.75	1.7281	1.3483	1.1555	1.0587	1.0207	1.0058	1.0014	1.0002	1	1	1
0.7	1.1490	1.0660	1.0260	1.0084	1.0021	1.0005	1.0001	1	1	1	1
0.65	1.0213	1.0080	1.0026	1.0006	1.0001	1	1	1	1	1	1
0.6	1.0019	1.0004	1.0002	1	1	1	1	1	1	1	1
0.55	1.0001	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.8505	22.2811	7.7263	4.4498	3.1015	2.3742	1.9325	1.6592	1.3726	1.1435	1.0404
0.95	9.2051	5.3101	3.6839	2.8133	2.2591	1.9185	1.6939	1.4126	1.1604	1.0442	1.0100
0.9	3.9328	3.0778	2.5183	2.1071	1.8869	1.6972	1.4191	1.1600	1.0425	1.0087	1.0015
0.85	2.6175	2.2215	1.9876	1.8550	1.6674	1.3794	1.1338	1.0334	1.0067	1.0010	1.0002
0.8	2.0211	1.9201	1.8035	1.5800	1.2852	1.0914	1.0213	1.0041	1.0007	1.0001	1
0.75	1.8541	1.6885	1.4176	1.1672	1.0466	1.0105	1.0022	1.0003	1	1	1
0.7	1.4574	1.2144	1.0722	1.0196	1.0042	1.0010	1.0001	1	1	1	1
0.65	1.0740	1.0235	1.0064	1.0013	1.0003	1.0001	1	1	1	1	1
0.6	1.0062	1.0017	1.0004	1.0001	1	1	1	1	1	1	1
0.55	1.0004	1.0001	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.9381	44.9505	11.3960	4.7835	2.8888	2.0858	1.6568	1.3280	1.1125	1.0279	1.0052
0.95	17.1883	6.4284	3.6416	2.5283	1.9675	1.6326	1.3193	1.1056	1.0261	1.0048	1.0008
0.9	3.9748	2.8245	2.1976	1.8501	1.5694	1.2661	1.0799	1.0189	1.0036	1.0007	1.0001
0.85	2.2866	1.9539	1.7228	1.4464	1.1789	1.0501	1.0110	1.0020	1.0003	1.0001	1
0.8	1.7844	1.5479	1.2698	1.0938	1.0243	1.0053	1.0009	1.0001	1	1	1
0.75	1.2965	1.1149	1.0352	1.0089	1.0020	1.0003	1.0001	1	1	1	1
0.7	1.0351	1.0111	1.0027	1.0007	1.0002	1	1	1	1	1	1
0.65	1.0029	1.0005	1.0001	1	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.3168	63.9779	19.7332	7.0012	3.3524	2.1150	1.5737	1.2353	1.0684	1.0153	1.0025

0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.8$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.5354	14.8772	5.0980	3.0927	2.2294	1.8281	1.5446	1.1287	1.0166	1.0021	1.0003
0.95	5.8882	3.5984	2.6395	1.9904	1.8271	1.5104	1.0846	1.0088	1.0011	1.0001	1
0.9	2.7887	2.1275	1.9287	1.7948	1.3531	1.0319	1.0030	1.0002	1	1	1
0.85	1.9604	1.8900	1.6471	1.1026	1.0066	1.0006	1	1	1	1	1
0.8	1.7327	1.1985	1.0130	1.0008	1.0001	1	1	1	1	1	1
0.75	1.0172	1.0018	1.0002	1	1	1	1	1	1	1	1
0.7	1.0004	1.0001	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.3586	39.3803	7.8971	3.1875	2.0635	1.6728	1.2089	1.0234	1.0029	1.0003	1
0.95	11.4171	4.0134	2.5026	1.8895	1.6024	1.1114	1.0097	1.0010	1.0001	1	1
0.9	2.6904	1.9951	1.8160	1.3710	1.0291	1.0022	1.0002	1	1	1	1
0.85	1.8733	1.5776	1.0710	1.0037	1.0003	1	1	1	1	1	1
0.8	1.0924	1.0072	1.0005	1.0001	1	1	1	1	1	1	1
0.75	1.0012	1.0002	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.7506	60.7027	16.5340	4.7590	2.2116	1.6003	1.1280	1.0140	1.0018	1.0001	1
0.95	31.2098	8.1607	2.9700	1.8987	1.4733	1.0559	1.0053	1.0005	1	1	1
0.9	3.3781	2.0945	1.7387	1.1957	1.0121	1.0010	1.0001	1	1	1	1
0.85	1.8115	1.3582	1.0252	1.0018	1.0001	1	1	1	1	1	1
0.8	1.0349	1.0029	1.0003	1	1	1	1	1	1	1	1
0.75	1.0006	1.0001	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2

1	199.8960	73.9285	25.6674	8.5731	2.9546	1.5912	1.0996	1.0108	1.0012	1.0001	1
0.95	54.1989	17.9996	5.6368	2.1674	1.4117	1.0406	1.0036	1.0004	1	1	1
0.9	7.9225	2.7782	1.7154	1.1363	1.0083	1.0008	1	1	1	1	1
0.85	1.8172	1.2642	1.0166	1.0011	1.0001	1	1	1	1	1	1
0.8	1.0233	1.0022	1.0001	1	1	1	1	1	1	1	1
0.75	1.0006	1.0001	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.3008	81.4207	32.6577	12.8547	4.7191	1.7759	1.0899	1.0096	1.0008	1.0001	1
0.95	71.9204	28.1312	10.6782	3.5986	1.4226	1.0336	1.0032	1.0003	1	1	1
0.9	17.5650	6.1804	2.0466	1.1096	1.0066	1.0006	1	1	1	1	1
0.85	2.4111	1.2202	1.0132	1.0008	1	1	1	1	1	1	1
0.8	1.0180	1.0016	1.0002	1	1	1	1	1	1	1	1
0.75	1.0004	1.0001	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.7025	84.1527	35.5455	14.7238	5.7146	2.0130	1.0897	1.0087	1.0008	1.0001	1
0.95	78.8165	32.8026	13.2623	4.7661	1.5184	1.0312	1.0028	1.0003	1	1	1
0.9	23.0584	8.8858	2.7322	1.1073	1.0062	1.0005	1	1	1	1	1
0.85	3.6077	1.2291	1.0116	1.0007	1.0001	1	1	1	1	1	1
0.8	1.0155	1.0015	1.0001	1	1	1	1	1	1	1	1
0.75	1.0004	1.0001	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S19 ARL of the EWMA TBEA schemes based on the statistic Z_1 when both variables A and T following the weibull distribution assuming a Clayton copula when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.9488	29.1557	9.2824	4.9763	3.4010	2.6037	2.1568	1.8518	1.6209	1.4204	1.2527
0.95	9.3045	4.9806	3.3907	2.6108	2.1561	1.8553	1.6212	1.4173	1.2515	1.1372	1.0633

0.9	3.3960	2.6135	2.1520	1.8563	1.6211	1.4205	1.2511	1.1343	1.0636	1.0265	1.0105
0.85	2.1553	1.8538	1.6230	1.4205	1.2482	1.1324	1.0610	1.0273	1.0110	1.0040	1.0014
0.8	1.6217	1.4185	1.2517	1.1324	1.0616	1.0259	1.0103	1.0038	1.0014	1.0005	1.0001
0.75	1.2483	1.1315	1.0618	1.0254	1.0102	1.0036	1.0012	1.0005	1.0002	1.0001	1.0001
0.7	1.0618	1.0258	1.0102	1.0037	1.0012	1.0004	1.0002	1	1	1	1
0.65	1.0101	1.0033	1.0013	1.0005	1.0002	1	1	1	1	1	1
0.6	1.0015	1.0004	1.0001	1	1	1	1	1	1	1	1
0.55	1.0002	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.4268	39.6919	11.5924	5.3391	3.3096	2.4041	1.9117	1.5865	1.3657	1.2068	1.1043
0.95	11.6729	5.3669	3.3128	2.4047	1.9082	1.5892	1.3621	1.2037	1.1033	1.0459	1.0192
0.9	3.3095	2.4019	1.9115	1.5869	1.3625	1.2030	1.1006	1.0449	1.0193	1.0076	1.0028
0.85	1.9117	1.5853	1.3624	1.2009	1.1015	1.0465	1.0177	1.0068	1.0024	1.0009	1.0003
0.8	1.3596	1.2008	1.0991	1.0448	1.0180	1.0066	1.0025	1.0008	1.0002	1.0001	1
0.75	1.0993	1.0446	1.0176	1.0071	1.0023	1.0009	1.0003	1.0001	1	1	1
0.7	1.0184	1.0068	1.0024	1.0009	1.0003	1.0001	1	1	1	1	1
0.65	1.0023	1.0006	1.0003	1.0001	1	1	1	1	1	1	1
0.6	1.0003	1.0001	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.1097	50.4862	15.4173	6.4156	3.5850	2.4164	1.8333	1.4943	1.2749	1.1430	1.0680
0.95	15.4722	6.4443	3.5921	2.4183	1.8363	1.4867	1.2754	1.1421	1.0669	1.0277	1.0114
0.9	3.5968	2.4243	1.8330	1.4891	1.2741	1.1404	1.0632	1.0268	1.0108	1.0040	1.0014
0.85	1.8311	1.4897	1.2709	1.1378	1.0649	1.0270	1.0105	1.0041	1.0012	1.0005	1.0001
0.8	1.2703	1.1371	1.0650	1.0277	1.0100	1.0037	1.0014	1.0004	1.0002	1.0001	1
0.75	1.0634	1.0266	1.0103	1.0034	1.0012	1.0005	1.0002	1.0001	1	1	1
0.7	1.0100	1.0039	1.0014	1.0004	1.0001	1.0001	1	1	1	1	1
0.65	1.0014	1.0003	1.0001	1.0001	1	1	1	1	1	1	1
0.6	1.0001	1.0001	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.8138	59.7718	20.2677	8.1748	4.1765	2.5928	1.8516	1.4603	1.2412	1.1187	1.0549
0.95	20.2338	8.2072	4.1745	2.5908	1.8548	1.4556	1.2393	1.1158	1.0521	1.0212	1.0083
0.9	4.2089	2.5954	1.8518	1.4543	1.2380	1.1141	1.0514	1.0208	1.0082	1.0027	1.0009
0.85	1.8506	1.4564	1.2382	1.1145	1.0521	1.0205	1.0082	1.0027	1.0009	1.0004	1.0001
0.8	1.2367	1.1139	1.0509	1.0215	1.0075	1.0026	1.0010	1.0003	1.0001	1	1
0.75	1.0490	1.0205	1.0078	1.0028	1.0009	1.0004	1.0001	1	1	1	1
0.7	1.0078	1.0030	1.0012	1.0003	1.0001	1.0001	1	1	1	1	1
0.65	1.0010	1.0001	1.0001	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1

0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.4988	67.7127	25.0197	10.4060	5.0817	2.9004	1.9481	1.4749	1.2301	1.1074	1.0463
0.95	25.0471	10.4615	5.1140	2.9133	1.9404	1.4720	1.2282	1.1066	1.0451	1.0187	1.0075
0.9	5.1226	2.9412	1.9552	1.4728	1.2285	1.1053	1.0446	1.0182	1.0065	1.0023	1.0007
0.85	1.9522	1.4687	1.2256	1.1028	1.0440	1.0181	1.0069	1.0023	1.0007	1.0003	1
0.8	1.2236	1.1036	1.0453	1.0176	1.0063	1.0023	1.0006	1.0002	1.0001	1	1
0.75	1.0439	1.0176	1.0062	1.0024	1.0008	1.0003	1.0001	1	1	1	1
0.7	1.0068	1.0022	1.0008	1.0003	1.0001	1	1	1	1	1	1
0.65	1.0009	1.0003	1.0000	1.0001	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.9976	71.3027	27.1634	11.7495	5.6748	3.1618	2.0343	1.4987	1.2345	1.1061	1.0456
0.95	27.5568	11.7791	5.6587	3.1621	2.0356	1.4933	1.2349	1.1049	1.0434	1.0175	1.0064
0.9	5.7347	3.1754	2.0296	1.4965	1.2304	1.1036	1.0440	1.0171	1.0063	1.0024	1.0008
0.85	2.0354	1.4901	1.2290	1.1039	1.0425	1.0167	1.0059	1.0019	1.0008	1.0004	1
0.8	1.2272	1.1012	1.0432	1.0172	1.0060	1.0019	1.0008	1.0003	1.0001	1	1
0.75	1.0402	1.0163	1.0062	1.0025	1.0007	1.0002	1.0001	1	1	1	1
0.7	1.0061	1.0022	1.0008	1.0002	1.0001	1.0001	1	1	1	1	1
0.65	1.0008	1.0002	1.0001	1	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.0120	13.6705	4.5769	2.7446	2.0399	1.6458	1.3232	1.1199	1.0375	1.0096	1.0021
0.95	4.5741	2.7458	2.0390	1.6470	1.3185	1.1203	1.0361	1.0091	1.0019	1.0004	1.0001
0.9	2.0410	1.6436	1.3185	1.1191	1.0351	1.0082	1.0019	1.0003	1.0001	1	1
0.85	1.3194	1.1143	1.0333	1.0080	1.0015	1.0003	1	1	1	1	1
0.8	1.0331	1.0077	1.0017	1.0002	1	1	1	1	1	1	1
0.75	1.0013	1.0003	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.2046	19.1106	4.9245	2.5985	1.8074	1.3860	1.1458	1.0458	1.0112	1.0026	1.0006

0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.2475	43.5887	11.7569	3.9781	1.8746	1.2543	1.0721	1.0187	1.0045	1.0010	1.0003
0.95	11.8259	4.0111	1.8760	1.2524	1.0718	1.0182	1.0042	1.0007	1.0001	1	1
0.9	1.8705	1.2483	1.0692	1.0178	1.0043	1.0009	1.0001	1	1	1	1
0.85	1.0673	1.0175	1.0040	1.0007	1.0001	1	1	1	1	1	1
0.8	1.0035	1.0006	1.0001	1	1	1	1	1	1	1	1
0.75	1.0001	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.8$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.4891	4.2659	2.0052	1.2338	1.0279	1.0033	1.0003	1	1	1	1
0.95	2.0063	1.2336	1.0256	1.0029	1.0002	1	1	1	1	1	1
0.9	1.0244	1.0026	1.0002	1	1	1	1	1	1	1	1
0.85	1.0002	1	1	1	1	1	1	1	1	1	1
0.8	1	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.0082	5.0873	1.8519	1.1099	1.0141	1.0015	1.0001	1	1	1	1
0.95	1.8492	1.1077	1.0131	1.0013	1.0001	1	1	1	1	1	1
0.9	1.0121	1.0013	1.0001	1	1	1	1	1	1	1	1
0.85	1.0001	1	1	1	1	1	1	1	1	1	1
0.8	1	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2

0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S20 ARL of the EWMA TBEA schemes based on the statistic Z_2 when both variables A and T following the weibull distribution assuming a Clayton copula when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.8757	24.3142	9.0653	5.2675	3.6453	2.7646	2.2077	1.8407	1.5642	1.3324	1.1644
0.95	10.7769	6.0430	4.1155	3.0989	2.4758	2.0576	1.7581	1.5082	1.2974	1.1408	1.0548
0.9	4.6293	3.4644	2.7498	2.2711	1.9416	1.6950	1.4640	1.2603	1.1171	1.0427	1.0129
0.85	3.0344	2.4975	2.1270	1.8637	1.6431	1.4226	1.2271	1.0990	1.0337	1.0093	1.0024
0.8	2.3203	2.0222	1.8054	1.5993	1.3819	1.1975	1.0798	1.0259	1.0070	1.0013	1.0004
0.75	1.9444	1.7580	1.5585	1.3442	1.1699	1.0633	1.0196	1.0046	1.0009	1.0002	1.0001
0.7	1.7188	1.5172	1.3065	1.1436	1.0503	1.0145	1.0032	1.0006	1.0002	1	1
0.65	1.4805	1.2710	1.1202	1.0389	1.0110	1.0025	1.0004	1.0001	1	1	1
0.6	1.2364	1.0987	1.0318	1.0085	1.0018	1.0003	1	1	1	1	1
0.55	1.0814	1.0238	1.0059	1.0013	1.0003	1	1	1	1	1	1
0.5	1.0192	1.0037	1.0008	1.0002	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.6430	35.8610	10.6965	5.0804	3.1855	2.2897	1.7887	1.4668	1.2475	1.1109	1.0418
0.95	14.6241	6.2584	3.7164	2.6185	2.0245	1.6470	1.3773	1.1931	1.0806	1.0288	1.0087
0.9	4.3817	2.9891	2.2724	1.8381	1.5304	1.2997	1.1450	1.0560	1.0179	1.0048	1.0013
0.85	2.5464	2.0377	1.6988	1.4405	1.2384	1.1043	1.0360	1.0115	1.0029	1.0008	1.0002
0.8	1.8773	1.5923	1.3629	1.1845	1.0751	1.0252	1.0070	1.0016	1.0004	1.0001	1
0.75	1.5081	1.2955	1.1374	1.0520	1.0160	1.0038	1.0009	1.0002	1.0001	1	1
0.7	1.2332	1.1004	1.0352	1.0103	1.0024	1.0005	1.0001	1	1	1	1
0.65	1.0713	1.0226	1.0060	1.0014	1.0003	1	1	1	1	1	1
0.6	1.0149	1.0036	1.0009	1.0002	1	1	1	1	1	1	1
0.55	1.0021	1.0005	1.0001	1	1	1	1	1	1	1	1
0.5	1.0003	1.0001	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.5630	48.2829	14.4384	6.0068	3.3383	2.2385	1.6801	1.3585	1.1684	1.0701	1.0240
0.95	22.4281	8.2154	4.1716	2.6494	1.9278	1.5179	1.2663	1.1188	1.0454	1.0143	1.0042
0.9	5.3140	3.1649	2.2211	1.7103	1.3957	1.1973	1.0826	1.0279	1.0088	1.0022	1.0006
0.85	2.5894	1.9459	1.5528	1.3012	1.1388	1.0551	1.0183	1.0046	1.0011	1.0003	1
0.8	1.7435	1.4385	1.2265	1.0954	1.0348	1.0098	1.0029	1.0007	1.0001	1	1
0.75	1.3410	1.1643	1.0657	1.0217	1.0058	1.0015	1.0003	1.0001	1	1	1
0.7	1.1169	1.0423	1.0126	1.0034	1.0008	1.0002	1	1	1	1	1
0.65	1.0257	1.0076	1.0018	1.0005	1	1	1	1	1	1	1
0.6	1.0045	1.0009	1.0003	1.0001	1	1	1	1	1	1	1

0.55	1.0006	1.0001	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.0212	58.0655	19.0158	7.5648	3.8027	2.3466	1.6703	1.3234	1.1416	1.0546	1.0191
0.95	31.7489	11.3930	5.1691	2.9262	1.9657	1.4855	1.2260	1.0956	1.0343	1.0108	1.0030
0.9	7.3492	3.7704	2.3644	1.7007	1.3504	1.1582	1.0605	1.0198	1.0063	1.0017	1.0004
0.85	2.9144	1.9950	1.5183	1.2497	1.1063	1.0371	1.0117	1.0031	1.0008	1.0002	1
0.8	1.7398	1.3819	1.1761	1.0705	1.0231	1.0064	1.0018	1.0004	1	1	1
0.75	1.2755	1.1213	1.0438	1.0134	1.0037	1.0009	1.0002	1	1	1	1
0.7	1.0798	1.0271	1.0083	1.0019	1.0005	1.0001	1	1	1	1	1
0.65	1.0167	1.0041	1.0010	1.0003	1.0001	1	1	1	1	1	1
0.6	1.0023	1.0006	1.0002	1	1	1	1	1	1	1	1
0.55	1.0003	1.0001	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.1169	65.0111	23.2944	9.4014	4.4922	2.5580	1.7089	1.3133	1.1302	1.0475	1.0161
0.95	39.8178	15.0203	6.5799	3.4041	2.0931	1.4944	1.2130	1.0843	1.0284	1.0089	1.0023
0.9	10.0460	4.7631	2.6747	1.7675	1.3440	1.1440	1.0514	1.0161	1.0047	1.0013	1.0004
0.85	3.5875	2.1748	1.5392	1.2366	1.0913	1.0305	1.0094	1.0028	1.0007	1.0002	1.0001
0.8	1.8280	1.3762	1.1554	1.0579	1.0178	1.0051	1.0013	1.0003	1	1	1
0.75	1.2592	1.1018	1.0343	1.0099	1.0028	1.0007	1.0001	1	1	1	1
0.7	1.0641	1.0202	1.0058	1.0014	1.0004	1.0001	1	1	1	1	1
0.65	1.0121	1.0031	1.0007	1.0001	1	1	1	1	1	1	1
0.6	1.0017	1.0004	1	1	1	1	1	1	1	1	1
0.55	1.0002	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.6514	67.6255	25.1035	10.3191	4.8916	2.7025	1.7442	1.3189	1.1307	1.0459	1.0148
0.95	43.0414	16.7547	7.3718	3.7079	2.1823	1.5156	1.2125	1.0823	1.0278	1.0082	1.0021
0.9	11.5420	5.3761	2.8966	1.8324	1.3507	1.1390	1.0502	1.0156	1.0042	1.0013	1.0003
0.85	4.0465	2.3275	1.5714	1.2422	1.0881	1.0292	1.0084	1.0025	1.0006	1.0001	1
0.8	1.9267	1.3915	1.1564	1.0538	1.0169	1.0049	1.0013	1.0002	1.0001	1	1
0.75	1.2646	1.0982	1.0325	1.0093	1.0026	1.0006	1	1	1	1	1
0.7	1.0617	1.0195	1.0052	1.0016	1.0003	1.0001	1	1	1	1	1
0.65	1.0110	1.0027	1.0008	1.0002	1	1	1	1	1	1	1
0.6	1.0014	1.0004	1	1	1	1	1	1	1	1	1
0.55	1.0002	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.1005	11.8660	4.8041	2.9957	2.1972	1.7376	1.3898	1.1622	1.0513	1.0119	1.0024

0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.0464	34.2560	7.6253	2.6468	1.4901	1.1513	1.0407	1.0100	1.0016	1.0003	1
0.95	12.0845	3.5572	1.7388	1.2298	1.0654	1.0154	1.0029	1.0004	1	1	1
0.9	2.1133	1.3443	1.1018	1.0256	1.0051	1.0007	1.0001	1	1	1	1
0.85	1.1581	1.0418	1.0092	1.0012	1.0002	1.0001	1	1	1	1	1
0.8	1.0146	1.0022	1.0004	1	1	1	1	1	1	1	1
0.75	1.0007	1.0001	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.8002	37.1827	8.5119	2.8018	1.5081	1.1492	1.0403	1.0089	1.0017	1.0003	1.0001
0.95	13.8800	3.9436	1.7927	1.2271	1.0612	1.0150	1.0029	1.0003	1.0001	1	1
0.9	2.2420	1.3515	1.0969	1.0224	1.0045	1.0008	1.0001	1	1	1	1
0.85	1.1501	1.0407	1.0081	1.0012	1.0001	1	1	1	1	1	1
0.8	1.0131	1.0026	1.0003	1	1	1	1	1	1	1	1
0.75	1.0007	1.0001	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.8$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	197.9852	4.2481	2.1188	1.3658	1.0589	1.0083	1.0010	1.0001	1	1	1
0.95	2.2653	1.5702	1.0965	1.0144	1.0013	1.0002	1	1	1	1	1
0.9	1.1662	1.0237	1.0026	1.0002	1	1	1	1	1	1	1
0.85	1.0050	1.0004	1	1	1	1	1	1	1	1	1
0.8	1	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2

0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.2698	14.5478	1.4499	1.0450	1.0060	1.0005	1	1	1	1	1
0.95	1.7826	1.0634	1.0079	1.0007	1	1	1	1	1	1	1
0.9	1.0110	1.0011	1.0001	1	1	1	1	1	1	1	1
0.85	1.0001	1	1	1	1	1	1	1	1	1	1
0.8	1	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S21 ARL of the EWMA TBEA schemes based on the statistic Z_3 when both variables A and T following the weibull distribution assuming a Clayton copula when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.5826	24.3394	9.0142	5.2292	3.6188	2.7314	2.1926	1.8308	1.5657	1.3332	1.1585
0.95	10.8860	6.2830	4.3250	3.2672	2.6121	2.1610	1.8538	1.6122	1.3837	1.1958	1.0775
0.9	4.6376	3.5905	2.9120	2.4383	2.0919	1.8491	1.6341	1.4114	1.2192	1.0894	1.0284
0.85	3.0219	2.5884	2.2464	2.0066	1.8214	1.6267	1.4176	1.2207	1.0928	1.0297	1.0071
0.8	2.2940	2.0821	1.9293	1.7817	1.5952	1.3856	1.2010	1.0852	1.0270	1.0066	1.0014
0.75	1.9696	1.8608	1.7149	1.5240	1.3221	1.1637	1.0661	1.0208	1.0053	1.0012	1.0002
0.7	1.7699	1.6003	1.4080	1.2329	1.1106	1.0427	1.0125	1.0038	1.0007	1.0001	1
0.65	1.4308	1.2633	1.1375	1.0623	1.0227	1.0072	1.0020	1.0004	1.0001	1	1
0.6	1.1343	1.0659	1.0287	1.0107	1.0037	1.0011	1.0003	1.0001	1	1	1
0.55	1.0252	1.0114	1.0043	1.0016	1.0004	1.0001	1	1	1	1	1
0.5	1.0035	1.0014	1.0004	1.0002	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.4709	36.5035	10.6578	5.0542	3.1404	2.2661	1.7772	1.4541	1.2339	1.0955	1.0314
0.95	15.6534	6.6862	3.9745	2.7836	2.1391	1.7357	1.4512	1.2391	1.1008	1.0326	1.0084
0.9	4.3780	3.1192	2.4161	1.9761	1.6671	1.4139	1.2202	1.0919	1.0298	1.0076	1.0015
0.85	2.5122	2.1077	1.8123	1.5666	1.3494	1.1796	1.0729	1.0232	1.0057	1.0013	1.0002
0.8	1.8707	1.6495	1.4400	1.2550	1.1228	1.0473	1.0147	1.0040	1.0009	1.0002	1
0.75	1.4657	1.2874	1.1517	1.0696	1.0274	1.0084	1.0023	1.0005	1.0001	1	1
0.7	1.1502	1.0763	1.0325	1.0122	1.0041	1.0010	1.0003	1	1	1	1
0.65	1.0296	1.0124	1.0051	1.0016	1.0005	1.0001	1	1	1	1	1
0.6	1.0042	1.0017	1.0006	1.0003	1.0001	1	1	1	1	1	1

0.55	1.0005	1.0002	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.6252	48.9867	14.6128	6.0692	3.3348	2.2239	1.6644	1.3408	1.1528	1.0545	1.0152
0.95	26.5075	9.4220	4.6259	2.8864	2.0711	1.6101	1.3255	1.1477	1.0528	1.0150	1.0033
0.9	5.5144	3.3880	2.4069	1.8644	1.5129	1.2746	1.1243	1.0444	1.0124	1.0029	1.0005
0.85	2.5364	2.0124	1.6540	1.3942	1.2062	1.0889	1.0315	1.0089	1.0019	1.0004	1
0.8	1.6991	1.4559	1.2625	1.1301	1.0549	1.0185	1.0054	1.0014	1.0003	1	1
0.75	1.2661	1.1437	1.0680	1.0265	1.0100	1.0026	1.0007	1.0002	1.0001	1	1
0.7	1.0637	1.0278	1.0114	1.0041	1.0014	1.0004	1.0001	1	1	1	1
0.65	1.0098	1.0044	1.0014	1.0004	1.0002	1.0001	1	1	1	1	1
0.6	1.0014	1.0006	1.0002	1.0001	1	1	1	1	1	1	1
0.55	1.0001	1.0001	1	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.6772	59.0134	19.2976	7.6718	3.8300	2.3331	1.6489	1.2994	1.1213	1.0395	1.0107
0.95	40.2125	14.0012	6.1013	3.2874	2.1406	1.5827	1.2746	1.1150	1.0374	1.0109	1.0024
0.9	8.2660	4.2535	2.6425	1.8772	1.4630	1.2210	1.0903	1.0300	1.0086	1.0018	1.0003
0.85	2.8853	2.0796	1.6090	1.3268	1.1544	1.0626	1.0205	1.0054	1.0011	1.0002	1
0.8	1.6511	1.3776	1.1952	1.0876	1.0354	1.0119	1.0030	1.0008	1.0002	1	1
0.75	1.1929	1.0958	1.0434	1.0170	1.0056	1.0018	1.0005	1.0001	1	1	1
0.7	1.0394	1.0165	1.0069	1.0020	1.0007	1.0002	1	1	1	1	1
0.65	1.0058	1.0024	1.0009	1.0003	1.0001	1	1	1	1	1	1
0.6	1.0008	1.0002	1.0001	1.0001	1	1	1	1	1	1	1
0.55	1.0001	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.0162	65.8179	23.6596	9.6430	4.5629	2.5552	1.6967	1.2915	1.1082	1.0342	1.0082
0.95	52.3014	19.3854	8.1728	4.0321	2.3619	1.6153	1.2649	1.0986	1.0306	1.0075	1.0015
0.9	12.7966	5.8930	3.1917	2.0256	1.4767	1.2051	1.0760	1.0238	1.0063	1.0013	1.0003
0.85	3.7747	2.3415	1.6585	1.3139	1.1339	1.0505	1.0160	1.0041	1.0010	1.0001	1
0.8	1.7117	1.3624	1.1704	1.0745	1.0269	1.0093	1.0023	1.0005	1.0001	1	1
0.75	1.1656	1.0778	1.0334	1.0121	1.0041	1.0010	1.0003	1.0001	1	1	1
0.7	1.0296	1.0133	1.0050	1.0016	1.0007	1.0002	1	1	1	1	1
0.65	1.0042	1.0018	1.0005	1.0003	1.0001	1	1	1	1	1	1
0.6	1.0005	1.0002	1.0001	1.0001	1	1	1	1	1	1	1
0.55	1.0001	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.5979	67.7207	25.3743	10.5393	4.9766	2.7150	1.7375	1.2977	1.1066	1.0324	1.0080
0.95	57.5808	21.8879	9.2943	4.4985	2.5181	1.6606	1.2658	1.0945	1.0288	1.0072	1.0015

0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.4965	30.1898	6.7195	2.6915	1.6205	1.2268	1.0702	1.0169	1.0030	1.0004	1
0.95	11.9845	3.8313	2.0199	1.3901	1.1335	1.0373	1.0079	1.0013	1.0001	1	1
0.9	2.2306	1.5053	1.1842	1.0571	1.0143	1.0029	1.0005	1.0001	1	1	1
0.85	1.1898	1.0616	1.0170	1.0034	1.0007	1.0001	1	1	1	1	1
0.8	1.0118	1.0027	1.0006	1.0001	1	1	1	1	1	1	1
0.75	1.0003	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.1909	36.9355	8.5749	3.0389	1.6482	1.2114	1.0596	1.0134	1.0023	1.0003	1.0001
0.95	17.8096	4.9736	2.1667	1.3807	1.1188	1.0319	1.0068	1.0010	1.0001	1	1
0.9	2.5471	1.5158	1.1639	1.0467	1.0107	1.0022	1.0004	1	1	1	1
0.85	1.1612	1.0469	1.0121	1.0028	1.0006	1.0001	1	1	1	1	1
0.8	1.0084	1.0020	1.0004	1.0001	1	1	1	1	1	1	1
0.75	1.0003	1.0001	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.6782	39.3077	9.6213	3.2629	1.6861	1.2081	1.0578	1.0132	1.0026	1.0003	1
0.95	21.0271	5.7090	2.2968	1.3936	1.1153	1.0295	1.0059	1.0009	1.0001	1	1
0.9	2.8447	1.5457	1.1620	1.0450	1.0098	1.0019	1.0002	1	1	1	1
0.85	1.1587	1.0456	1.0115	1.0023	1.0004	1.0001	1	1	1	1	1
0.8	1.0079	1.0020	1.0004	1.0001	1	1	1	1	1	1	1
0.75	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.8$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2

0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.2099	33.0988	4.8090	1.4360	1.0639	1.0089	1.0011	1.0001	1	1	1
0.95	10.9393	1.8601	1.0894	1.0116	1.0014	1.0001	1	1	1	1	1
0.9	1.0702	1.0072	1.0008	1.0001	1	1	1	1	1	1	1
0.85	1.0002	1	1	1	1	1	1	1	1	1	1
0.8	1	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.0957	36.5124	5.8398	1.4789	1.0627	1.0091	1.0009	1.0001	1	1	1
0.95	14.8833	2.0879	1.0891	1.0111	1.0012	1.0001	1	1	1	1	1
0.9	1.0688	1.0069	1.0007	1.0001	1	1	1	1	1	1	1
0.85	1.0002	1	1	1	1	1	1	1	1	1	1
0.8	1	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S22 ARL of the EWMA TBEA schemes based on the statistic Z_1 when both variables A and T following the weibull distribution assuming a Frank copula when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.9468	34.9523	10.7382	5.6017	3.7565	2.8658	2.3531	2.0279	1.7945	1.5807	1.3875
0.95	10.7757	5.6201	3.7683	2.8742	2.3513	2.0281	1.7928	1.5813	1.3879	1.2300	1.1245
0.9	3.7634	2.8684	2.3486	2.0280	1.7954	1.5808	1.3861	1.2284	1.1260	1.0649	1.0319
0.85	2.3462	2.0258	1.7942	1.5816	1.3881	1.2280	1.1240	1.0637	1.0325	1.0150	1.0073
0.8	1.7983	1.5867	1.3891	1.2259	1.1240	1.0639	1.0315	1.0154	1.0072	1.0034	1.0015
0.75	1.3853	1.2285	1.1229	1.0630	1.0312	1.0150	1.0076	1.0031	1.0016	1.0007	1.0003
0.7	1.1209	1.0628	1.0310	1.0160	1.0074	1.0033	1.0017	1.0008	1.0003	1.0001	1.0001
0.65	1.0312	1.0152	1.0074	1.0036	1.0017	1.0007	1.0003	1.0001	1.0001	1	1
0.6	1.0078	1.0035	1.0017	1.0009	1.0004	1.0001	1.0001	1	1	1	1

0.55	1.0016	1.0009	1.0003	1.0002	1.0001	1	1	1	1	1	1
0.5	1.0003	1.0002	1.0001	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.2235	50.8386	15.3393	6.6450	3.9083	2.7655	2.1672	1.8035	1.5370	1.3278	1.1866
0.95	15.3426	6.6611	3.9244	2.7614	2.1727	1.8055	1.5364	1.3313	1.1855	1.0971	1.0491
0.9	3.9011	2.7666	2.1756	1.8024	1.5341	1.3272	1.1860	1.0983	1.0488	1.0234	1.0112
0.85	2.1678	1.8079	1.5331	1.3270	1.1838	1.0955	1.0480	1.0237	1.0111	1.0055	1.0023
0.8	1.5347	1.3258	1.1809	1.0960	1.0466	1.0245	1.0112	1.0056	1.0026	1.0010	1.0005
0.75	1.1816	1.0961	1.0487	1.0235	1.0112	1.0054	1.0025	1.0011	1.0004	1.0002	1.0001
0.7	1.0488	1.0237	1.0117	1.0056	1.0025	1.0011	1.0005	1.0002	1.0001	1	1
0.65	1.0116	1.0059	1.0029	1.0012	1.0007	1.0004	1.0001	1.0001	1	1	1
0.6	1.0027	1.0013	1.0005	1.0002	1.0001	1.0001	1	1	1	1	1
0.55	1.0005	1.0002	1.0001	1.0001	1	1	1	1	1	1	1
0.5	1.0001	1.0001	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.8659	66.8705	22.7270	9.1166	4.6936	2.9731	2.1814	1.7315	1.4411	1.2514	1.1360
0.95	22.6886	9.1458	4.6893	2.9815	2.1829	1.7351	1.4402	1.2493	1.1335	1.0682	1.0325
0.9	4.6955	2.9865	2.1791	1.7319	1.4433	1.2492	1.1327	1.0666	1.0337	1.0158	1.0076
0.85	2.1873	1.7315	1.4422	1.2471	1.1315	1.0665	1.0330	1.0160	1.0075	1.0034	1.0018
0.8	1.4404	1.2467	1.1302	1.0652	1.0325	1.0159	1.0080	1.0034	1.0016	1.0008	1.0003
0.75	1.1304	1.0666	1.0323	1.0165	1.0076	1.0033	1.0015	1.0006	1.0002	1.0001	1.0000
0.7	1.0333	1.0160	1.0078	1.0037	1.0018	1.0009	1.0004	1.0001	1.0001	1.0000	1.0000
0.65	1.0077	1.0037	1.0020	1.0009	1.0003	1.0001	1.0001	1.0000	1.0000	1.0000	1.0000
0.6	1.0018	1.0008	1.0003	1.0002	1.0001	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
0.55	1.0004	1.0001	1.0001	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
0.5	1.0001	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.5768	79.2804	31.0859	13.0576	6.2139	3.5310	2.3429	1.7488	1.4104	1.2237	1.1147
0.95	31.0596	13.0981	6.2410	3.5374	2.3417	1.7425	1.4147	1.2188	1.1136	1.0554	1.0268
0.9	6.2687	3.5400	2.3489	1.7427	1.4096	1.2180	1.1108	1.0544	1.0268	1.0125	1.0058
0.85	2.3462	1.7427	1.4079	1.2167	1.1091	1.0553	1.0270	1.0129	1.0063	1.0026	1.0011
0.8	1.4088	1.2173	1.1119	1.0560	1.0264	1.0133	1.0058	1.0027	1.0010	1.0004	1.0002
0.75	1.1082	1.0561	1.0274	1.0128	1.0064	1.0027	1.0012	1.0004	1.0002	1.0001	1
0.7	1.0265	1.0125	1.0061	1.0028	1.0014	1.0006	1.0003	1.0001	1.0001	1	1
0.65	1.0063	1.0031	1.0013	1.0006	1.0003	1.0001	1.0001	1	1	1	1
0.6	1.0014	1.0005	1.0003	1.0001	1	1	1	1	1	1	1
0.55	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.9675	87.4798	38.4005	17.5872	8.4688	4.5006	2.6942	1.8529	1.4260	1.2154	1.1054
0.95	38.2425	17.4924	8.5003	4.4941	2.7034	1.8390	1.4251	1.2135	1.1023	1.0512	1.0236

0.9	8.5713	4.5199	2.7070	1.8427	1.4296	1.2113	1.1047	1.0501	1.0249	1.0114	1.0054
0.85	2.7086	1.8423	1.4221	1.2101	1.1034	1.0504	1.0247	1.0114	1.0050	1.0024	1.0011
0.8	1.4239	1.2106	1.1040	1.0501	1.0247	1.0119	1.0054	1.0023	1.0009	1.0004	1.0002
0.75	1.1004	1.0494	1.0244	1.0116	1.0054	1.0024	1.0014	1.0005	1.0002	1.0001	1
0.7	1.0239	1.0110	1.0054	1.0028	1.0014	1.0005	1.0002	1.0001	1	1	1
0.65	1.0056	1.0026	1.0011	1.0006	1.0003	1.0001	1	1	1	1	1
0.6	1.0015	1.0005	1.0003	1.0001	1	1	1	1	1	1	1
0.55	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.4346	90.3427	41.4271	19.6896	9.7156	5.1151	2.9403	1.9310	1.4436	1.2145	1.1046
0.95	41.1092	19.7126	9.8074	5.1516	2.9623	1.9349	1.4493	1.2149	1.1022	1.0483	1.0224
0.9	9.8484	5.1358	2.9663	1.9385	1.4442	1.2128	1.1017	1.0483	1.0235	1.0107	1.0050
0.85	2.9933	1.9343	1.4385	1.2146	1.1014	1.0481	1.0237	1.0109	1.0052	1.0022	1.0010
0.8	1.4461	1.2149	1.1000	1.0478	1.0228	1.0107	1.0054	1.0025	1.0012	1.0004	1.0002
0.75	1.1000	1.0483	1.0232	1.0109	1.0053	1.0028	1.0010	1.0004	1.0001	1.0001	1
0.7	1.0226	1.0118	1.0051	1.0027	1.0012	1.0006	1.0002	1.0001	1.0001	1	1
0.65	1.0053	1.0026	1.0013	1.0005	1.0002	1.0001	1	1	1	1	1
0.6	1.0012	1.0005	1.0001	1.0002	1	1	1	1	1	1	1
0.55	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.3271	21.2849	6.3286	3.5761	2.5566	2.0594	1.7616	1.4590	1.2112	1.0873	1.0336
0.95	6.3024	3.5720	2.5555	2.0607	1.7613	1.4610	1.2115	1.0855	1.0344	1.0136	1.0054
0.9	2.5567	2.0614	1.7659	1.4626	1.2118	1.0839	1.0343	1.0144	1.0055	1.0022	1.0011
0.85	1.7673	1.4614	1.2113	1.0845	1.0336	1.0137	1.0066	1.0026	1.0015	1.0004	1.0003
0.8	1.2090	1.0833	1.0353	1.0145	1.0064	1.0027	1.0010	1.0005	1.0002	1.0001	1
0.75	1.0343	1.0142	1.0065	1.0029	1.0015	1.0007	1.0002	1.0001	1	1	1
0.7	1.0065	1.0031	1.0014	1.0006	1.0002	1.0002	1.0001	1	1	1	1
0.65	1.0015	1.0006	1.0002	1.0001	1	1	1	1	1	1	1
0.6	1.0003	1.0002	1.0000	1.0001	1	1	1	1	1	1	1
0.55	1.0001	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.9705	38.3493	8.6419	3.8390	2.4775	1.8960	1.5172	1.2381	1.0954	1.0384	1.0152
0.95	8.6779	3.8382	2.4835	1.8960	1.5154	1.2367	1.0964	1.0368	1.0156	1.0061	1.0029
0.9	2.4743	1.8968	1.5182	1.2350	1.0965	1.0375	1.0157	1.0070	1.0028	1.0012	1.0006
0.85	1.5148	1.2364	1.0950	1.0379	1.0164	1.0069	1.0030	1.0012	1.0006	1.0002	1.0001
0.8	1.0931	1.0377	1.0160	1.0069	1.0035	1.0013	1.0004	1.0002	1.0001	1	1
0.75	1.0158	1.0071	1.0034	1.0014	1.0007	1.0003	1.0001	1.0001	1	1	1
0.7	1.0032	1.0015	1.0008	1.0003	1.0002	1.0001	1	1	1	1	1
0.65	1.0007	1.0003	1.0001	1	1	1	1	1	1	1	1

0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.4112	56.5300	14.3631	4.9689	2.6775	1.8608	1.4292	1.1818	1.0695	1.0279	1.0115
0.95	14.4092	4.9734	2.6771	1.8626	1.4267	1.1788	1.0708	1.0278	1.0114	1.0053	1.0019
0.9	2.6729	1.8620	1.4256	1.1766	1.0696	1.0279	1.0111	1.0049	1.0019	1.0008	1.0004
0.85	1.4219	1.1756	1.0689	1.0276	1.0115	1.0051	1.0021	1.0010	1.0004	1.0001	1
0.8	1.0708	1.0283	1.0122	1.0054	1.0024	1.0010	1.0004	1.0002	1.0001	1	1
0.75	1.0119	1.0059	1.0024	1.0011	1.0005	1.0002	1.0001	1	1	1	1
0.7	1.0026	1.0013	1.0005	1.0002	1.0001	1.0001	1	1	1	1	1
0.65	1.0005	1.0003	1.0001	1	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.1423	70.9176	22.7773	7.4830	3.2446	1.9372	1.4008	1.1552	1.0610	1.0233	1.0101
0.95	22.7614	7.5891	3.2503	1.9379	1.3995	1.1536	1.0589	1.0245	1.0100	1.0037	1.0018
0.9	3.2581	1.9367	1.3976	1.1550	1.0597	1.0245	1.0101	1.0043	1.0018	1.0008	1.0002
0.85	1.3952	1.1538	1.0592	1.0246	1.0102	1.0045	1.0017	1.0009	1.0004	1.0001	1.0001
0.8	1.0585	1.0232	1.0105	1.0046	1.0019	1.0009	1.0004	1.0001	1.0000	1.0001	1
0.75	1.0108	1.0046	1.0021	1.0011	1.0004	1.0001	1.0001	1	1	1	1
0.7	1.0022	1.0009	1.0004	1.0002	1.0001	1	1	1	1	1	1
0.65	1.0004	1.0002	1.0001	1	1	1	1	1	1	1	1
0.6	1.0001	1.0001	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.6109	81.1331	30.9757	11.4707	4.4763	2.1782	1.4127	1.1485	1.0558	1.0219	1.0086
0.95	30.8101	11.5366	4.4836	2.1721	1.4108	1.1471	1.0545	1.0209	1.0091	1.0038	1.0018
0.9	4.5072	2.1829	1.4127	1.1449	1.0533	1.0218	1.0092	1.0041	1.0017	1.0007	1.0002
0.85	1.4098	1.1443	1.0548	1.0215	1.0094	1.0042	1.0018	1.0007	1.0003	1.0001	1
0.8	1.0540	1.0221	1.0091	1.0041	1.0020	1.0008	1.0004	1.0002	1	1	1
0.75	1.0096	1.0041	1.0019	1.0008	1.0003	1.0001	1.0001	1	1	1	1
0.7	1.0020	1.0008	1.0004	1.0002	1	1	1	1	1	1	1
0.65	1.0003	1.0002	1	1	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.8924	84.5058	34.5963	13.6781	5.3725	2.3885	1.4448	1.1483	1.0540	1.0201	1.0088

0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.7313	54.5962	7.8034	1.9759	1.1540	1.0300	1.0095	1.0033	1.0016	1.0004	1.0002
0.95	7.8186	1.9764	1.1527	1.0300	1.0096	1.0037	1.0017	1.0006	1.0001	1	1
0.9	1.1524	1.0296	1.0097	1.0040	1.0015	1.0006	1.0002	1.0001	1	1	1
0.85	1.0105	1.0037	1.0016	1.0007	1.0003	1.0001	1	1	1	1	1
0.8	1.0017	1.0006	1.0003	1.0001	1	1	1	1	1	1	1
0.75	1.0004	1.0001	1.0001	1	1	1	1	1	1	1	1
0.7	1.0001	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.9054	68.4770	15.5845	2.4612	1.1487	1.0289	1.0090	1.0034	1.0013	1.0004	1.0002
0.95	15.5724	2.4698	1.1450	1.0288	1.0095	1.0034	1.0013	1.0006	1.0002	1.0001	1
0.9	1.1451	1.0276	1.0093	1.0043	1.0016	1.0005	1.0002	1.0001	1	1	1
0.85	1.0099	1.0044	1.0015	1.0007	1.0002	1.0001	1	1	1	1	1
0.8	1.0018	1.0008	1.0003	1.0001	1	1	1	1	1	1	1
0.75	1.0003	1.0001	1.0001	1	1	1	1	1	1	1	1
0.7	1.0001	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.8372	74.7518	20.2367	3.1681	1.1540	1.0284	1.0093	1.0034	1.0011	1.0005	1.0001
0.95	20.2974	3.1956	1.1524	1.0277	1.0093	1.0033	1.0014	1.0006	1.0002	1.0001	1.0001
0.9	1.1508	1.0280	1.0091	1.0038	1.0015	1.0006	1.0002	1.0001	1.0000	1	1
0.85	1.0090	1.0039	1.0016	1.0006	1.0002	1.0001	1.0001	1	1	1	1
0.8	1.0015	1.0005	1.0003	1.0001	1	1	1	1	1	1	1
0.75	1.0003	1.0001	1.0001	1	1	1	1	1	1	1	1
0.7	1.0001	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S23 ARL of the EWMA TBFA schemes based on the statistic Z_2 when both variables A and T following the weibull distribution assuming a Frank copula when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in$

{0.2,0.5,0.8}

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.4058	30.2703	10.7534	6.0770	4.1349	3.0960	2.4548	2.0309	1.7298	1.4782	1.2635
0.95	13.0216	7.0229	4.7129	3.4982	2.7683	2.2734	1.9383	1.6803	1.4436	1.2401	1.1058
0.9	5.3300	3.9282	3.0923	2.5328	2.1385	1.8651	1.6388	1.4146	1.2173	1.0909	1.0322
0.85	3.4210	2.8072	2.3591	2.0400	1.8161	1.6079	1.3841	1.1931	1.0791	1.0261	1.0085
0.8	2.5895	2.2232	1.9646	1.7787	1.5787	1.3572	1.1752	1.0684	1.0223	1.0069	1.0018
0.75	2.1182	1.9165	1.7475	1.5517	1.3307	1.1551	1.0582	1.0195	1.0059	1.0015	1.0004
0.7	1.8787	1.7223	1.5226	1.3003	1.1367	1.0501	1.0161	1.0045	1.0013	1.0003	1.0001
0.65	1.6962	1.4943	1.2793	1.1228	1.0426	1.0133	1.0041	1.0011	1.0002	1.0001	1
0.6	1.4667	1.2542	1.1057	1.0367	1.0108	1.0034	1.0009	1.0002	1.0001	1	1
0.55	1.2313	1.0928	1.0308	1.0099	1.0027	1.0008	1.0001	1	1	1	1
0.5	1.0807	1.0260	1.0082	1.0018	1.0005	1.0002	1.0001	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.5579	51.1573	15.7281	6.8218	4.0004	2.7696	2.1044	1.7021	1.4125	1.2089	1.0906
0.95	22.8262	8.8620	4.8527	3.2184	2.4099	1.9284	1.6008	1.3470	1.1697	1.0694	1.0243
0.9	5.9013	3.7737	2.7529	2.1742	1.7988	1.5221	1.2909	1.1328	1.0542	1.0187	1.0064
0.85	3.1479	2.4296	2.0002	1.7058	1.4505	1.2389	1.1055	1.0389	1.0129	1.0044	1.0014
0.8	2.2117	1.8777	1.6234	1.3847	1.1941	1.0812	1.0290	1.0098	1.0030	1.0010	1.0002
0.75	1.7892	1.5548	1.3269	1.1568	1.0602	1.0217	1.0071	1.0022	1.0006	1.0002	1.0001
0.7	1.4900	1.2744	1.1230	1.0485	1.0159	1.0050	1.0016	1.0004	1.0002	1.0001	1
0.65	1.2262	1.0965	1.0346	1.0117	1.0037	1.0012	1.0005	1.0001	1	1	1
0.6	1.0740	1.0260	1.0087	1.0026	1.0009	1.0003	1.0001	1	1	1	1
0.55	1.0196	1.0064	1.0021	1.0006	1.0001	1	1	1	1	1	1
0.5	1.0048	1.0015	1.0005	1.0001	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.0908	68.1857	23.9569	9.6961	4.8381	2.9621	2.0922	1.6129	1.3248	1.1516	1.0605
0.95	38.2916	14.3087	6.5106	3.6799	2.4929	1.8717	1.4974	1.2557	1.1127	1.0448	1.0154
0.9	9.0786	4.7411	2.9909	2.1714	1.7148	1.4043	1.2009	1.0824	1.0315	1.0108	1.0039
0.85	3.6709	2.5537	1.9583	1.5902	1.3262	1.1527	1.0610	1.0222	1.0074	1.0025	1.0008
0.8	2.2415	1.7992	1.4943	1.2642	1.1163	1.0445	1.0155	1.0048	1.0016	1.0006	1.0002
0.75	1.6786	1.4098	1.2074	1.0882	1.0326	1.0112	1.0036	1.0013	1.0004	1.0001	1
0.7	1.3353	1.1581	1.0635	1.0221	1.0076	1.0025	1.0007	1.0002	1.0001	1	1
0.65	1.1215	1.0460	1.0168	1.0053	1.0017	1.0006	1.0001	1.0001	1	1	1
0.6	1.0331	1.0109	1.0039	1.0010	1.0004	1.0001	1	1	1	1	1
0.55	1.0080	1.0028	1.0007	1.0003	1.0001	1	1	1	1	1	1
0.5	1.0020	1.0008	1.0001	1.0001	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.1023	78.5484	31.7502	13.4189	6.3533	3.4723	2.2209	1.6085	1.2947	1.1277	1.0493
0.95	51.6050	21.3073	9.4283	4.7613	2.8273	1.9338	1.4703	1.2218	1.0915	1.0349	1.0121

0.9	14.4980	6.8063	3.7224	2.3850	1.7233	1.3673	1.1656	1.0677	1.0246	1.0082	1.0028
0.85	5.0946	3.0188	2.0655	1.5690	1.2794	1.1222	1.0471	1.0168	1.0055	1.0017	1.0006
0.8	2.5416	1.8378	1.4471	1.2126	1.0871	1.0321	1.0124	1.0036	1.0014	1.0003	1.0001
0.75	1.6724	1.3482	1.1592	1.0645	1.0235	1.0074	1.0021	1.0008	1.0002	1.0001	1
0.7	1.2738	1.1182	1.0447	1.0157	1.0052	1.0018	1.0004	1.0002	1.0001	1	1
0.65	1.0861	1.0322	1.0111	1.0038	1.0012	1.0004	1.0002	1	1	1	1
0.6	1.0222	1.0077	1.0024	1.0008	1.0003	1.0001	1	1	1	1	1
0.55	1.0055	1.0018	1.0005	1.0001	1.0001	1	1	1	1	1	1
0.5	1.0011	1.0005	1.0001	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.8609	85.2179	37.6983	17.0764	8.2224	4.2724	2.4910	1.6692	1.2923	1.1187	1.0446
0.95	61.6658	27.6801	12.8158	6.3789	3.4076	2.1122	1.4958	1.2110	1.0850	1.0309	1.0110
0.9	20.4868	9.7324	4.9694	2.8290	1.8431	1.3699	1.1539	1.0577	1.0211	1.0072	1.0022
0.85	7.5145	3.9986	2.3754	1.6271	1.2744	1.1101	1.0395	1.0143	1.0049	1.0017	1.0006
0.8	3.2379	2.0434	1.4667	1.1989	1.0765	1.0285	1.0099	1.0033	1.0010	1.0003	1.0001
0.75	1.7874	1.3527	1.1436	1.0547	1.0197	1.0063	1.0026	1.0007	1.0003	1.0001	1
0.7	1.2571	1.1048	1.0375	1.0132	1.0042	1.0014	1.0004	1.0002	1.0001	1	1
0.65	1.0732	1.0258	1.0091	1.0031	1.0010	1.0002	1.0001	1	1	1	1
0.6	1.0176	1.0061	1.0020	1.0007	1.0001	1.0001	1	1	1	1	1
0.55	1.0042	1.0015	1.0005	1.0002	1	1	1	1	1	1	1
0.5	1.0010	1.0003	1.0001	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.8297	87.6592	39.6828	18.5606	9.0856	4.6722	2.6423	1.7138	1.2930	1.1171	1.0439
0.95	64.8898	30.0431	14.3761	7.1822	3.8138	2.2364	1.5274	1.2103	1.0800	1.0294	1.0104
0.9	22.9750	11.2610	5.7205	3.1481	1.9337	1.3833	1.1538	1.0578	1.0197	1.0068	1.0022
0.85	8.8290	4.6348	2.6321	1.6931	1.2772	1.1049	1.0386	1.0133	1.0044	1.0016	1.0005
0.8	3.7710	2.2110	1.5083	1.1997	1.0757	1.0275	1.0088	1.0027	1.0011	1.0003	1.0001
0.75	1.9024	1.3686	1.1416	1.0514	1.0185	1.0065	1.0022	1.0008	1.0003	1.0001	1
0.7	1.2612	1.0987	1.0357	1.0120	1.0042	1.0012	1.0005	1.0001	1	1	1
0.65	1.0678	1.0242	1.0080	1.0024	1.0009	1.0003	1.0001	1	1	1	1
0.6	1.0171	1.0052	1.0019	1.0005	1.0003	1	1	1	1	1	1
0.55	1.0038	1.0012	1.0005	1.0001	1	1	1	1	1	1	1
0.5	1.0010	1.0003	1.0001	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.9978	20.2921	7.1962	4.2146	2.9605	2.2759	1.8838	1.5795	1.2758	1.0919	1.0250
0.95	8.3315	4.7707	3.3141	2.5358	2.0581	1.7844	1.4891	1.2052	1.0616	1.0160	1.0036
0.9	3.6835	2.8104	2.2510	1.9336	1.7007	1.3978	1.1480	1.0413	1.0098	1.0023	1.0006
0.85	2.4754	2.0656	1.8565	1.6210	1.3131	1.1009	1.0262	1.0066	1.0017	1.0004	1.0001
0.8	1.9620	1.7926	1.5353	1.2354	1.0678	1.0166	1.0043	1.0008	1.0002	1.0001	1
0.75	1.7330	1.4459	1.1692	1.0460	1.0108	1.0025	1.0007	1.0002	1	1	1
0.7	1.3550	1.1201	1.0304	1.0071	1.0017	1.0006	1.0002	1	1	1	1
0.65	1.0809	1.0193	1.0046	1.0014	1.0004	1.0001	1	1	1	1	1

0.6	1.0128	1.0031	1.0008	1.0002	1.0001	1	1	1	1	1	1
0.55	1.0021	1.0006	1.0002	1	1	1	1	1	1	1	1
0.5	1.0004	1.0001	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.7256	41.9200	10.1775	4.3732	2.7067	1.9940	1.5714	1.2486	1.0793	1.0214	1.0057
0.95	13.9786	5.3503	3.1318	2.2389	1.7838	1.4242	1.1601	1.0477	1.0122	1.0032	1.0008
0.9	3.6236	2.5170	1.9713	1.6245	1.3024	1.0984	1.0274	1.0070	1.0020	1.0005	1.0001
0.85	2.1665	1.8101	1.4938	1.2008	1.0590	1.0162	1.0042	1.0012	1.0003	1.0001	1
0.8	1.6819	1.3610	1.1249	1.0349	1.0093	1.0027	1.0008	1.0001	1.0001	1	1
0.75	1.2448	1.0761	1.0204	1.0052	1.0014	1.0004	1.0001	1	1	1	1
0.7	1.0451	1.0124	1.0031	1.0009	1.0003	1	1	1	1	1	1
0.65	1.0069	1.0023	1.0007	1.0002	1.0001	1	1	1	1	1	1
0.6	1.0014	1.0004	1.0001	1	1	1	1	1	1	1	1
0.55	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.5957	62.4287	18.2623	6.1784	3.0250	1.9813	1.4718	1.1767	1.0537	1.0146	1.0037
0.95	28.0384	8.8351	3.8052	2.3304	1.6990	1.3147	1.1037	1.0293	1.0078	1.0023	1.0006
0.9	4.9616	2.7687	1.9514	1.5086	1.1999	1.0602	1.0159	1.0040	1.0014	1.0003	1.0001
0.85	2.2357	1.7221	1.3531	1.1196	1.0337	1.0095	1.0026	1.0008	1.0003	1.0001	1
0.8	1.5456	1.2261	1.0700	1.0184	1.0051	1.0016	1.0005	1.0001	1	1	1
0.75	1.1385	1.0392	1.0101	1.0033	1.0011	1.0003	1	1	1	1	1
0.7	1.0220	1.0060	1.0020	1.0006	1.0003	1	1	1	1	1	1
0.65	1.0038	1.0012	1.0003	1.0001	1	1	1	1	1	1	1
0.6	1.0008	1.0002	1	1	1	1	1	1	1	1	1
0.55	1.0001	1.0001	1	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.9748	74.6501	27.0037	9.6649	3.9250	2.1217	1.4381	1.1484	1.0427	1.0113	1.0031
0.95	42.0494	15.1738	5.7264	2.7325	1.7182	1.2718	1.0842	1.0222	1.0069	1.0017	1.0005
0.9	8.5879	3.6670	2.0898	1.4674	1.1614	1.0463	1.0122	1.0034	1.0010	1.0003	1.0001
0.85	2.6334	1.7318	1.2940	1.0917	1.0262	1.0075	1.0019	1.0005	1.0003	1.0001	1
0.8	1.4930	1.1819	1.0500	1.0144	1.0039	1.0012	1.0004	1.0001	1	1	1
0.75	1.1043	1.0282	1.0080	1.0025	1.0009	1.0003	1.0001	1	1	1	1
0.7	1.0153	1.0045	1.0014	1.0004	1.0002	1	1	1	1	1	1
0.65	1.0031	1.0009	1.0002	1.0001	1	1	1	1	1	1	1
0.6	1.0004	1.0002	1.0001	1	1	1	1	1	1	1	1
0.55	1.0001	1	1	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.2274	82.5001	34.2420	13.7806	5.5576	2.4925	1.4714	1.1382	1.0370	1.0100	1.0027

0.65	1.0001	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.3495	50.1596	7.4331	2.3471	1.5037	1.0686	1.0101	1.0023	1.0005	1.0002	1
0.95	11.7791	2.8334	1.7767	1.1528	1.0213	1.0037	1.0009	1.0003	1.0001	1	1
0.9	1.9776	1.3484	1.0416	1.0072	1.0018	1.0005	1.0002	1	1	1	1
0.85	1.0964	1.0131	1.0025	1.0009	1.0003	1	1	1	1	1	1
0.8	1.0051	1.0013	1.0005	1.0001	1	1	1	1	1	1	1
0.75	1.0007	1.0003	1.0001	1	1	1	1	1	1	1	1
0.7	1.0001	1.0001	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.0710	66.2606	16.6548	3.1156	1.4570	1.0575	1.0098	1.0018	1.0005	1.0002	1
0.95	25.8667	4.7960	1.8031	1.1254	1.0175	1.0036	1.0008	1.0003	1.0001	1	1
0.9	2.2009	1.2869	1.0355	1.0062	1.0015	1.0005	1.0001	1	1	1	1
0.85	1.0736	1.0108	1.0025	1.0008	1.0002	1.0001	1	1	1	1	1
0.8	1.0039	1.0011	1.0004	1.0002	1	1	1	1	1	1	1
0.75	1.0006	1.0002	1.0001	1	1	1	1	1	1	1	1
0.7	1.0001	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.5795	76.5371	25.8241	5.8469	1.5049	1.0522	1.0082	1.0019	1.0005	1.0001	1
0.95	38.0584	10.4871	2.1592	1.1143	1.0159	1.0031	1.0008	1.0003	1.0001	1	1
0.9	3.5819	1.2764	1.0301	1.0057	1.0013	1.0005	1.0001	1	1	1	1
0.85	1.0654	1.0100	1.0023	1.0008	1.0002	1.0001	1	1	1	1	1
0.8	1.0034	1.0011	1.0004	1.0001	1	1	1	1	1	1	1
0.75	1.0007	1.0003	1	1	1	1	1	1	1	1	1
0.7	1.0001	1.0001	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2

1	201.1229	80.4268	29.6864	7.8376	1.5957	1.0504	1.0083	1.0018	1.0004	1.0001	1
0.95	43.5082	13.8179	2.7035	1.1136	1.0151	1.0030	1.0006	1.0002	1	1	1
0.9	5.2684	1.3009	1.0297	1.0056	1.0011	1.0004	1.0001	1	1	1	1
0.85	1.0636	1.0093	1.0021	1.0007	1.0002	1.0001	1	1	1	1	1
0.8	1.0039	1.0011	1.0004	1.0002	1.0001	1	1	1	1	1	1
0.75	1.0006	1.0002	1.0001	1	1	1	1	1	1	1	1
0.7	1.0001	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S24 ARL of the EWMA TBEA schemes based on the statistic Z_3 when both variables A and T following the weibull distribution assuming a Frank copula when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.9272	29.7760	10.5461	5.9410	4.0419	3.0273	2.4044	1.9911	1.6996	1.4564	1.2454
0.95	12.9311	7.1935	4.8855	3.6463	2.8860	2.3672	2.0099	1.7534	1.5293	1.3027	1.1360
0.9	5.2358	4.0263	3.2413	2.6930	2.2863	1.9921	1.7817	1.5765	1.3490	1.1661	1.0610
0.85	3.3702	2.8704	2.4730	2.1669	1.9513	1.7873	1.5992	1.3781	1.1843	1.0689	1.0208
0.8	2.5366	2.2556	2.0518	1.9119	1.7715	1.5927	1.3730	1.1842	1.0704	1.0222	1.0058
0.75	2.0837	1.9729	1.8727	1.7407	1.5481	1.3343	1.1637	1.0632	1.0207	1.0058	1.0013
0.7	1.9175	1.8186	1.6604	1.4657	1.2668	1.1264	1.0509	1.0169	1.0052	1.0015	1.0004
0.65	1.7107	1.5285	1.3376	1.1834	1.0844	1.0349	1.0136	1.0043	1.0011	1.0003	1.0001
0.6	1.3479	1.2048	1.1054	1.0502	1.0202	1.0090	1.0029	1.0009	1.0002	1.0001	1
0.55	1.1044	1.0548	1.0270	1.0121	1.0048	1.0018	1.0006	1.0002	1	1	1
0.5	1.0270	1.0140	1.0066	1.0030	1.0012	1.0005	1.0002	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.2904	50.4140	15.3264	6.6321	3.8774	2.6838	2.0459	1.6569	1.3799	1.1804	1.0659
0.95	24.3050	9.3838	5.0999	3.3837	2.5151	2.0077	1.6728	1.4058	1.1998	1.0759	1.0232
0.9	5.7408	3.8567	2.8846	2.3022	1.9269	1.6488	1.4023	1.2026	1.0805	1.0241	1.0068
0.85	3.0203	2.4652	2.0929	1.8289	1.5969	1.3638	1.1797	1.0708	1.0227	1.0063	1.0014
0.8	2.1518	1.9172	1.7206	1.5058	1.2946	1.1432	1.0560	1.0189	1.0060	1.0014	1.0004
0.75	1.7684	1.5726	1.3733	1.2023	1.0959	1.0395	1.0144	1.0049	1.0013	1.0004	1.0001
0.7	1.3847	1.2289	1.1216	1.0572	1.0249	1.0102	1.0034	1.0011	1.0004	1.0001	1
0.65	1.1187	1.0632	1.0319	1.0141	1.0057	1.0020	1.0006	1.0003	1.0001	1	1
0.6	1.0302	1.0157	1.0072	1.0033	1.0013	1.0004	1.0001	1.0001	1	1	1
0.55	1.0072	1.0038	1.0017	1.0006	1.0003	1.0001	1	1	1	1	1
0.5	1.0017	1.0009	1.0003	1.0002	1.0001	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.7842	67.3996	23.4734	9.3846	4.6829	2.8600	2.0231	1.5707	1.2825	1.1159	1.0383
0.95	45.0184	16.3057	7.2504	3.9855	2.6394	1.9663	1.5627	1.2981	1.1273	1.0424	1.0122

0.9	9.4802	5.0190	3.2200	2.3494	1.8544	1.5234	1.2781	1.1203	1.0419	1.0123	1.0030
0.85	3.4806	2.5794	2.0612	1.7159	1.4430	1.2303	1.0991	1.0358	1.0107	1.0029	1.0007
0.8	2.1324	1.8141	1.5539	1.3317	1.1686	1.0709	1.0269	1.0085	1.0025	1.0007	1.0002
0.75	1.5915	1.3766	1.2121	1.1056	1.0457	1.0183	1.0063	1.0022	1.0004	1.0001	1
0.7	1.2110	1.1157	1.0590	1.0271	1.0118	1.0039	1.0015	1.0005	1.0001	1	1
0.65	1.0570	1.0288	1.0140	1.0061	1.0024	1.0008	1.0003	1.0001	1	1	1
0.6	1.0142	1.0068	1.0031	1.0015	1.0005	1.0002	1	1	1	1	1
0.55	1.0034	1.0016	1.0007	1.0003	1.0002	1.0001	1	1	1	1	1
0.5	1.0007	1.0003	1.0002	1.0001	1.0001	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.9058	77.7678	31.0445	13.1682	6.1593	3.3586	2.1452	1.5598	1.2512	1.0939	1.0289
0.95	64.3143	25.8357	11.2158	5.4693	3.1297	2.0811	1.5470	1.2522	1.0990	1.0310	1.0082
0.9	17.4639	8.0769	4.3107	2.6879	1.9153	1.4866	1.2292	1.0922	1.0305	1.0082	1.0020
0.85	5.1312	3.1701	2.2284	1.7095	1.3847	1.1806	1.0741	1.0250	1.0075	1.0022	1.0005
0.8	2.3453	1.8395	1.4958	1.2668	1.1234	1.0510	1.0194	1.0061	1.0018	1.0005	1.0001
0.75	1.5144	1.3014	1.1586	1.0757	1.0334	1.0127	1.0050	1.0013	1.0002	1.0001	1
0.7	1.1579	1.0821	1.0397	1.0182	1.0076	1.0028	1.0010	1.0003	1	1	1
0.65	1.0403	1.0203	1.0095	1.0041	1.0016	1.0006	1.0002	1	1	1	1
0.6	1.0095	1.0048	1.0022	1.0009	1.0004	1.0002	1	1	1	1	1
0.55	1.0022	1.0012	1.0004	1.0002	1.0001	1	1	1	1	1	1
0.5	1.0004	1.0002	1.0001	1.0001	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.9158	84.3507	36.7218	16.7131	8.0061	4.1288	2.3928	1.6110	1.2426	1.0853	1.0244
0.95	77.5833	34.3452	15.6641	7.5631	4.0003	2.3468	1.5943	1.2464	1.0883	1.0264	1.0071
0.9	27.4205	12.8221	6.3690	3.4856	2.1403	1.5183	1.2140	1.0787	1.0246	1.0066	1.0015
0.85	8.8077	4.6471	2.7369	1.8305	1.3861	1.1617	1.0623	1.0206	1.0061	1.0015	1.0004
0.8	3.0506	2.0393	1.5208	1.2419	1.1078	1.0426	1.0149	1.0049	1.0014	1.0004	1.0001
0.75	1.5446	1.2801	1.1379	1.0614	1.0266	1.0100	1.0035	1.0013	1.0003	1.0001	1
0.7	1.1354	1.0690	1.0332	1.0151	1.0061	1.0023	1.0008	1.0002	1.0001	1	1
0.65	1.0318	1.0168	1.0078	1.0033	1.0013	1.0005	1.0002	1	1	1	1
0.6	1.0077	1.0040	1.0015	1.0007	1.0002	1.0001	1	1	1	1	1
0.55	1.0018	1.0010	1.0004	1.0001	1.0001	1	1	1	1	1	1
0.5	1.0003	1.0002	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.5771	86.8003	39.0527	18.2027	8.8664	4.5698	2.5782	1.6465	1.2480	1.0826	1.0230
0.95	82.8825	38.0131	17.7830	8.7734	4.5264	2.5535	1.6511	1.2464	1.0854	1.0255	1.0063
0.9	32.0444	15.3917	7.6302	4.0147	2.3408	1.5602	1.2162	1.0764	1.0234	1.0066	1.0017
0.85	11.2565	5.7765	3.1810	1.9647	1.4116	1.1620	1.0590	1.0192	1.0058	1.0015	1.0004
0.8	3.7201	2.2741	1.5807	1.2526	1.1014	1.0402	1.0145	1.0049	1.0012	1.0003	1.0001
0.75	1.6101	1.2958	1.1364	1.0595	1.0246	1.0092	1.0030	1.0009	1.0003	1.0000	1
0.7	1.1332	1.0660	1.0316	1.0135	1.0059	1.0022	1.0007	1.0002	1.0001	1	1
0.65	1.0309	1.0156	1.0073	1.0031	1.0012	1.0004	1.0002	1	1	1	1
0.6	1.0068	1.0037	1.0014	1.0007	1.0003	1.0001	1	1	1	1	1

0.55	1.0016	1.0007	1.0003	1.0001	1.0001	1	1	1	1	1	1
0.5	1.0004	1.0001	1.0001	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.3496	21.3503	7.4281	4.3043	3.0115	2.3110	1.8941	1.6205	1.3341	1.1207	1.0321
0.95	8.7608	5.1032	3.5678	2.7360	2.1964	1.8818	1.6552	1.3664	1.1339	1.0343	1.0068
0.9	3.7896	2.9871	2.4410	2.0596	1.8614	1.6572	1.3595	1.1265	1.0308	1.0058	1.0011
0.85	2.5328	2.1549	1.9685	1.8330	1.6161	1.3071	1.1001	1.0233	1.0047	1.0008	1.0002
0.8	2.0070	1.9130	1.7674	1.4984	1.2098	1.0616	1.0148	1.0033	1.0005	1.0001	1
0.75	1.8251	1.6050	1.3109	1.1116	1.0313	1.0089	1.0021	1.0005	1.0001	1	1
0.7	1.3270	1.1325	1.0476	1.0164	1.0049	1.0016	1.0004	1.0002	1	1	1
0.65	1.0530	1.0221	1.0084	1.0027	1.0012	1.0004	1.0001	1	1	1	1
0.6	1.0105	1.0047	1.0017	1.0007	1.0002	1.0001	1	1	1	1	1
0.55	1.0025	1.0011	1.0004	1.0001	1.0001	1	1	1	1	1	1
0.5	1.0007	1.0002	1.0001	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.4120	45.0596	11.3014	4.6930	2.8232	2.0474	1.6316	1.3140	1.1031	1.0261	1.0049
0.95	17.4668	6.2424	3.5391	2.4807	1.9465	1.6118	1.2963	1.0935	1.0210	1.0042	1.0005
0.9	3.8568	2.7558	2.1606	1.8349	1.5397	1.2412	1.0690	1.0151	1.0032	1.0006	1.0001
0.85	2.2372	1.9449	1.7145	1.4070	1.1533	1.0421	1.0100	1.0021	1.0004	1.0001	1
0.8	1.7771	1.4969	1.2244	1.0746	1.0214	1.0059	1.0015	1.0003	1	1	1
0.75	1.2336	1.0919	1.0337	1.0114	1.0038	1.0013	1.0002	1.0001	1	1	1
0.7	1.0375	1.0152	1.0061	1.0022	1.0007	1.0002	1.0001	1	1	1	1
0.65	1.0076	1.0034	1.0013	1.0005	1.0001	1.0001	1	1	1	1	1
0.6	1.0017	1.0007	1.0002	1.0001	1.0001	1	1	1	1	1	1
0.55	1.0003	1.0002	1.0001	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.2656	64.7602	20.0779	7.0041	3.2986	2.0902	1.5617	1.2246	1.0651	1.0143	1.0025
0.95	39.7300	12.2815	4.8747	2.7398	1.9352	1.5037	1.2010	1.0543	1.0118	1.0019	1.0004
0.9	5.9002	3.2266	2.2404	1.7648	1.4061	1.1432	1.0383	1.0080	1.0014	1.0004	1
0.85	2.3447	1.9063	1.5693	1.2557	1.0802	1.0210	1.0052	1.0010	1.0003	1	1
0.8	1.6199	1.3168	1.1200	1.0391	1.0121	1.0031	1.0009	1.0002	1.0001	1	1
0.75	1.1249	1.0496	1.0176	1.0067	1.0019	1.0006	1.0002	1.0001	1	1	1
0.7	1.0213	1.0089	1.0037	1.0015	1.0003	1.0002	1.0001	1	1	1	1
0.65	1.0043	1.0020	1.0006	1.0003	1.0001	1	1	1	1	1	1
0.6	1.0010	1.0005	1.0002	1.0001	1	1	1	1	1	1	1
0.55	1.0002	1	1	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.8596	76.1525	28.7323	10.9360	4.5456	2.3533	1.5573	1.1940	1.0505	1.0105	1.0018

0.95	60.2897	22.4349	8.4458	3.6856	2.1058	1.4715	1.1623	1.0418	1.0080	1.0013	1.0003
0.9	13.0489	5.1878	2.6873	1.8014	1.3483	1.1089	1.0266	1.0055	1.0012	1.0003	1
0.85	2.9511	1.9979	1.5093	1.1994	1.0591	1.0151	1.0038	1.0008	1.0002	1	1
0.8	1.5545	1.2447	1.0876	1.0281	1.0080	1.0026	1.0008	1.0001	1	1	1
0.75	1.0942	1.0363	1.0137	1.0050	1.0017	1.0005	1.0001	1	1	1	1
0.7	1.0171	1.0067	1.0027	1.0012	1.0002	1.0001	1	1	1	1	1
0.65	1.0033	1.0015	1.0004	1.0002	1.0001	1	1	1	1	1	1
0.6	1.0008	1.0003	1.0002	1	1	1	1	1	1	1	1
0.55	1.0001	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.5912	83.2960	35.2411	14.9283	6.4153	2.9331	1.6399	1.1837	1.0448	1.0096	1.0013
0.95	75.8644	32.0382	13.3584	5.6750	2.6088	1.5281	1.1497	1.0357	1.0067	1.0013	1.0002
0.9	24.0892	9.8118	4.1318	2.0807	1.3467	1.0958	1.0223	1.0047	1.0009	1.0001	1
0.85	5.3293	2.5516	1.5540	1.1763	1.0499	1.0134	1.0030	1.0008	1.0002	1	1
0.8	1.6096	1.2234	1.0741	1.0244	1.0073	1.0021	1.0006	1.0002	1	1	1
0.75	1.0809	1.0321	1.0115	1.0043	1.0014	1.0004	1.0001	1	1	1	1
0.7	1.0141	1.0063	1.0024	1.0008	1.0003	1.0001	1	1	1	1	1
0.65	1.0029	1.0014	1.0005	1.0002	1.0001	1	1	1	1	1	1
0.6	1.0006	1.0003	1.0001	1	1	1	1	1	1	1	1
0.55	1.0001	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.4684	85.5066	37.4124	16.5346	7.3195	3.3165	1.7079	1.1879	1.0427	1.0087	1.0012
0.95	81.2952	35.8860	15.7076	6.8295	3.0214	1.5833	1.1491	1.0322	1.0067	1.0011	1.0001
0.9	29.0569	12.5787	5.2431	2.3555	1.3720	1.0918	1.0212	1.0043	1.0008	1.0002	1
0.85	7.4867	3.1706	1.6481	1.1814	1.0472	1.0124	1.0027	1.0007	1.0002	1	1
0.8	1.7257	1.2296	1.0715	1.0231	1.0063	1.0020	1.0005	1.0001	1	1	1
0.75	1.0763	1.0300	1.0107	1.0038	1.0013	1.0004	1.0001	1	1	1	1
0.7	1.0131	1.0056	1.0026	1.0009	1.0002	1.0001	1	1	1	1	1
0.65	1.0027	1.0013	1.0005	1.0001	1	1	1	1	1	1	1
0.6	1.0007	1.0003	1.0001	1	1	1	1	1	1	1	1
0.55	1.0001	1.0001	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.8$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.5243	12.2255	4.4557	2.7784	2.0053	1.7254	1.3097	1.0559	1.0078	1.0010	1.0001
0.95	5.0838	3.1780	2.3177	1.9188	1.6951	1.2095	1.0272	1.0031	1.0003	1	1
0.9	2.4709	1.9911	1.8887	1.5363	1.0747	1.0077	1.0007	1	1	1	1
0.85	1.9461	1.7489	1.1542	1.0147	1.0018	1.0002	1	1	1	1	1
0.8	1.1827	1.0243	1.0052	1.0011	1.0002	1	1	1	1	1	1
0.75	1.0095	1.0027	1.0008	1.0002	1	1	1	1	1	1	1
0.7	1.0017	1.0005	1.0001	1	1	1	1	1	1	1	1

0.65	1.0003	1.0001	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.5131	37.1387	6.2808	2.7935	1.9094	1.5050	1.1025	1.0130	1.0014	1.0001	1
0.95	8.7772	3.4017	2.1984	1.8194	1.3491	1.0417	1.0048	1.0004	1	1	1
0.9	2.3365	1.9393	1.6565	1.1035	1.0096	1.0010	1.0001	1	1	1	1
0.85	1.7658	1.1710	1.0176	1.0027	1.0003	1.0001	1	1	1	1	1
0.8	1.0266	1.0066	1.0018	1.0006	1.0001	1	1	1	1	1	1
0.75	1.0037	1.0010	1.0003	1	1	1	1	1	1	1	1
0.7	1.0008	1.0003	1	1	1	1	1	1	1	1	1
0.65	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.2809	59.3817	14.8444	3.7607	1.9777	1.4220	1.0712	1.0092	1.0007	1.0001	1
0.95	32.4227	6.1942	2.4848	1.7841	1.2443	1.0260	1.0030	1.0002	1	1	1
0.9	2.7303	1.9451	1.5205	1.0585	1.0060	1.0006	1.0001	1	1	1	1
0.85	1.6229	1.0886	1.0116	1.0019	1.0003	1.0001	1	1	1	1	1
0.8	1.0188	1.0050	1.0012	1.0002	1.0001	1	1	1	1	1	1
0.75	1.0030	1.0009	1.0002	1.0001	1	1	1	1	1	1	1
0.7	1.0006	1.0002	1	1	1	1	1	1	1	1	1
0.65	1.0001	1	1	1	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.9455	72.5398	24.3423	7.1361	2.3451	1.3934	1.0591	1.0072	1.0008	1	1
0.95	55.9487	17.1028	4.0958	1.8423	1.1970	1.0207	1.0019	1.0001	1	1	1
0.9	5.6984	2.1420	1.4419	1.0444	1.0048	1.0004	1	1	1	1	1
0.85	1.5372	1.0651	1.0090	1.0016	1.0004	1.0001	1	1	1	1	1
0.8	1.0159	1.0045	1.0011	1.0003	1	1	1	1	1	1	1
0.75	1.0028	1.0008	1.0003	1	1	1	1	1	1	1	1
0.7	1.0006	1.0001	1	1	1	1	1	1	1	1	1
0.65	1.0001	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2

1	200.8240	80.6506	31.4088	11.3931	3.5795	1.4286	1.0526	1.0069	1.0006	1.0001	1
0.95	73.6880	28.2211	9.1903	2.4651	1.1802	1.0171	1.0016	1.0002	1	1	1
0.9	18.0805	4.2283	1.4344	1.0381	1.0040	1.0003	1	1	1	1	1
0.85	1.5454	1.0565	1.0081	1.0015	1.0002	1	1	1	1	1	1
0.8	1.0143	1.0043	1.0010	1.0003	1	1	1	1	1	1	1
0.75	1.0022	1.0008	1.0002	1.0001	1	1	1	1	1	1	1
0.7	1.0004	1.0001	1	1	1	1	1	1	1	1	1
0.65	1.0001	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.9876	83.2309	34.3726	13.3110	4.4020	1.4994	1.0522	1.0061	1.0005	1.0001	1
0.95	79.7697	32.6262	11.8885	3.2885	1.1899	1.0169	1.0016	1.0001	1	1	1
0.9	24.7806	7.0568	1.5519	1.0340	1.0036	1.0004	1	1	1	1	1
0.85	1.7510	1.0535	1.0073	1.0013	1.0002	1	1	1	1	1	1
0.8	1.0139	1.0039	1.0010	1.0003	1.0001	1	1	1	1	1	1
0.75	1.0021	1.0007	1.0002	1	1	1	1	1	1	1	1
0.7	1.0005	1.0001	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S25 ARL of the EWMA TBEA schemes based on the statistic Z_1 when variable A following the weibull distribution and T following the gamma distribution assuming a Clayton copula when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.7691	25.7692	8.5697	4.6999	3.2485	2.5142	2.0820	1.7793	1.5398	1.3515	1.2090
0.95	8.5466	4.7072	3.2468	2.5137	2.0769	1.7756	1.5397	1.3525	1.2115	1.1152	1.0583
0.9	3.2508	2.5089	2.0783	1.7750	1.5437	1.3502	1.2098	1.1129	1.0566	1.0278	1.0122
0.85	2.0799	1.7768	1.5394	1.3512	1.2081	1.1140	1.0570	1.0272	1.0121	1.0049	1.0022
0.8	1.5398	1.3514	1.2072	1.1122	1.0564	1.0265	1.0117	1.0053	1.0018	1.0008	1.0002
0.75	1.2073	1.1118	1.0559	1.0264	1.0118	1.0048	1.0019	1.0008	1.0004	1.0001	1
0.7	1.0550	1.0257	1.0112	1.0044	1.0018	1.0007	1.0002	1.0001	1.0001	1	1
0.65	1.0111	1.0050	1.0018	1.0006	1.0003	1.0001	1.0001	1	1	1	1
0.6	1.0018	1.0007	1.0002	1.0001	1	1	1	1	1	1	1
0.55	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.1170	31.4134	9.5129	4.6813	2.9898	2.2106	1.7712	1.4818	1.2898	1.1618	1.0832
0.95	9.5274	4.6622	2.9985	2.2139	1.7711	1.4850	1.2872	1.1625	1.0837	1.0404	1.0181

0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.2341	51.9252	17.6551	7.4173	3.7861	2.3290	1.6559	1.3271	1.1621	1.0781	1.0365
0.95	17.8211	7.4550	3.7964	2.3335	1.6565	1.3304	1.1619	1.0782	1.0353	1.0153	1.0071
0.9	3.8020	2.3312	1.6646	1.3311	1.1624	1.0772	1.0341	1.0154	1.0069	1.0027	1.0011
0.85	1.6650	1.3294	1.1618	1.0784	1.0346	1.0149	1.0062	1.0026	1.0011	1.0004	1.0002
0.8	1.1608	1.0771	1.0351	1.0151	1.0061	1.0024	1.0010	1.0004	1.0002	1.0001	1
0.75	1.0349	1.0153	1.0060	1.0023	1.0009	1.0003	1.0001	1	1	1	1
0.7	1.0057	1.0023	1.0008	1.0004	1.0001	1	1	1	1	1	1
0.65	1.0011	1.0002	1.0001	1	1	1	1	1	1	1	1
0.6	1.0001	1.0001	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.2916	14.6147	4.9116	2.9215	2.1534	1.7385	1.4068	1.1820	1.0707	1.0274	1.0105
0.95	4.9134	2.9290	2.1511	1.7383	1.4089	1.1810	1.0706	1.0267	1.0096	1.0040	1.0014
0.9	2.1548	1.7361	1.4064	1.1790	1.0707	1.0257	1.0096	1.0034	1.0011	1.0003	1.0002
0.85	1.4054	1.1786	1.0694	1.0254	1.0095	1.0029	1.0010	1.0003	1.0001	1	1
0.8	1.0686	1.0256	1.0092	1.0029	1.0009	1.0004	1.0001	1	1	1	1
0.75	1.0088	1.0028	1.0008	1.0002	1.0001	1	1	1	1	1	1
0.7	1.0009	1.0003	1.0001	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.9039	19.7137	5.2542	2.7628	1.9113	1.4590	1.1981	1.0778	1.0297	1.0113	1.0040
0.95	5.2518	2.7535	1.9053	1.4598	1.1944	1.0757	1.0283	1.0101	1.0035	1.0015	1.0005
0.9	1.9080	1.4601	1.1958	1.0766	1.0276	1.0102	1.0038	1.0009	1.0004	1.0001	1
0.85	1.1959	1.0755	1.0275	1.0092	1.0031	1.0011	1.0003	1.0001	1	1	1
0.8	1.0279	1.0096	1.0031	1.0011	1.0003	1.0001	1	1	1	1	1
0.75	1.0031	1.0009	1.0002	1.0001	1	1	1	1	1	1	1
0.7	1.0003	1.0001	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.8647	26.2630	6.3193	2.8541	1.8145	1.3576	1.1404	1.0523	1.0208	1.0075	1.0029

0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.8$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.8566	4.9467	2.2172	1.4253	1.0951	1.0267	1.0087	1.0030	1.0008	1.0003	1.0001
0.95	2.2201	1.4220	1.0928	1.0274	1.0079	1.0024	1.0007	1.0002	1.0001	1	1
0.9	1.0924	1.0263	1.0087	1.0023	1.0006	1.0001	1	1	1	1	1
0.85	1.0081	1.0022	1.0007	1.0002	1.0001	1	1	1	1	1	1
0.8	1.0005	1.0001	1.0001	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.0360	5.7259	2.0298	1.2263	1.0551	1.0173	1.0057	1.0016	1.0005	1.0002	1.0001
0.95	2.0314	1.2270	1.0564	1.0175	1.0052	1.0016	1.0004	1.0001	1.0001	1	1
0.9	1.0564	1.0161	1.0045	1.0014	1.0005	1.0001	1	1	1	1	1
0.85	1.0047	1.0013	1.0004	1.0001	1	1	1	1	1	1	1
0.8	1.0003	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.6187	8.3818	1.9794	1.1824	1.0456	1.0136	1.0043	1.0015	1.0004	1.0001	1
0.95	1.9831	1.1802	1.0472	1.0144	1.0042	1.0012	1.0004	1.0001	1	1	1
0.9	1.0467	1.0139	1.0042	1.0012	1.0003	1.0001	1	1	1	1	1
0.85	1.0038	1.0010	1.0003	1	1	1	1	1	1	1	1
0.8	1.0003	1.0001	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2

1	199.8327	13.3795	2.0369	1.1637	1.0415	1.0132	1.0043	1.0011	1.0003	1.0002	1
0.95	2.0391	1.1645	1.0416	1.0128	1.0039	1.0011	1.0003	1.0001	1	1	1
0.9	1.0417	1.0121	1.0038	1.0010	1.0002	1.0001	1	1	1	1	1
0.85	1.0034	1.0011	1.0002	1.0001	1	1	1	1	1	1	1
0.8	1.0003	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.4716	19.3720	2.2672	1.1575	1.0395	1.0121	1.0038	1.0014	1.0004	1.0001	1
0.95	2.2773	1.1570	1.0403	1.0121	1.0037	1.0011	1.0003	1.0001	1	1	1
0.9	1.0392	1.0113	1.0032	1.0009	1.0002	1.0001	1	1	1	1	1
0.85	1.0035	1.0010	1.0003	1.0001	1	1	1	1	1	1	1
0.8	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.9337	22.4830	2.4890	1.1559	1.0385	1.0121	1.0037	1.0013	1.0003	1.0001	1.0001
0.95	2.4940	1.1603	1.0394	1.0119	1.0034	1.0012	1.0003	1.0002	1	1	1
0.9	1.0395	1.0116	1.0035	1.0009	1.0003	1.0001	1	1	1	1	1
0.85	1.0035	1.0008	1.0003	1	1	1	1	1	1	1	1
0.8	1.0003	1.0001	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S26 ARL of the EWMA TBFA schemes based on the statistic Z_2 when variable A following the weibull distribution and T following the gamma distribution assuming a Clayton copula when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.3300	21.3886	8.3095	4.8925	3.4246	2.6146	2.1021	1.7433	1.4693	1.2679	1.1386
0.95	9.7175	5.6042	3.8673	2.9305	2.3539	1.9577	1.6590	1.4107	1.2345	1.1195	1.0557

0.9	4.3294	3.2655	2.6068	2.1771	1.8554	1.5919	1.3659	1.2025	1.1013	1.0455	1.0205
0.85	2.8701	2.3816	2.0436	1.7740	1.5292	1.3240	1.1750	1.0836	1.0382	1.0160	1.0063
0.8	2.2241	1.9469	1.7066	1.4781	1.2848	1.1517	1.0726	1.0315	1.0122	1.0048	1.0019
0.75	1.8731	1.6507	1.4326	1.2523	1.1305	1.0601	1.0243	1.0101	1.0034	1.0014	1.0004
0.7	1.5998	1.3935	1.2222	1.1117	1.0511	1.0205	1.0080	1.0027	1.0010	1.0004	1.0001
0.65	1.3488	1.1941	1.0949	1.0418	1.0162	1.0058	1.0020	1.0006	1.0003	1.0001	1
0.6	1.1679	1.0792	1.0344	1.0137	1.0048	1.0017	1.0005	1.0002	1.0001	1	1
0.55	1.0671	1.0277	1.0102	1.0036	1.0012	1.0004	1.0001	1	1	1	1
0.5	1.0220	1.0084	1.0029	1.0009	1.0003	1.0001	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.8742	25.9357	8.1299	4.1673	2.7373	2.0277	1.6059	1.3389	1.1812	1.0874	1.0394
0.95	10.6099	5.0166	3.1626	2.3010	1.7964	1.4745	1.2654	1.1359	1.0626	1.0278	1.0123
0.9	3.6629	2.6001	2.0182	1.6360	1.3755	1.2059	1.0997	1.0454	1.0191	1.0085	1.0030
0.85	2.2471	1.8161	1.5151	1.2977	1.1578	1.0768	1.0319	1.0132	1.0055	1.0018	1.0006
0.8	1.6687	1.4163	1.2325	1.1180	1.0548	1.0227	1.0091	1.0033	1.0013	1.0003	1.0002
0.75	1.3334	1.1804	1.0888	1.0398	1.0164	1.0063	1.0023	1.0007	1.0003	1.0001	1
0.7	1.1389	1.0649	1.0276	1.0106	1.0038	1.0014	1.0004	1.0001	1.0001	1	1
0.65	1.0475	1.0183	1.0073	1.0028	1.0009	1.0002	1.0001	1	1	1	1
0.6	1.0130	1.0052	1.0015	1.0005	1.0001	1.0001	1	1	1	1	1
0.55	1.0032	1.0009	1.0003	1.0001	1	1	1	1	1	1	1
0.5	1.0006	1.0002	1.0001	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.6797	31.6272	9.1903	4.2425	2.5881	1.8449	1.4579	1.2371	1.1171	1.0544	1.0235
0.95	13.4265	5.4308	3.0901	2.1139	1.6197	1.3361	1.1723	1.0816	1.0362	1.0156	1.0064
0.9	3.7283	2.4466	1.8215	1.4591	1.2456	1.1220	1.0559	1.0238	1.0094	1.0038	1.0015
0.85	2.0583	1.6189	1.3442	1.1802	1.0872	1.0376	1.0152	1.0064	1.0022	1.0009	1.0002
0.8	1.4646	1.2565	1.1295	1.0583	1.0258	1.0100	1.0040	1.0014	1.0005	1.0002	1
0.75	1.1899	1.0924	1.0411	1.0167	1.0067	1.0024	1.0007	1.0002	1.0001	1	1
0.7	1.0632	1.0273	1.0099	1.0037	1.0013	1.0006	1.0002	1.0001	1	1	1
0.65	1.0171	1.0068	1.0022	1.0008	1.0003	1.0001	1	1	1	1	1
0.6	1.0042	1.0012	1.0004	1.0001	1	1	1	1	1	1	1
0.55	1.0010	1.0002	1.0001	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.6151	37.9787	10.8693	4.6242	2.6103	1.7966	1.4016	1.1954	1.0914	1.0419	1.0176
0.95	17.3823	6.3425	3.2765	2.0940	1.5567	1.2788	1.1349	1.0617	1.0278	1.0115	1.0049
0.9	4.1972	2.5096	1.7639	1.3959	1.1992	1.0925	1.0410	1.0164	1.0070	1.0026	1.0010
0.85	2.0364	1.5449	1.2784	1.1384	1.0624	1.0272	1.0107	1.0041	1.0016	1.0004	1.0003
0.8	1.3924	1.1980	1.0940	1.0416	1.0173	1.0059	1.0025	1.0008	1.0003	1.0001	1
0.75	1.1379	1.0629	1.0263	1.0104	1.0040	1.0014	1.0005	1.0002	1	1	1
0.7	1.0405	1.0168	1.0061	1.0021	1.0008	1.0003	1.0001	1	1	1	1
0.65	1.0098	1.0039	1.0014	1.0004	1.0002	1	1	1	1	1	1
0.6	1.0023	1.0007	1.0002	1.0001	1	1	1	1	1	1	1

0.55	1.0004	1.0001	1.0001	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.4047	42.9131	12.8634	5.2665	2.7840	1.8152	1.3861	1.1804	1.0827	1.0369	1.0158
0.95	21.8404	7.7272	3.6521	2.1707	1.5562	1.2623	1.1229	1.0558	1.0243	1.0097	1.0039
0.9	5.0036	2.7059	1.7913	1.3795	1.1784	1.0816	1.0352	1.0139	1.0052	1.0021	1.0008
0.85	2.1326	1.5432	1.2590	1.1200	1.0524	1.0225	1.0090	1.0034	1.0012	1.0004	1.0002
0.8	1.3703	1.1764	1.0795	1.0346	1.0131	1.0048	1.0021	1.0006	1.0002	1.0001	1.0001
0.75	1.1179	1.0522	1.0218	1.0082	1.0032	1.0010	1.0003	1.0001	1.0001	1	1
0.7	1.0326	1.0133	1.0047	1.0015	1.0006	1.0003	1.0001	1	1	1	1
0.65	1.0079	1.0028	1.0010	1.0003	1.0001	1.0001	1	1	1	1	1
0.6	1.0016	1.0005	1.0001	1.0001	1	1	1	1	1	1	1
0.55	1.0003	1.0001	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.6913	45.6223	13.9702	5.6045	2.8965	1.8412	1.3882	1.1807	1.0819	1.0347	1.0152
0.95	24.0955	8.4932	3.8974	2.2460	1.5656	1.2622	1.1209	1.0520	1.0222	1.0089	1.0032
0.9	5.5410	2.8640	1.8303	1.3843	1.1736	1.0779	1.0339	1.0142	1.0053	1.0021	1.0009
0.85	2.2320	1.5627	1.2539	1.1187	1.0505	1.0202	1.0079	1.0031	1.0011	1.0004	1.0002
0.8	1.3742	1.1737	1.0754	1.0318	1.0130	1.0051	1.0017	1.0006	1.0002	1.0001	1
0.75	1.1144	1.0490	1.0193	1.0078	1.0027	1.0010	1.0002	1.0001	1	1	1
0.7	1.0304	1.0122	1.0041	1.0016	1.0005	1.0002	1	1	1	1	1
0.65	1.0071	1.0026	1.0007	1.0003	1.0001	1	1	1	1	1	1
0.6	1.0012	1.0004	1.0001	1.0001	1	1	1	1	1	1	1
0.55	1.0003	1.0001	1	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.2214	12.5754	5.0972	3.1633	2.3067	1.8334	1.4321	1.1964	1.0856	1.0370	1.0148
0.95	5.7922	3.5352	2.5426	2.0326	1.6417	1.3148	1.1400	1.0612	1.0243	1.0092	1.0038
0.9	2.7929	2.2063	1.8517	1.4839	1.2249	1.1000	1.0407	1.0157	1.0056	1.0021	1.0007
0.85	2.0123	1.6937	1.3588	1.1627	1.0689	1.0278	1.0104	1.0036	1.0013	1.0003	1.0001
0.8	1.5412	1.2628	1.1161	1.0474	1.0182	1.0058	1.0021	1.0007	1.0002	1	1
0.75	1.1921	1.0787	1.0312	1.0112	1.0037	1.0011	1.0003	1.0001	1	1	1
0.7	1.0546	1.0212	1.0073	1.0021	1.0007	1.0002	1	1	1	1	1
0.65	1.0133	1.0044	1.0013	1.0003	1.0001	1	1	1	1	1	1
0.6	1.0027	1.0009	1.0002	1.0001	1	1	1	1	1	1	1
0.55	1.0004	1.0001	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.4892	15.0089	4.5158	2.5267	1.7537	1.3309	1.1384	1.0566	1.0230	1.0086	1.0034

0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.7162	34.6473	8.1646	2.7641	1.4968	1.1632	1.0620	1.0232	1.0092	1.0034	1.0011
0.95	13.2706	3.8815	1.7585	1.2424	1.0885	1.0333	1.0120	1.0042	1.0016	1.0004	1.0002
0.9	2.2039	1.3542	1.1265	1.0469	1.0169	1.0059	1.0019	1.0007	1.0003	1.0001	1
0.85	1.1768	1.0652	1.0238	1.0083	1.0027	1.0006	1.0003	1.0001	1	1	1
0.8	1.0335	1.0120	1.0040	1.0012	1.0003	1.0001	1	1	1	1	1
0.75	1.0059	1.0016	1.0005	1.0002	1	1	1	1	1	1	1
0.7	1.0006	1.0002	1.0001	1	1	1	1	1	1	1	1
0.65	1.0001	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.8$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.6398	5.2478	2.4738	1.7452	1.1714	1.0564	1.0200	1.0068	1.0024	1.0006	1.0003
0.95	2.7218	1.9734	1.2727	1.0811	1.0271	1.0096	1.0027	1.0010	1.0002	1.0001	1
0.9	1.4545	1.1181	1.0389	1.0137	1.0040	1.0013	1.0004	1.0001	1	1	1
0.85	1.0570	1.0194	1.0059	1.0019	1.0004	1.0001	1	1	1	1	1
0.8	1.0090	1.0027	1.0006	1.0002	1.0001	1	1	1	1	1	1
0.75	1.0011	1.0003	1.0001	1	1	1	1	1	1	1	1
0.7	1.0001	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.6997	4.6638	1.9382	1.1937	1.0552	1.0182	1.0066	1.0022	1.0006	1.0002	1.0001
0.95	2.1732	1.2759	1.0742	1.0233	1.0077	1.0025	1.0007	1.0003	1	1	1
0.9	1.0983	1.0319	1.0103	1.0032	1.0009	1.0003	1.0001	1	1	1	1
0.85	1.0132	1.0042	1.0009	1.0003	1.0001	1	1	1	1	1	1
0.8	1.0015	1.0004	1.0001	1	1	1	1	1	1	1	1
0.75	1.0002	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2

0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S27 ARL of the EWMA TBEA schemes based on the statistic Z_3 when variable A following the weibull distribution and T following the gamma distribution assuming a Clayton copula when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.0646	20.2247	7.8565	4.6651	3.2685	2.5046	2.0219	1.6759	1.4172	1.2329	1.1178
0.95	9.4041	5.5834	3.9139	2.9827	2.4031	2.0078	1.7046	1.4528	1.2580	1.1342	1.0643
0.9	4.1882	3.2733	2.6740	2.2613	1.9595	1.6980	1.4617	1.2714	1.1391	1.0663	1.0285
0.85	2.7729	2.3875	2.1112	1.8923	1.6697	1.4455	1.2608	1.1376	1.0635	1.0283	1.0112
0.8	2.1544	1.9821	1.8101	1.6073	1.4017	1.2315	1.1198	1.0557	1.0229	1.0092	1.0034
0.75	1.8719	1.7077	1.5110	1.3221	1.1789	1.0889	1.0393	1.0169	1.0064	1.0026	1.0008
0.7	1.5597	1.3721	1.2231	1.1193	1.0573	1.0258	1.0101	1.0040	1.0016	1.0005	1.0001
0.65	1.2292	1.1278	1.0652	1.0292	1.0125	1.0052	1.0020	1.0006	1.0002	1	1
0.6	1.0583	1.0271	1.0127	1.0054	1.0021	1.0007	1.0003	1.0002	1.0001	1	1
0.55	1.0100	1.0043	1.0018	1.0007	1.0003	1.0001	1	1	1	1	1
0.5	1.0015	1.0006	1.0002	1.0001	1.0001	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.3907	24.2113	7.3954	3.8547	2.5534	1.9008	1.5159	1.2793	1.1419	1.0653	1.0290
0.95	10.0022	4.9200	3.1547	2.3119	1.8090	1.4818	1.2695	1.1366	1.0629	1.0269	1.0110
0.9	3.4358	2.5642	2.0394	1.6781	1.4128	1.2304	1.1168	1.0529	1.0229	1.0093	1.0034
0.85	2.1290	1.7989	1.5307	1.3217	1.1767	1.0857	1.0395	1.0158	1.0063	1.0024	1.0009
0.8	1.5766	1.3726	1.2188	1.1153	1.0554	1.0241	1.0097	1.0039	1.0014	1.0003	1.0001
0.75	1.2205	1.1218	1.0618	1.0287	1.0120	1.0048	1.0018	1.0009	1.0002	1.0001	1
0.7	1.0552	1.0266	1.0117	1.0051	1.0020	1.0007	1.0003	1.0001	1	1	1
0.65	1.0093	1.0041	1.0019	1.0006	1.0003	1.0001	1.0001	1	1	1	1
0.6	1.0011	1.0006	1.0003	1.0001	1	1	1	1	1	1	1
0.55	1.0001	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.9957	29.3339	8.1745	3.8138	2.3564	1.7118	1.3704	1.1842	1.0867	1.0385	1.0152
0.95	12.5571	5.2761	3.0494	2.0986	1.6014	1.3221	1.1640	1.0777	1.0333	1.0139	1.0051
0.9	3.4402	2.3877	1.8148	1.4678	1.2537	1.1275	1.0600	1.0254	1.0104	1.0043	1.0016
0.85	1.8965	1.5580	1.3242	1.1738	1.0858	1.0387	1.0163	1.0064	1.0025	1.0009	1.0003
0.8	1.3400	1.1932	1.1010	1.0477	1.0206	1.0089	1.0031	1.0011	1.0005	1.0002	1.0001
0.75	1.0947	1.0462	1.0207	1.0092	1.0040	1.0014	1.0005	1.0001	1	1	1
0.7	1.0184	1.0079	1.0039	1.0014	1.0006	1.0002	1.0001	1	1	1	1
0.65	1.0025	1.0009	1.0003	1.0002	1.0001	1	1	1	1	1	1
0.6	1.0002	1.0001	1.0001	1	1	1	1	1	1	1	1

0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.4028	34.9878	9.5858	4.0358	2.3327	1.6418	1.3108	1.1484	1.0659	1.0281	1.0116
0.95	16.5780	6.1554	3.1825	2.0501	1.5289	1.2648	1.1246	1.0553	1.0232	1.0092	1.0034
0.9	3.7917	2.3969	1.7323	1.3859	1.1958	1.0930	1.0399	1.0176	1.0064	1.0025	1.0010
0.85	1.8166	1.4594	1.2485	1.1226	1.0582	1.0258	1.0110	1.0035	1.0015	1.0006	1.0002
0.8	1.2510	1.1304	1.0640	1.0293	1.0132	1.0051	1.0016	1.0007	1.0002	1.0001	1
0.75	1.0570	1.0277	1.0123	1.0053	1.0022	1.0008	1.0003	1.0002	1.0001	1	1
0.7	1.0098	1.0045	1.0019	1.0008	1.0003	1.0001	1	1	1	1	1
0.65	1.0012	1.0004	1.0001	1.0001	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.2782	40.5336	11.4232	4.5117	2.4105	1.6351	1.2912	1.1316	1.0587	1.0238	1.0093
0.95	21.5044	7.5777	3.5384	2.1058	1.5124	1.2465	1.1113	1.0490	1.0195	1.0076	1.0026
0.9	4.5132	2.5743	1.7500	1.3629	1.1728	1.0783	1.0328	1.0141	1.0054	1.0018	1.0008
0.85	1.8556	1.4345	1.2177	1.1040	1.0452	1.0199	1.0077	1.0030	1.0010	1.0004	1.0002
0.8	1.2191	1.1106	1.0513	1.0238	1.0090	1.0038	1.0015	1.0005	1.0001	1	1
0.75	1.0450	1.0208	1.0097	1.0040	1.0013	1.0006	1.0002	1.0001	1	1	1
0.7	1.0070	1.0029	1.0013	1.0005	1.0003	1.0001	1	1	1	1	1
0.65	1.0009	1.0004	1.0001	1.0001	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.3240	42.5762	12.3506	4.7822	2.4889	1.6453	1.2948	1.1305	1.0549	1.0230	1.0091
0.95	24.0459	8.4178	3.7949	2.1761	1.5245	1.2368	1.1063	1.0462	1.0177	1.0073	1.0027
0.9	5.0152	2.7242	1.7835	1.3643	1.1692	1.0753	1.0314	1.0120	1.0051	1.0019	1.0006
0.85	1.9171	1.4420	1.2119	1.0999	1.0439	1.0192	1.0069	1.0027	1.0009	1.0004	1.0001
0.8	1.2155	1.1053	1.0478	1.0211	1.0092	1.0034	1.0014	1.0004	1.0001	1.0001	1
0.75	1.0422	1.0190	1.0087	1.0037	1.0013	1.0005	1.0002	1.0001	1	1	1
0.7	1.0062	1.0027	1.0013	1.0004	1.0002	1	1	1	1	1	1
0.65	1.0010	1.0003	1.0001	1.0001	1.0001	1	1	1	1	1	1
0.6	1.0001	1.0001	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.4663	11.4278	4.7032	2.9461	2.1867	1.7037	1.3759	1.1816	1.0821	1.0347	1.0139

0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.0628	30.5222	5.9253	2.1226	1.3756	1.1475	1.0585	1.0230	1.0084	1.0029	1.0010
0.95	10.4505	3.2692	1.6331	1.2252	1.0859	1.0335	1.0124	1.0044	1.0014	1.0005	1.0002
0.9	1.8706	1.2758	1.1033	1.0383	1.0153	1.0051	1.0018	1.0006	1.0002	1.0001	1
0.85	1.0897	1.0320	1.0121	1.0042	1.0017	1.0004	1.0001	1	1	1	1
0.8	1.0068	1.0022	1.0008	1.0003	1.0001	1	1	1	1	1	1
0.75	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.2737	33.4713	6.6464	2.1853	1.3808	1.1429	1.0567	1.0216	1.0079	1.0027	1.0009
0.95	12.0541	3.6023	1.6574	1.2229	1.0834	1.0313	1.0117	1.0043	1.0014	1.0005	1.0002
0.9	1.9377	1.2737	1.1014	1.0363	1.0135	1.0052	1.0017	1.0005	1.0003	1.0001	1
0.85	1.0861	1.0303	1.0110	1.0038	1.0011	1.0004	1.0001	1.0001	1	1	1
0.8	1.0060	1.0021	1.0007	1.0002	1.0001	1	1	1	1	1	1
0.75	1.0003	1.0000	1.0001	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.8$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	197.5921	4.5203	2.2715	1.4785	1.1472	1.0507	1.0177	1.0062	1.0022	1.0008	1.0002
0.95	2.4605	1.8671	1.2048	1.0619	1.0208	1.0068	1.0022	1.0006	1.0001	1.0001	1
0.9	1.2525	1.0516	1.0163	1.0057	1.0018	1.0005	1.0001	1	1	1	1
0.85	1.0074	1.0020	1.0005	1.0002	1.0001	1	1	1	1	1	1
0.8	1.0001	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2

0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.4898	19.1375	1.5012	1.1085	1.0337	1.0120	1.0039	1.0010	1.0003	1.0002	1
0.95	2.2364	1.1049	1.0307	1.0097	1.0031	1.0010	1.0002	1.0001	1	1	1
0.9	1.0157	1.0044	1.0015	1.0004	1.0001	1	1	1	1	1	1
0.85	1.0004	1.0001	1	1	1	1	1	1	1	1	1
0.8	1	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S28 ARL of the EWMA TBEA schemes based on the statistic Z_1 when variable A following the weibull distribution and T following the gamma distribution assuming a Frank copula when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.1765	27.4273	9.0717	4.9287	3.3866	2.6097	2.1571	1.8534	1.6093	1.4042	1.2498
0.95	9.0394	4.9431	3.3768	2.6112	2.1575	1.8501	1.6092	1.4010	1.2473	1.1450	1.0794
0.9	3.3791	2.6119	2.1571	1.8512	1.6084	1.4036	1.2504	1.1420	1.0779	1.0409	1.0209
0.85	2.1608	1.8502	1.6069	1.4031	1.2500	1.1440	1.0774	1.0403	1.0209	1.0102	1.0048
0.8	1.6064	1.4044	1.2477	1.1415	1.0757	1.0406	1.0201	1.0103	1.0049	1.0023	1.0009
0.75	1.2471	1.1425	1.0758	1.0405	1.0210	1.0101	1.0050	1.0021	1.0010	1.0005	1.0002
0.7	1.0770	1.0399	1.0202	1.0102	1.0052	1.0025	1.0009	1.0006	1.0004	1.0001	1
0.65	1.0207	1.0098	1.0047	1.0024	1.0009	1.0005	1.0003	1.0001	1	1	1
0.6	1.0051	1.0022	1.0014	1.0005	1.0002	1.0002	1.0001	1	1	1	1
0.55	1.0011	1.0006	1.0002	1.0001	1.0001	1.0001	1	1	1	1	1
0.5	1.0002	1.0002	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.4113	35.3851	10.5864	5.0419	3.1872	2.3440	1.8688	1.5549	1.3399	1.1984	1.1092
0.95	10.5489	5.0673	3.2070	2.3484	1.8664	1.5530	1.3400	1.1964	1.1100	1.0574	1.0309
0.9	3.2049	2.3493	1.8659	1.5514	1.3420	1.1973	1.1077	1.0579	1.0297	1.0145	1.0073
0.85	1.8648	1.5574	1.3438	1.1970	1.1075	1.0560	1.0287	1.0146	1.0072	1.0035	1.0015
0.8	1.3388	1.1992	1.1086	1.0564	1.0290	1.0146	1.0071	1.0037	1.0014	1.0007	1.0003
0.75	1.1071	1.0567	1.0295	1.0145	1.0070	1.0033	1.0016	1.0006	1.0003	1.0001	1.0001
0.7	1.0294	1.0143	1.0070	1.0031	1.0017	1.0007	1.0004	1.0003	1.0001	1	1
0.65	1.0066	1.0033	1.0015	1.0006	1.0003	1.0002	1.0001	1	1	1	1
0.6	1.0017	1.0007	1.0003	1.0001	1.0001	1.0001	1	1	1	1	1

0.55	1.0005	1.0001	1.0001	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.1508	43.2659	13.1375	5.7202	3.3157	2.2817	1.7593	1.4460	1.2554	1.1423	1.0735
0.95	13.1805	5.7320	3.3130	2.2822	1.7517	1.4453	1.2539	1.1406	1.0755	1.0386	1.0189
0.9	3.3090	2.2845	1.7523	1.4435	1.2532	1.1402	1.0735	1.0381	1.0192	1.0092	1.0042
0.85	1.7575	1.4467	1.2530	1.1386	1.0725	1.0376	1.0193	1.0086	1.0040	1.0020	1.0009
0.8	1.2545	1.1415	1.0730	1.0377	1.0181	1.0095	1.0044	1.0018	1.0008	1.0004	1.0002
0.75	1.0738	1.0385	1.0189	1.0095	1.0041	1.0021	1.0009	1.0005	1.0002	1.0001	1
0.7	1.0193	1.0089	1.0047	1.0023	1.0009	1.0004	1.0001	1.0001	1	1	1
0.65	1.0043	1.0021	1.0012	1.0005	1.0002	1.0001	1	1	1	1	1
0.6	1.0009	1.0004	1.0002	1.0001	1	1	1	1	1	1	1
0.55	1.0003	1.0001	1.0001	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.7995	51.7789	16.6980	6.8701	3.6658	2.3591	1.7365	1.4069	1.2213	1.1182	1.0604
0.95	16.6552	6.9144	3.6651	2.3675	1.7441	1.4062	1.2225	1.1176	1.0614	1.0311	1.0155
0.9	3.6658	2.3620	1.7373	1.4070	1.2218	1.1160	1.0602	1.0308	1.0146	1.0069	1.0032
0.85	1.7403	1.4089	1.2222	1.1170	1.0593	1.0302	1.0155	1.0073	1.0034	1.0016	1.0007
0.8	1.2209	1.1183	1.0605	1.0316	1.0151	1.0074	1.0033	1.0016	1.0009	1.0004	1.0002
0.75	1.0598	1.0293	1.0154	1.0075	1.0037	1.0017	1.0007	1.0004	1.0002	1.0001	1
0.7	1.0152	1.0074	1.0035	1.0016	1.0007	1.0004	1.0001	1	1	1	1
0.65	1.0038	1.0016	1.0009	1.0004	1.0002	1.0001	1	1	1	1	1
0.6	1.0008	1.0004	1.0002	1.0001	1	1	1	1	1	1	1
0.55	1.0001	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.1724	58.8514	20.4355	8.5617	4.2606	2.5513	1.7795	1.4076	1.2091	1.1083	1.0560
0.95	20.4436	8.5316	4.2641	2.5508	1.7889	1.4027	1.2115	1.1078	1.0552	1.0282	1.0136
0.9	4.2672	2.5575	1.7775	1.4040	1.2117	1.1073	1.0550	1.0275	1.0130	1.0062	1.0031
0.85	1.7832	1.4022	1.2126	1.1062	1.0561	1.0268	1.0135	1.0065	1.0033	1.0012	1.0007
0.8	1.2081	1.1080	1.0541	1.0268	1.0126	1.0063	1.0029	1.0015	1.0007	1.0003	1.0001
0.75	1.0546	1.0277	1.0131	1.0066	1.0029	1.0016	1.0007	1.0004	1.0001	1.0001	1
0.7	1.0130	1.0066	1.0029	1.0014	1.0005	1.0003	1.0001	1.0001	1	1	1
0.65	1.0030	1.0013	1.0008	1.0003	1.0002	1.0001	1	1	1	1	1
0.6	1.0006	1.0003	1.0001	1.0001	1	1	1	1	1	1	1
0.55	1.0002	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.3576	61.6631	22.2440	9.3698	4.6542	2.7074	1.8264	1.4172	1.2121	1.1070	1.0529
0.95	22.3657	9.4287	4.6411	2.7005	1.8191	1.4156	1.2101	1.1064	1.0542	1.0268	1.0138

0.9	4.6551	2.6950	1.8286	1.4163	1.2081	1.1058	1.0530	1.0258	1.0128	1.0059	1.0029
0.85	1.8237	1.4157	1.2083	1.1047	1.0539	1.0266	1.0124	1.0066	1.0027	1.0012	1.0006
0.8	1.2107	1.1060	1.0538	1.0264	1.0133	1.0065	1.0030	1.0014	1.0006	1.0003	1.0001
0.75	1.0520	1.0259	1.0124	1.0059	1.0029	1.0012	1.0007	1.0003	1.0001	1.0001	1
0.7	1.0128	1.0064	1.0028	1.0014	1.0006	1.0003	1.0001	1	1	1	1
0.65	1.0028	1.0014	1.0006	1.0003	1.0002	1.0001	1	1	1	1	1
0.6	1.0007	1.0002	1.0001	1	1	1	1	1	1	1	1
0.55	1.0001	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.7601	16.2304	5.3503	3.1495	2.2953	1.8559	1.5143	1.2577	1.1171	1.0503	1.0222
0.95	5.3505	3.1521	2.2951	1.8596	1.5136	1.2570	1.1162	1.0511	1.0214	1.0090	1.0040
0.9	2.2944	1.8567	1.5141	1.2570	1.1163	1.0503	1.0222	1.0093	1.0039	1.0015	1.0006
0.85	1.5111	1.2584	1.1159	1.0504	1.0217	1.0090	1.0035	1.0014	1.0004	1.0002	1.0001
0.8	1.1174	1.0508	1.0210	1.0090	1.0037	1.0017	1.0007	1.0003	1.0001	1.0001	1
0.75	1.0218	1.0084	1.0039	1.0015	1.0007	1.0002	1.0002	1	1	1	1
0.7	1.0040	1.0015	1.0006	1.0003	1.0002	1.0001	1	1	1	1	1
0.65	1.0007	1.0003	1.0002	1	1	1	1	1	1	1	1
0.6	1.0001	1.0001	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.5387	22.7589	5.8976	3.0122	2.0576	1.5693	1.2710	1.1212	1.0514	1.0227	1.0088
0.95	5.8811	3.0077	2.0577	1.5692	1.2736	1.1191	1.0519	1.0224	1.0091	1.0037	1.0014
0.9	2.0595	1.5657	1.2728	1.1189	1.0519	1.0221	1.0093	1.0037	1.0014	1.0007	1.0002
0.85	1.2706	1.1184	1.0518	1.0221	1.0093	1.0037	1.0018	1.0007	1.0002	1	1
0.8	1.0506	1.0215	1.0095	1.0045	1.0016	1.0006	1.0004	1.0001	1	1	1
0.75	1.0091	1.0040	1.0019	1.0008	1.0003	1.0001	1	1	1	1	1
0.7	1.0014	1.0007	1.0002	1.0001	1	1	1	1	1	1	1
0.65	1.0004	1.0001	1	1	1	1	1	1	1	1	1
0.6	1.0001	1.0001	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.7660	31.8219	7.5054	3.2070	1.9928	1.4669	1.2048	1.0890	1.0373	1.0161	1.0061
0.95	7.4894	3.2063	1.9894	1.4655	1.2043	1.0877	1.0380	1.0155	1.0065	1.0026	1.0011
0.9	1.9866	1.4656	1.2062	1.0874	1.0369	1.0158	1.0065	1.0027	1.0013	1.0004	1.0001
0.85	1.2046	1.0890	1.0369	1.0156	1.0069	1.0026	1.0010	1.0005	1.0002	1.0001	1
0.8	1.0369	1.0161	1.0064	1.0028	1.0012	1.0005	1.0001	1.0001	1	1	1
0.75	1.0067	1.0029	1.0011	1.0006	1.0002	1.0001	1	1	1	1	1
0.7	1.0011	1.0005	1.0002	1.0001	1	1	1	1	1	1	1
0.65	1.0002	1.0001	1.0001	1	1	1	1	1	1	1	1

0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.6234	41.2178	10.0805	3.7155	2.0292	1.4269	1.1794	1.0760	1.0308	1.0128	1.0055
0.95	10.1367	3.7053	2.0345	1.4276	1.1808	1.0750	1.0308	1.0128	1.0056	1.0021	1.0010
0.9	2.0372	1.4316	1.1789	1.0753	1.0318	1.0138	1.0054	1.0023	1.0008	1.0004	1.0001
0.85	1.1793	1.0760	1.0310	1.0128	1.0056	1.0024	1.0011	1.0005	1.0002	1	1
0.8	1.0303	1.0139	1.0053	1.0023	1.0010	1.0003	1.0001	1	1	1	1
0.75	1.0055	1.0026	1.0009	1.0004	1.0001	1.0001	1.0001	1	1	1	1
0.7	1.0010	1.0004	1.0002	1.0001	1	1	1	1	1	1	1
0.65	1.0002	1	1	1	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.2188	49.3948	13.5029	4.6055	2.1978	1.4344	1.1700	1.0687	1.0285	1.0119	1.0045
0.95	13.5622	4.6021	2.1897	1.4342	1.1714	1.0713	1.0284	1.0119	1.0049	1.0021	1.0008
0.9	2.1893	1.4328	1.1717	1.0696	1.0291	1.0117	1.0047	1.0021	1.0009	1.0003	1.0001
0.85	1.1697	1.0668	1.0282	1.0117	1.0049	1.0022	1.0007	1.0003	1.0001	1	1
0.8	1.0295	1.0120	1.0050	1.0021	1.0011	1.0003	1.0001	1.0001	1	1	1
0.75	1.0053	1.0022	1.0007	1.0003	1.0001	1.0001	1	1	1	1	1
0.7	1.0009	1.0003	1.0001	1.0001	1	1	1	1	1	1	1
0.65	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.3936	52.4182	15.2647	5.1959	2.3111	1.4512	1.1694	1.0688	1.0291	1.0116	1.0046
0.95	15.2557	5.1747	2.3195	1.4524	1.1677	1.0686	1.0280	1.0116	1.0047	1.0017	1.0007
0.9	2.3160	1.4548	1.1696	1.0671	1.0280	1.0116	1.0044	1.0020	1.0011	1.0003	1.0002
0.85	1.1692	1.0669	1.0274	1.0115	1.0050	1.0018	1.0010	1.0004	1.0001	1	1
0.8	1.0280	1.0112	1.0045	1.0019	1.0009	1.0003	1.0002	1.0001	1	1	1
0.75	1.0047	1.0021	1.0009	1.0003	1.0002	1.0000	1.0001	1	1	1	1
0.7	1.0011	1.0004	1.0002	1.0001	1	1	1	1	1	1	1
0.65	1.0001	1.0001	1.0001	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.8$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2

0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.5511	19.0481	2.1620	1.1676	1.0483	1.0170	1.0059	1.0019	1.0007	1.0003	1.0001
0.95	2.1670	1.1701	1.0474	1.0168	1.0067	1.0022	1.0010	1.0003	1.0001	1	1
0.9	1.0476	1.0172	1.0063	1.0018	1.0008	1.0003	1.0001	1	1	1	1
0.85	1.0059	1.0026	1.0011	1.0003	1.0001	1.0001	1	1	1	1	1
0.8	1.0009	1.0004	1.0001	1.0001	1.0001	1	1	1	1	1	1
0.75	1.0001	1.0001	1.0001	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.8539	21.8026	2.3052	1.1670	1.0457	1.0159	1.0053	1.0020	1.0008	1.0001	1.0001
0.95	2.3088	1.1650	1.0471	1.0166	1.0063	1.0022	1.0008	1.0002	1.0001	1.0001	1
0.9	1.0469	1.0157	1.0060	1.0022	1.0007	1.0003	1.0001	1	1	1	1
0.85	1.0063	1.0026	1.0010	1.0004	1.0001	1	1	1	1	1	1
0.8	1.0010	1.0004	1.0002	1	1	1	1	1	1	1	1
0.75	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S29 ARL of the EWMA TBEA schemes based on the statistic Z_2 when variable A following the weibull distribution and T following the gamma distribution assuming a Frank copula when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.4893	23.1872	8.8534	5.2037	3.6182	2.7503	2.2044	1.8347	1.5457	1.3220	1.1714
0.95	10.4629	5.9578	4.0880	3.0940	2.4738	2.0595	1.7444	1.4875	1.2854	1.1480	1.0709
0.9	4.5917	3.4492	2.7494	2.2767	1.9484	1.6798	1.4414	1.2545	1.1298	1.0607	1.0266
0.85	3.0298	2.4996	2.1357	1.8648	1.6244	1.3997	1.2258	1.1145	1.0526	1.0219	1.0095
0.8	2.3234	2.0329	1.8019	1.5735	1.3662	1.1991	1.0970	1.0432	1.0188	1.0079	1.0030
0.75	1.9556	1.7440	1.5303	1.3238	1.1782	1.0853	1.0362	1.0154	1.0063	1.0022	1.0010
0.7	1.7011	1.4888	1.2924	1.1547	1.0715	1.0316	1.0126	1.0051	1.0017	1.0006	1.0002
0.65	1.4511	1.2631	1.1345	1.0618	1.0267	1.0112	1.0040	1.0015	1.0005	1.0001	1.0001
0.6	1.2351	1.1178	1.0530	1.0224	1.0089	1.0033	1.0013	1.0004	1.0001	1	1

0.55	1.1019	1.0451	1.0191	1.0068	1.0026	1.0010	1.0002	1.0001	1.0001	1	1
0.5	1.0382	1.0159	1.0055	1.0019	1.0008	1.0002	1.0001	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.8836	30.1676	9.2397	4.6327	2.9860	2.1850	1.7195	1.4151	1.2239	1.1108	1.0518
0.95	12.3524	5.6373	3.4757	2.4886	1.9358	1.5780	1.3318	1.1736	1.0853	1.0396	1.0169
0.9	4.0412	2.8298	2.1791	1.7574	1.4689	1.2622	1.1377	1.0649	1.0284	1.0120	1.0052
0.85	2.4327	1.9588	1.6286	1.3816	1.2097	1.1038	1.0484	1.0205	1.0086	1.0033	1.0015
0.8	1.8018	1.5225	1.3072	1.1655	1.0809	1.0363	1.0156	1.0059	1.0023	1.0009	1.0003
0.75	1.4392	1.2482	1.1280	1.0595	1.0267	1.0115	1.0043	1.0016	1.0006	1.0003	1.0001
0.7	1.2014	1.0991	1.0456	1.0190	1.0080	1.0029	1.0011	1.0005	1.0001	1	1
0.65	1.0743	1.0325	1.0140	1.0059	1.0022	1.0006	1.0003	1.0001	1	1	1
0.6	1.0242	1.0095	1.0036	1.0012	1.0005	1.0002	1.0001	1	1	1	1
0.55	1.0068	1.0027	1.0010	1.0004	1.0001	1.0001	1	1	1	1	1
0.5	1.0016	1.0006	1.0003	1.0001	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.2338	38.4706	11.3862	5.0086	2.9385	2.0454	1.5677	1.3046	1.1518	1.0729	1.0339
0.95	17.0685	6.6243	3.5760	2.3744	1.7796	1.4320	1.2257	1.1122	1.0512	1.0232	1.0095
0.9	4.4378	2.8010	2.0223	1.5954	1.3280	1.1687	1.0820	1.0377	1.0159	1.0066	1.0026
0.85	2.3180	1.7831	1.4566	1.2483	1.1239	1.0582	1.0256	1.0106	1.0043	1.0016	1.0007
0.8	1.6160	1.3535	1.1865	1.0912	1.0416	1.0181	1.0076	1.0026	1.0010	1.0003	1.0002
0.75	1.2709	1.1364	1.0658	1.0286	1.0127	1.0048	1.0019	1.0007	1.0002	1.0001	1.0001
0.7	1.1026	1.0474	1.0204	1.0081	1.0032	1.0011	1.0005	1.0002	1.0001	1	1
0.65	1.0333	1.0140	1.0056	1.0024	1.0009	1.0004	1.0001	1	1	1	1
0.6	1.0094	1.0038	1.0014	1.0004	1.0002	1	1	1	1	1	1
0.55	1.0023	1.0009	1.0003	1.0001	1.0001	1	1	1	1	1	1
0.5	1.0006	1.0003	1.0001	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.1918	46.4106	14.2093	5.8028	3.1319	2.0361	1.5223	1.2616	1.1269	1.0588	1.0267
0.95	23.3556	8.3550	4.0462	2.4586	1.7457	1.3795	1.1900	1.0899	1.0412	1.0182	1.0077
0.9	5.4699	3.0261	2.0231	1.5402	1.2766	1.1349	1.0624	1.0277	1.0120	1.0052	1.0020
0.85	2.4225	1.7510	1.3918	1.2003	1.0939	1.0427	1.0189	1.0081	1.0027	1.0013	1.0006
0.8	1.5570	1.2897	1.1431	1.0670	1.0286	1.0126	1.0049	1.0018	1.0007	1.0003	1.0002
0.75	1.2120	1.1027	1.0476	1.0206	1.0078	1.0031	1.0013	1.0005	1.0002	1	1
0.7	1.0720	1.0321	1.0139	1.0054	1.0021	1.0007	1.0003	1.0001	1.0001	1	1
0.65	1.0226	1.0090	1.0034	1.0013	1.0004	1.0002	1.0001	1	1	1	1
0.6	1.0062	1.0023	1.0010	1.0002	1.0001	1	1	1	1	1	1
0.55	1.0014	1.0005	1.0001	1.0001	1.0001	1	1	1	1	1	1
0.5	1.0003	1.0001	1.0001	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.2385	52.9233	17.1790	6.9259	3.4624	2.1160	1.5259	1.2439	1.1150	1.0519	1.0228
0.95	29.2724	10.6508	4.7994	2.6610	1.7807	1.3649	1.1739	1.0803	1.0364	1.0153	1.0065

0.9	6.9850	3.5034	2.1314	1.5455	1.2603	1.1221	1.0547	1.0236	1.0098	1.0041	1.0017
0.85	2.6923	1.7994	1.3840	1.1827	1.0824	1.0370	1.0156	1.0068	1.0024	1.0009	1.0004
0.8	1.5602	1.2679	1.1269	1.0583	1.0253	1.0106	1.0042	1.0016	1.0007	1.0002	1.0001
0.75	1.1907	1.0888	1.0389	1.0165	1.0070	1.0028	1.0009	1.0003	1.0002	1	1
0.7	1.0617	1.0258	1.0114	1.0043	1.0018	1.0006	1.0002	1.0001	1	1	1
0.65	1.0168	1.0069	1.0028	1.0011	1.0005	1.0001	1.0001	1	1	1	1
0.6	1.0047	1.0017	1.0008	1.0001	1.0001	1	1	1	1	1	1
0.55	1.0011	1.0003	1.0002	1	1	1	1	1	1	1	1
0.5	1.0003	1.0001	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.0294	55.3600	18.5978	7.4978	3.6838	2.1764	1.5350	1.2458	1.1138	1.0526	1.0223
0.95	32.0258	11.8361	5.2779	2.8053	1.8112	1.3756	1.1701	1.0777	1.0336	1.0151	1.0062
0.9	7.8946	3.8251	2.2318	1.5608	1.2591	1.1188	1.0527	1.0229	1.0099	1.0043	1.0015
0.85	2.9144	1.8554	1.3932	1.1780	1.0806	1.0360	1.0153	1.0063	1.0026	1.0012	1.0004
0.8	1.5896	1.2691	1.1238	1.0545	1.0239	1.0096	1.0041	1.0015	1.0007	1.0003	1.0001
0.75	1.1856	1.0867	1.0361	1.0151	1.0063	1.0028	1.0008	1.0004	1.0001	1	1
0.7	1.0581	1.0245	1.0101	1.0042	1.0015	1.0007	1.0003	1.0001	1.0001	1	1
0.65	1.0163	1.0063	1.0026	1.0009	1.0004	1.0002	1.0001	1	1	1	1
0.6	1.0038	1.0016	1.0006	1.0003	1.0002	1	1	1	1	1	1
0.55	1.0008	1.0004	1.0001	1	1	1	1	1	1	1	1
0.5	1.0003	1.0001	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.7181	14.7119	5.8246	3.5714	2.5628	2.0221	1.6453	1.3282	1.1417	1.0542	1.0214
0.95	6.6809	3.9972	2.8474	2.2229	1.8508	1.5190	1.2456	1.1011	1.0387	1.0155	1.0053
0.9	3.1510	2.4299	2.0203	1.7181	1.4055	1.1802	1.0724	1.0274	1.0098	1.0033	1.0013
0.85	2.1757	1.8873	1.6021	1.3133	1.1308	1.0517	1.0182	1.0062	1.0024	1.0008	1.0002
0.8	1.7836	1.4860	1.2332	1.0930	1.0354	1.0126	1.0044	1.0016	1.0004	1.0002	1
0.75	1.3870	1.1714	1.0675	1.0245	1.0085	1.0028	1.0009	1.0003	1.0001	1	1
0.7	1.1235	1.0473	1.0165	1.0062	1.0016	1.0005	1.0001	1	1	1	1
0.65	1.0316	1.0108	1.0037	1.0011	1.0004	1.0001	1	1	1	1	1
0.6	1.0072	1.0022	1.0006	1.0002	1.0001	1	1	1	1	1	1
0.55	1.0015	1.0004	1.0002	1	1	1	1	1	1	1	1
0.5	1.0002	1.0001	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.3398	20.1879	5.6530	2.9980	2.0513	1.5459	1.2424	1.0985	1.0374	1.0135	1.0049
0.95	7.0403	3.4701	2.3110	1.7400	1.3668	1.1553	1.0609	1.0229	1.0080	1.0030	1.0009
0.9	2.5874	1.9474	1.5250	1.2403	1.0961	1.0366	1.0132	1.0050	1.0016	1.0005	1.0002
0.85	1.7018	1.3644	1.1545	1.0588	1.0225	1.0081	1.0027	1.0009	1.0003	1.0001	1.0001
0.8	1.2403	1.0985	1.0364	1.0131	1.0044	1.0015	1.0005	1.0002	1	1	1
0.75	1.0611	1.0234	1.0076	1.0025	1.0008	1.0003	1.0001	1	1	1	1
0.7	1.0128	1.0044	1.0017	1.0005	1.0002	1.0001	1	1	1	1	1
0.65	1.0024	1.0008	1.0003	1.0001	1	1	1	1	1	1	1

0.6	1.0005	1.0002	1.0001	1	1	1	1	1	1	1	1
0.55	1.0001	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.6751	29.6488	7.0549	3.0755	1.9177	1.4083	1.1657	1.0631	1.0233	1.0086	1.0032
0.95	9.8739	3.7715	2.2194	1.5708	1.2455	1.0968	1.0367	1.0134	1.0048	1.0016	1.0006
0.9	2.5724	1.7820	1.3644	1.1492	1.0582	1.0214	1.0074	1.0032	1.0009	1.0003	1.0001
0.85	1.5107	1.2249	1.0899	1.0338	1.0118	1.0042	1.0013	1.0004	1.0002	1	1
0.8	1.1366	1.0521	1.0189	1.0065	1.0025	1.0008	1.0003	1.0001	1	1	1
0.75	1.0306	1.0118	1.0036	1.0013	1.0004	1.0001	1	1	1	1	1
0.7	1.0057	1.0019	1.0007	1.0003	1.0001	1	1	1	1	1	1
0.65	1.0011	1.0004	1.0001	1	1	1	1	1	1	1	1
0.6	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.4149	39.0428	9.4283	3.4802	1.9089	1.3552	1.1370	1.0514	1.0183	1.0072	1.0028
0.95	14.7466	4.6825	2.3058	1.5218	1.2026	1.0778	1.0288	1.0102	1.0039	1.0012	1.0005
0.9	2.8538	1.7517	1.3050	1.1180	1.0436	1.0158	1.0056	1.0019	1.0006	1.0002	1.0001
0.85	1.4434	1.1761	1.0660	1.0253	1.0093	1.0029	1.0012	1.0004	1.0001	1	1
0.8	1.1025	1.0376	1.0143	1.0051	1.0014	1.0006	1.0002	1	1	1	1
0.75	1.0213	1.0080	1.0024	1.0008	1.0003	1.0001	1.0001	1	1	1	1
0.7	1.0037	1.0012	1.0005	1.0002	1.0001	1	1	1	1	1	1
0.65	1.0008	1.0003	1.0001	1	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.9242	46.8334	12.4983	4.2090	1.9974	1.3448	1.1230	1.0456	1.0170	1.0060	1.0022
0.95	20.3400	6.2382	2.5675	1.5290	1.1886	1.0685	1.0251	1.0090	1.0034	1.0012	1.0004
0.9	3.4856	1.8169	1.2881	1.1058	1.0384	1.0140	1.0047	1.0018	1.0005	1.0002	1
0.85	1.4353	1.1589	1.0586	1.0211	1.0075	1.0024	1.0009	1.0004	1	1	1
0.8	1.0885	1.0324	1.0106	1.0038	1.0013	1.0005	1.0002	1.0001	1	1	1
0.75	1.0173	1.0065	1.0021	1.0006	1.0003	1.0001	1	1	1	1	1
0.7	1.0030	1.0012	1.0004	1.0001	1	1	1	1	1	1	1
0.65	1.0007	1.0002	1.0001	1	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.5095	50.1752	14.1602	4.6906	2.0893	1.3480	1.1222	1.0454	1.0170	1.0061	1.0023

0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.8436	15.0111	2.2135	1.2121	1.0555	1.0172	1.0051	1.0018	1.0005	1.0002	1.0001
0.95	2.7725	1.3095	1.0736	1.0230	1.0074	1.0024	1.0008	1.0002	1.0001	1	1
0.9	1.0994	1.0294	1.0102	1.0029	1.0008	1.0002	1.0001	1	1	1	1
0.85	1.0122	1.0042	1.0013	1.0003	1.0001	1	1	1	1	1	1
0.8	1.0016	1.0006	1.0002	1	1	1	1	1	1	1	1
0.75	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.0369	22.7908	2.4862	1.1958	1.0492	1.0158	1.0053	1.0018	1.0005	1.0002	1
0.95	3.6194	1.2908	1.0660	1.0211	1.0067	1.0021	1.0006	1.0002	1.0001	1	1
0.9	1.0907	1.0281	1.0095	1.0032	1.0008	1.0003	1.0001	1	1	1	1
0.85	1.0111	1.0038	1.0012	1.0003	1.0001	1.0001	1	1	1	1	1
0.8	1.0015	1.0005	1.0002	1	1	1	1	1	1	1	1
0.75	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.7	1.0001	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.5203	26.3133	2.7272	1.1944	1.0486	1.0154	1.0049	1.0013	1.0004	1.0002	1
0.95	4.3142	1.2922	1.0655	1.0201	1.0067	1.0018	1.0007	1.0003	1.0001	1	1
0.9	1.0887	1.0259	1.0087	1.0026	1.0010	1.0003	1.0001	1	1	1	1
0.85	1.0105	1.0032	1.0011	1.0004	1.0001	1	1	1	1	1	1
0.8	1.0015	1.0005	1.0002	1	1	1	1	1	1	1	1
0.75	1.0003	1.0001	1	1	1	1	1	1	1	1	1
0.7	1.0001	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S30 ARL of the EWMA TBEA schemes based on the statistic Z_3 when variable A following the weibull distribution and T following the gamma distribution assuming a Frank copula when

$\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.2497	21.9732	8.4508	4.9761	3.4670	2.6381	2.1267	1.7674	1.4888	1.2738	1.1394
0.95	10.1587	5.9631	4.1533	3.1527	2.5287	2.1073	1.7990	1.5383	1.3171	1.1631	1.0770
0.9	4.4339	3.4693	2.8235	2.3691	2.0497	1.7956	1.5526	1.3397	1.1789	1.0848	1.0370
0.85	2.9268	2.5062	2.1987	1.9720	1.7700	1.5499	1.3373	1.1819	1.0864	1.0378	1.0153
0.8	2.2450	2.0551	1.8941	1.7157	1.5106	1.3093	1.1648	1.0782	1.0345	1.0143	1.0056
0.75	1.9460	1.8066	1.6319	1.4334	1.2536	1.1315	1.0623	1.0267	1.0115	1.0047	1.0017
0.7	1.6863	1.4994	1.3197	1.1821	1.0939	1.0438	1.0190	1.0077	1.0033	1.0013	1.0005
0.65	1.3348	1.2005	1.1114	1.0559	1.0264	1.0111	1.0050	1.0020	1.0007	1.0003	1.0002
0.6	1.1062	1.0578	1.0295	1.0148	1.0068	1.0029	1.0013	1.0003	1.0002	1.0001	1
0.55	1.0280	1.0146	1.0071	1.0033	1.0016	1.0007	1.0003	1.0001	1	1	1
0.5	1.0065	1.0037	1.0019	1.0008	1.0003	1.0001	1.0001	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.0022	28.3530	8.5897	4.3409	2.8042	2.0647	1.6289	1.3469	1.1781	1.0832	1.0354
0.95	11.9453	5.6098	3.4800	2.5114	1.9622	1.5959	1.3410	1.1762	1.0822	1.0350	1.0145
0.9	3.8097	2.8022	2.2219	1.8228	1.5297	1.3019	1.1555	1.0741	1.0317	1.0137	1.0052
0.85	2.3041	1.9537	1.6672	1.4300	1.2453	1.1250	1.0577	1.0250	1.0098	1.0042	1.0013
0.8	1.7213	1.4960	1.3056	1.1725	1.0877	1.0400	1.0176	1.0068	1.0029	1.0011	1.0003
0.75	1.3191	1.1906	1.1026	1.0523	1.0243	1.0107	1.0046	1.0018	1.0007	1.0002	1.0001
0.7	1.0986	1.0520	1.0268	1.0130	1.0060	1.0022	1.0011	1.0004	1.0002	1	1
0.65	1.0252	1.0131	1.0067	1.0031	1.0012	1.0006	1.0003	1	1	1	1
0.6	1.0058	1.0029	1.0014	1.0008	1.0002	1.0002	1	1	1	1	1
0.55	1.0013	1.0006	1.0004	1.0001	1.0001	1.0001	1	1	1	1	1
0.5	1.0003	1.0002	1.0001	1.0001	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.6809	36.1103	10.3310	4.5715	2.7097	1.9013	1.4842	1.2389	1.1110	1.0499	1.0210
0.95	16.8133	6.5589	3.5942	2.3857	1.7820	1.4294	1.2207	1.1042	1.0462	1.0185	1.0075
0.9	4.1015	2.7436	2.0404	1.6207	1.3511	1.1827	1.0854	1.0383	1.0157	1.0068	1.0021
0.85	2.1482	1.7399	1.4502	1.2533	1.1303	1.0619	1.0272	1.0110	1.0044	1.0017	1.0006
0.8	1.4753	1.2875	1.1589	1.0806	1.0374	1.0165	1.0072	1.0030	1.0012	1.0004	1.0001
0.75	1.1538	1.0846	1.0435	1.0208	1.0098	1.0042	1.0018	1.0008	1.0003	1	1
0.7	1.0401	1.0212	1.0103	1.0050	1.0021	1.0007	1.0004	1.0001	1	1	1
0.65	1.0096	1.0050	1.0027	1.0011	1.0005	1.0002	1.0001	1	1	1	1
0.6	1.0022	1.0011	1.0005	1.0002	1.0001	1.0001	1	1	1	1	1
0.55	1.0004	1.0002	1.0001	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.7728	43.8766	12.8666	5.2382	2.8240	1.8664	1.4222	1.1982	1.0883	1.0375	1.0157
0.95	23.8486	8.5359	4.0883	2.4625	1.7305	1.3654	1.1777	1.0798	1.0345	1.0134	1.0056

0.9	5.0700	2.9716	2.0305	1.5537	1.2840	1.1381	1.0620	1.0267	1.0109	1.0042	1.0018
0.85	2.1573	1.6646	1.3709	1.1923	1.0947	1.0432	1.0184	1.0074	1.0028	1.0010	1.0004
0.8	1.3825	1.2148	1.1132	1.0559	1.0266	1.0115	1.0049	1.0018	1.0007	1.0004	1.0001
0.75	1.1108	1.0574	1.0293	1.0139	1.0057	1.0033	1.0010	1.0004	1.0002	1.0001	1
0.7	1.0274	1.0132	1.0077	1.0031	1.0016	1.0006	1.0002	1.0001	1	1	1
0.65	1.0060	1.0032	1.0016	1.0006	1.0003	1.0002	1	1	1	1	1
0.6	1.0015	1.0007	1.0004	1.0001	1	1	1	1	1	1	1
0.55	1.0003	1.0001	1.0001	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.6973	49.6927	15.4982	6.1645	3.0977	1.9145	1.4130	1.1856	1.0783	1.0337	1.0130
0.95	31.3926	11.1442	4.8929	2.6760	1.7569	1.3524	1.1618	1.0686	1.0290	1.0116	1.0044
0.9	6.7496	3.4684	2.1485	1.5530	1.2605	1.1211	1.0536	1.0220	1.0089	1.0037	1.0013
0.85	2.3573	1.6863	1.3466	1.1685	1.0799	1.0360	1.0151	1.0059	1.0024	1.0009	1.0003
0.8	1.3568	1.1878	1.0942	1.0457	1.0203	1.0088	1.0037	1.0013	1.0006	1.0003	1.0001
0.75	1.0884	1.0457	1.0233	1.0111	1.0046	1.0019	1.0009	1.0003	1.0001	1.0001	1
0.7	1.0218	1.0115	1.0051	1.0026	1.0011	1.0006	1.0001	1.0002	1	1	1
0.65	1.0052	1.0028	1.0010	1.0006	1.0003	1.0001	1.0001	1	1	1	1
0.6	1.0011	1.0005	1.0003	1.0002	1.0001	1	1	1	1	1	1
0.55	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.8559	52.4500	16.8832	6.7030	3.2562	1.9535	1.4138	1.1799	1.0751	1.0322	1.0134
0.95	34.9380	12.4522	5.4167	2.8319	1.7956	1.3579	1.1543	1.0673	1.0280	1.0109	1.0041
0.9	7.8298	3.8403	2.2504	1.5774	1.2583	1.1178	1.0515	1.0215	1.0083	1.0032	1.0012
0.85	2.5302	1.7381	1.3521	1.1648	1.0747	1.0329	1.0146	1.0049	1.0022	1.0007	1.0003
0.8	1.3592	1.1845	1.0893	1.0424	1.0198	1.0087	1.0037	1.0014	1.0007	1.0002	1.0001
0.75	1.0843	1.0439	1.0220	1.0111	1.0045	1.0017	1.0008	1.0003	1.0001	1.0001	1
0.7	1.0195	1.0102	1.0050	1.0024	1.0010	1.0004	1.0002	1.0001	1	1	1
0.65	1.0044	1.0023	1.0010	1.0005	1.0002	1.0001	1	1	1	1	1
0.6	1.0011	1.0005	1.0002	1.0001	1.0001	1	1	1	1	1	1
0.55	1.0002	1.0001	1.0001	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.8282	14.2221	5.6216	3.4468	2.4776	1.9586	1.5967	1.3078	1.1326	1.0532	1.0219
0.95	6.5213	4.0328	2.9060	2.2710	1.9008	1.5912	1.3005	1.1260	1.0493	1.0187	1.0072
0.9	3.0847	2.4563	2.0705	1.8219	1.5361	1.2584	1.1055	1.0401	1.0143	1.0052	1.0019
0.85	2.1291	1.9374	1.7217	1.4222	1.1879	1.0716	1.0261	1.0089	1.0030	1.0011	1.0004
0.8	1.7984	1.5337	1.2665	1.1102	1.0396	1.0141	1.0048	1.0014	1.0005	1.0002	1
0.75	1.2866	1.1258	1.0497	1.0184	1.0065	1.0021	1.0009	1.0002	1.0001	1	1
0.7	1.0484	1.0202	1.0085	1.0027	1.0007	1.0003	1.0001	1	1	1	1
0.65	1.0093	1.0039	1.0014	1.0006	1.0002	1.0001	1	1	1	1	1

0.6	1.0017	1.0009	1.0004	1.0002	1	1	1	1	1	1	1
0.55	1.0003	1.0001	1.0001	1.0001	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.9409	19.4674	5.4392	2.8767	1.9802	1.5026	1.2190	1.0856	1.0331	1.0124	1.0045
0.95	6.9488	3.5090	2.3633	1.7966	1.4121	1.1755	1.0661	1.0233	1.0088	1.0029	1.0012
0.9	2.5076	1.9700	1.5899	1.2828	1.1135	1.0403	1.0145	1.0049	1.0018	1.0006	1.0003
0.85	1.6618	1.3568	1.1528	1.0588	1.0208	1.0071	1.0024	1.0008	1.0002	1.0001	1
0.8	1.1576	1.0661	1.0250	1.0093	1.0036	1.0010	1.0004	1.0002	1	1	1
0.75	1.0261	1.0107	1.0042	1.0016	1.0005	1.0002	1	1	1	1	1
0.7	1.0051	1.0022	1.0008	1.0002	1.0001	1	1	1	1	1	1
0.65	1.0009	1.0004	1.0002	1	1	1	1	1	1	1	1
0.6	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.9457	28.3994	6.7236	2.9458	1.8464	1.3653	1.1444	1.0541	1.0193	1.0070	1.0025
0.95	10.1837	3.9013	2.2874	1.6274	1.2676	1.1009	1.0377	1.0134	1.0046	1.0017	1.0005
0.9	2.4893	1.8099	1.3942	1.1609	1.0588	1.0213	1.0075	1.0023	1.0008	1.0003	1.0001
0.85	1.4335	1.1929	1.0775	1.0273	1.0093	1.0034	1.0010	1.0004	1.0001	1	1
0.8	1.0759	1.0320	1.0120	1.0046	1.0014	1.0005	1.0001	1	1	1	1
0.75	1.0126	1.0053	1.0024	1.0009	1.0003	1.0001	1	1	1	1	1
0.7	1.0027	1.0011	1.0005	1.0002	1.0001	1	1	1	1	1	1
0.65	1.0007	1.0002	1.0001	1	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.6248	37.6632	9.0288	3.3365	1.8452	1.3203	1.1189	1.0427	1.0155	1.0060	1.0021
0.95	16.3284	5.0873	2.4373	1.5739	1.2192	1.0809	1.0288	1.0099	1.0037	1.0011	1.0004
0.9	2.7686	1.7658	1.3244	1.1213	1.0437	1.0154	1.0054	1.0019	1.0006	1.0002	1.0001
0.85	1.3505	1.1451	1.0567	1.0206	1.0072	1.0021	1.0007	1.0003	1.0001	1	1
0.8	1.0555	1.0222	1.0088	1.0031	1.0009	1.0004	1.0001	1	1	1	1
0.75	1.0098	1.0039	1.0016	1.0005	1.0002	1.0001	1	1	1	1	1
0.7	1.0021	1.0009	1.0003	1.0001	1.0001	1	1	1	1	1	1
0.65	1.0004	1.0001	1	1	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.9535	44.5082	11.8545	4.0050	1.9205	1.3047	1.1055	1.0378	1.0138	1.0047	1.0018

0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.0882	13.4119	2.5079	1.3873	1.0835	1.0236	1.0078	1.0026	1.0006	1.0002	1.0001
0.95	3.0527	1.6522	1.1233	1.0279	1.0079	1.0022	1.0006	1.0002	1.0001	1	1
0.9	1.1323	1.0291	1.0078	1.0021	1.0004	1.0001	1	1	1	1	1
0.85	1.0103	1.0028	1.0008	1.0001	1	1	1	1	1	1	1
0.8	1.0015	1.0005	1.0001	1	1	1	1	1	1	1	1
0.75	1.0004	1.0001	1	1	1	1	1	1	1	1	1
0.7	1.0001	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.2628	22.7403	2.9603	1.3374	1.0718	1.0203	1.0061	1.0020	1.0007	1.0001	1.0001
0.95	4.3702	1.5925	1.0990	1.0235	1.0068	1.0020	1.0005	1.0001	1.0001	1	1
0.9	1.1047	1.0247	1.0064	1.0015	1.0004	1.0001	1	1	1	1	1
0.85	1.0087	1.0027	1.0006	1.0001	1	1	1	1	1	1	1
0.8	1.0016	1.0006	1.0001	1	1	1	1	1	1	1	1
0.75	1.0003	1.0001	1	1	1	1	1	1	1	1	1
0.7	1.0001	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.3895	31.2411	4.1041	1.3354	1.0672	1.0196	1.0060	1.0018	1.0006	1.0002	1.0001
0.95	8.4950	1.6346	1.0892	1.0207	1.0058	1.0017	1.0006	1.0001	1	1	1
0.9	1.0936	1.0221	1.0056	1.0013	1.0003	1.0001	1	1	1	1	1
0.85	1.0077	1.0022	1.0006	1.0001	1	1	1	1	1	1	1
0.8	1.0011	1.0004	1.0001	1	1	1	1	1	1	1	1
0.75	1.0002	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2

1	201.6296	34.8779	4.9488	1.3497	1.0659	1.0185	1.0059	1.0018	1.0005	1.0002	1.0001
0.95	11.6375	1.7372	1.0895	1.0207	1.0053	1.0018	1.0005	1.0001	1	1	1
0.9	1.0905	1.0216	1.0056	1.0014	1.0003	1	1	1	1	1	1
0.85	1.0076	1.0024	1.0005	1.0002	1	1	1	1	1	1	1
0.8	1.0015	1.0004	1.0001	1	1	1	1	1	1	1	1
0.75	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S31 ARL of the EWMA TBEA schemes based on the statistic Z_1 when variable A following the normal distribution and T following the gamma distribution assuming a Clayton copula when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.6165	29.8538	9.6228	5.1677	3.5071	2.6887	2.2202	1.9037	1.6648	1.4626	1.2898
0.95	9.6604	5.1660	3.5076	2.6920	2.2107	1.9024	1.6655	1.4622	1.2892	1.1615	1.0822
0.9	3.5063	2.6907	2.2124	1.9053	1.6671	1.4638	1.2881	1.1607	1.0830	1.0396	1.0175
0.85	2.2163	1.9025	1.6644	1.4605	1.2900	1.1640	1.0831	1.0396	1.0166	1.0069	1.0026
0.8	1.6672	1.4632	1.2904	1.1644	1.0825	1.0386	1.0167	1.0065	1.0024	1.0010	1.0003
0.75	1.2893	1.1627	1.0813	1.0370	1.0164	1.0064	1.0023	1.0008	1.0004	1.0001	1
0.7	1.0842	1.0380	1.0158	1.0060	1.0022	1.0007	1.0002	1.0001	1	1	1
0.65	1.0157	1.0062	1.0022	1.0006	1.0003	1.0001	1	1	1	1	1
0.6	1.0020	1.0006	1.0002	1	1	1	1	1	1	1	1
0.55	1.0002	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.0490	39.3783	11.9111	5.5357	3.4256	2.4798	1.9619	1.6327	1.4016	1.2343	1.1238
0.95	11.8792	5.5523	3.4153	2.4784	1.9637	1.6341	1.3952	1.2330	1.1221	1.0611	1.0281
0.9	3.4184	2.4776	1.9600	1.6354	1.4008	1.2313	1.1220	1.0610	1.0263	1.0122	1.0044
0.85	1.9624	1.6344	1.3980	1.2327	1.1226	1.0598	1.0268	1.0113	1.0046	1.0016	1.0008
0.8	1.4010	1.2325	1.1229	1.0581	1.0260	1.0107	1.0043	1.0016	1.0006	1.0001	1
0.75	1.1210	1.0594	1.0264	1.0110	1.0042	1.0014	1.0005	1.0001	1.0001	1	1
0.7	1.0261	1.0104	1.0034	1.0015	1.0003	1.0001	1	1	1	1	1
0.65	1.0035	1.0012	1.0004	1.0002	1	1	1	1	1	1	1
0.6	1.0004	1.0001	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.3292	49.0395	15.4580	6.6066	3.6951	2.4979	1.8835	1.5278	1.3031	1.1635	1.0831
0.95	15.4147	6.6186	3.7083	2.4876	1.8914	1.5311	1.3048	1.1637	1.0809	1.0372	1.0164

0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.5739	18.3033	5.7046	3.2933	2.3766	1.9411	1.6344	1.3155	1.1312	1.0539	1.0215
0.95	5.6875	3.2947	2.3783	1.9428	1.6344	1.3187	1.1313	1.0545	1.0216	1.0087	1.0033
0.9	2.3763	1.9448	1.6343	1.3166	1.1301	1.0544	1.0208	1.0084	1.0028	1.0009	1.0004
0.85	1.6352	1.3151	1.1309	1.0530	1.0203	1.0077	1.0026	1.0009	1.0003	1.0001	1
0.8	1.1320	1.0533	1.0201	1.0068	1.0024	1.0009	1.0002	1.0001	1	1	1
0.75	1.0205	1.0077	1.0024	1.0008	1.0003	1.0001	1	1	1	1	1
0.7	1.0022	1.0007	1.0002	1.0001	1	1	1	1	1	1	1
0.65	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.2924	28.6301	7.0111	3.3310	2.2222	1.7252	1.3508	1.1409	1.0578	1.0234	1.0093
0.95	7.0256	3.3349	2.2278	1.7192	1.3480	1.1431	1.0580	1.0230	1.0090	1.0034	1.0012
0.9	2.2207	1.7173	1.3476	1.1415	1.0578	1.0226	1.0080	1.0029	1.0010	1.0003	1.0001
0.85	1.3450	1.1419	1.0574	1.0219	1.0084	1.0030	1.0009	1.0003	1.0001	1	1
0.8	1.0572	1.0218	1.0080	1.0027	1.0009	1.0003	1.0001	1	1	1	1
0.75	1.0078	1.0028	1.0007	1.0003	1.0001	1	1	1	1	1	1
0.7	1.0009	1.0002	1.0001	1	1	1	1	1	1	1	1
0.65	1.0001	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.9726	40.3570	9.9904	3.8776	2.2504	1.6219	1.2584	1.1024	1.0418	1.0168	1.0069
0.95	10.0272	3.9055	2.2607	1.6249	1.2567	1.1035	1.0412	1.0163	1.0062	1.0022	1.0008
0.9	2.2579	1.6219	1.2589	1.1040	1.0400	1.0160	1.0061	1.0022	1.0006	1.0003	1.0001
0.85	1.2568	1.1015	1.0407	1.0150	1.0054	1.0020	1.0007	1.0001	1	1	1
0.8	1.0397	1.0154	1.0051	1.0016	1.0005	1.0001	1.0001	1	1	1	1
0.75	1.0051	1.0016	1.0004	1.0001	1.0001	1	1	1	1	1	1
0.7	1.0004	1.0002	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.5697	50.3235	14.2650	5.1501	2.4852	1.5980	1.2247	1.0890	1.0358	1.0137	1.0058

0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.8120	12.0063	2.7449	1.7829	1.1265	1.0343	1.0113	1.0037	1.0012	1.0005	1.0001
0.95	2.7437	1.7852	1.1242	1.0338	1.0107	1.0037	1.0012	1.0004	1.0001	1.0001	1
0.9	1.1242	1.0326	1.0095	1.0028	1.0012	1.0003	1.0001	1	1	1	1
0.85	1.0103	1.0029	1.0009	1.0003	1.0001	1	1	1	1	1	1
0.8	1.0010	1.0002	1.0001	1	1	1	1	1	1	1	1
0.75	1.0001	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.5844	24.0128	3.3233	1.7007	1.0994	1.0290	1.0099	1.0030	1.0010	1.0004	1.0001
0.95	3.3121	1.7029	1.1015	1.0279	1.0092	1.0029	1.0010	1.0003	1.0001	1	1
0.9	1.1019	1.0279	1.0083	1.0026	1.0009	1.0002	1.0001	1.0001	1	1	1
0.85	1.0083	1.0027	1.0008	1.0002	1	1	1	1	1	1	1
0.8	1.0008	1.0002	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.5465	36.6001	5.6151	1.6884	1.0914	1.0268	1.0090	1.0028	1.0009	1.0004	1.0001
0.95	5.6216	1.6879	1.0905	1.0266	1.0086	1.0024	1.0008	1.0003	1.0001	1	1
0.9	1.0912	1.0251	1.0081	1.0024	1.0008	1.0002	1.0001	1	1	1	1
0.85	1.0081	1.0023	1.0008	1.0002	1	1	1	1	1	1	1
0.8	1.0006	1.0002	1.0001	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2

1	199.6418	47.2610	10.2194	1.8151	1.0881	1.0257	1.0083	1.0027	1.0008	1.0003	1.0001
0.95	10.2418	1.8210	1.0868	1.0242	1.0084	1.0027	1.0007	1.0003	1	1	1
0.9	1.0882	1.0237	1.0080	1.0024	1.0007	1.0002	1.0001	1	1	1	1
0.85	1.0073	1.0023	1.0007	1.0001	1	1	1	1	1	1	1
0.8	1.0006	1.0001	1.0001	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.1042	51.9586	12.8776	2.0775	1.0865	1.0249	1.0084	1.0027	1.0009	1.0002	1.0001
0.95	12.8988	2.0873	1.0871	1.0250	1.0075	1.0027	1.0007	1.0002	1	1	1
0.9	1.0875	1.0231	1.0077	1.0022	1.0007	1.0002	1	1	1	1	1
0.85	1.0075	1.0022	1.0007	1.0001	1	1	1	1	1	1	1
0.8	1.0005	1.0002	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S32 ARL of the EWMA TBEA schemes based on the statistic Z_2 when variable A following the normal distribution and T following the gamma distribution assuming a Clayton copula when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.3072	22.8673	8.6417	5.0608	3.5276	2.6807	2.1504	1.7791	1.4975	1.2909	1.1562
0.95	10.1821	5.7982	3.9864	3.0115	2.4081	2.0018	1.6954	1.4422	1.2550	1.1317	1.0639
0.9	4.4564	3.3532	2.6754	2.2202	1.8953	1.6251	1.3972	1.2249	1.1136	1.0522	1.0239
0.85	2.9443	2.4343	2.0841	1.8152	1.5687	1.3579	1.1973	1.0999	1.0434	1.0191	1.0081
0.8	2.2680	1.9846	1.7481	1.5178	1.3205	1.1746	1.0840	1.0375	1.0148	1.0061	1.0021
0.75	1.9128	1.6943	1.4761	1.2862	1.1521	1.0723	1.0308	1.0119	1.0043	1.0018	1.0005
0.7	1.6429	1.4312	1.2540	1.1322	1.0608	1.0242	1.0105	1.0036	1.0012	1.0005	1.0001
0.65	1.3913	1.2245	1.1146	1.0511	1.0212	1.0074	1.0026	1.0009	1.0003	1.0001	1
0.6	1.1986	1.0980	1.0420	1.0166	1.0062	1.0021	1.0006	1.0002	1.0001	1	1
0.55	1.0848	1.0346	1.0132	1.0044	1.0014	1.0004	1.0001	1.0001	1	1	1
0.5	1.0293	1.0108	1.0035	1.0011	1.0003	1.0001	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.8753	29.8820	9.0950	4.5314	2.9108	2.1171	1.6722	1.3846	1.2064	1.1041	1.0481
0.95	12.0686	5.4968	3.3871	2.4240	1.8886	1.5367	1.3045	1.1610	1.0769	1.0335	1.0144

0.9	3.9373	2.7564	2.1190	1.7133	1.4322	1.2422	1.1229	1.0573	1.0246	1.0096	1.0037
0.85	2.3725	1.9156	1.5844	1.3483	1.1900	1.0946	1.0418	1.0181	1.0066	1.0024	1.0010
0.8	1.7510	1.4846	1.2836	1.1475	1.0699	1.0299	1.0113	1.0045	1.0015	1.0004	1.0002
0.75	1.3965	1.2253	1.1139	1.0509	1.0206	1.0081	1.0028	1.0009	1.0003	1	1
0.7	1.1758	1.0862	1.0374	1.0145	1.0052	1.0017	1.0005	1.0002	1	1	1
0.65	1.0644	1.0276	1.0108	1.0037	1.0010	1.0002	1.0001	1	1	1	1
0.6	1.0183	1.0065	1.0023	1.0007	1.0002	1.0001	1	1	1	1	1
0.55	1.0045	1.0014	1.0004	1.0001	1	1	1	1	1	1	1
0.5	1.0007	1.0003	1.0001	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.8397	38.1656	11.0863	4.8643	2.8616	1.9883	1.5378	1.2827	1.1411	1.0666	1.0296
0.95	16.5641	6.4197	3.4692	2.3128	1.7295	1.3987	1.2088	1.1014	1.0457	1.0195	1.0079
0.9	4.2976	2.7114	1.9657	1.5542	1.3010	1.1561	1.0735	1.0322	1.0126	1.0047	1.0017
0.85	2.2566	1.7334	1.4226	1.2259	1.1119	1.0500	1.0213	1.0084	1.0029	1.0010	1.0004
0.8	1.5733	1.3252	1.1697	1.0807	1.0347	1.0141	1.0044	1.0016	1.0007	1.0002	1.0001
0.75	1.2465	1.1236	1.0550	1.0235	1.0084	1.0029	1.0009	1.0003	1.0001	1.0001	1
0.7	1.0877	1.0393	1.0150	1.0051	1.0019	1.0006	1.0001	1	1	1	1
0.65	1.0259	1.0100	1.0036	1.0013	1.0004	1.0001	1	1	1	1	1
0.6	1.0061	1.0019	1.0007	1.0002	1.0001	1	1	1	1	1	1
0.55	1.0011	1.0003	1.0001	1	1	1	1	1	1	1	1
0.5	1.0002	1.0001	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.8142	45.7501	13.6978	5.6196	3.0138	1.9657	1.4878	1.2453	1.1153	1.0520	1.0222
0.95	22.4896	7.9986	3.8680	2.3583	1.6908	1.3516	1.1728	1.0791	1.0349	1.0151	1.0056
0.9	5.1664	2.8952	1.9621	1.4975	1.2531	1.1212	1.0558	1.0228	1.0090	1.0035	1.0014
0.85	2.3193	1.6906	1.3645	1.1831	1.0846	1.0363	1.0146	1.0058	1.0020	1.0008	1.0002
0.8	1.5076	1.2636	1.1267	1.0580	1.0246	1.0097	1.0032	1.0012	1.0005	1.0002	1.0001
0.75	1.1916	1.0896	1.0392	1.0155	1.0060	1.0018	1.0005	1.0002	1	1	1
0.7	1.0608	1.0258	1.0099	1.0032	1.0010	1.0003	1.0001	1	1	1	1
0.65	1.0164	1.0059	1.0018	1.0006	1.0002	1	1	1	1	1	1
0.6	1.0036	1.0011	1.0003	1.0001	1	1	1	1	1	1	1
0.55	1.0006	1.0002	1.0001	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.1087	52.2205	16.6920	6.6251	3.2986	2.0287	1.4819	1.2280	1.1061	1.0469	1.0202
0.95	28.3749	10.0939	4.5198	2.5233	1.7198	1.3377	1.1572	1.0705	1.0305	1.0121	1.0051
0.9	6.5681	3.2998	2.0450	1.4895	1.2348	1.1079	1.0470	1.0195	1.0073	1.0029	1.0012
0.85	2.5400	1.7221	1.3503	1.1649	1.0743	1.0312	1.0123	1.0045	1.0017	1.0006	1.0002
0.8	1.5064	1.2443	1.1126	1.0485	1.0195	1.0074	1.0029	1.0008	1.0003	1.0000	1.0001
0.75	1.1686	1.0770	1.0312	1.0116	1.0042	1.0015	1.0003	1.0002	1	1	1
0.7	1.0502	1.0200	1.0072	1.0026	1.0007	1.0002	1	1	1	1	1
0.65	1.0121	1.0043	1.0012	1.0004	1.0001	1	1	1	1	1	1
0.6	1.0023	1.0007	1.0002	1.0001	1	1	1	1	1	1	1

0.55	1.0005	1.0001	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.7389	54.8415	18.1266	7.1900	3.5146	2.0821	1.4932	1.2284	1.1043	1.0453	1.0194
0.95	31.1533	11.3307	4.9605	2.6632	1.7430	1.3398	1.1557	1.0704	1.0293	1.0118	1.0046
0.9	7.4180	3.5793	2.1218	1.5118	1.2350	1.1045	1.0447	1.0183	1.0072	1.0028	1.0010
0.85	2.7180	1.7628	1.3571	1.1618	1.0703	1.0293	1.0112	1.0045	1.0015	1.0005	1.0003
0.8	1.5322	1.2440	1.1099	1.0468	1.0184	1.0066	1.0023	1.0009	1.0004	1.0001	1
0.75	1.1662	1.0733	1.0297	1.0115	1.0044	1.0013	1.0005	1.0001	1	1	1
0.7	1.0473	1.0185	1.0069	1.0023	1.0008	1.0002	1.0001	1	1	1	1
0.65	1.0108	1.0037	1.0013	1.0005	1.0001	1	1	1	1	1	1
0.6	1.0022	1.0007	1.0001	1.0001	1	1	1	1	1	1	1
0.55	1.0003	1.0001	1	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.8429	13.7496	5.3884	3.3064	2.3938	1.9043	1.4939	1.2366	1.1062	1.0462	1.0187
0.95	6.1378	3.6927	2.6414	2.1085	1.7158	1.3724	1.1749	1.0766	1.0320	1.0124	1.0046
0.9	2.9209	2.2777	1.9226	1.5567	1.2780	1.1260	1.0544	1.0221	1.0085	1.0030	1.0011
0.85	2.0746	1.7704	1.4302	1.2073	1.0915	1.0378	1.0145	1.0048	1.0018	1.0006	1.0002
0.8	1.6303	1.3301	1.1526	1.0637	1.0259	1.0095	1.0031	1.0009	1.0003	1.0001	1
0.75	1.2462	1.1095	1.0439	1.0170	1.0060	1.0020	1.0007	1.0002	1	1	1
0.7	1.0804	1.0312	1.0106	1.0036	1.0012	1.0005	1.0001	1	1	1	1
0.65	1.0206	1.0073	1.0022	1.0007	1.0002	1.0001	1	1	1	1	1
0.6	1.0045	1.0015	1.0003	1.0002	1	1	1	1	1	1	1
0.55	1.0008	1.0002	1	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.9840	19.3279	5.2216	2.7700	1.9084	1.4240	1.1855	1.0813	1.0337	1.0136	1.0052
0.95	6.4339	3.1879	2.1604	1.6014	1.2748	1.1210	1.0503	1.0192	1.0078	1.0028	1.0009
0.9	2.4031	1.8085	1.3985	1.1795	1.0769	1.0308	1.0118	1.0040	1.0015	1.0005	1.0002
0.85	1.5633	1.2636	1.1174	1.0474	1.0179	1.0062	1.0023	1.0007	1.0002	1.0001	1
0.8	1.1700	1.0725	1.0275	1.0099	1.0037	1.0010	1.0004	1.0001	1	1	1
0.75	1.0461	1.0169	1.0057	1.0018	1.0007	1.0002	1	1	1	1	1
0.7	1.0092	1.0029	1.0010	1.0003	1.0001	1	1	1	1	1	1
0.65	1.0017	1.0004	1.0001	1	1	1	1	1	1	1	1
0.6	1.0002	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.2347	28.5074	6.5298	2.7878	1.7423	1.3090	1.1317	1.0542	1.0220	1.0085	1.0029

0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.8$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.1567	5.7254	2.6141	1.9137	1.2229	1.0653	1.0230	1.0074	1.0025	1.0008	1.0003
0.95	2.9258	2.0449	1.3785	1.0997	1.0322	1.0107	1.0038	1.0011	1.0004	1.0002	1.0001
0.9	1.6512	1.1539	1.0480	1.0153	1.0051	1.0014	1.0004	1.0001	1	1	1
0.85	1.0716	1.0231	1.0075	1.0022	1.0005	1.0002	1	1	1	1	1
0.8	1.0111	1.0034	1.0009	1.0003	1	1	1	1	1	1	1
0.75	1.0014	1.0004	1.0001	1	1	1	1	1	1	1	1
0.7	1.0001	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.2384	6.9447	2.2776	1.3410	1.0866	1.0272	1.0088	1.0031	1.0010	1.0003	1.0001
0.95	2.5213	1.5624	1.1190	1.0390	1.0126	1.0035	1.0013	1.0004	1.0001	1	1
0.9	1.1784	1.0506	1.0167	1.0054	1.0016	1.0004	1.0002	1	1	1	1
0.85	1.0224	1.0071	1.0023	1.0005	1.0002	1	1	1	1	1	1
0.8	1.0030	1.0010	1.0001	1.0001	1	1	1	1	1	1	1
0.75	1.0004	1.0002	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.6340	14.0623	2.3124	1.2493	1.0659	1.0217	1.0075	1.0025	1.0008	1.0002	1.0001
0.95	2.6940	1.3877	1.0901	1.0272	1.0092	1.0033	1.0008	1.0003	1.0001	1	1
0.9	1.1256	1.0367	1.0118	1.0037	1.0014	1.0003	1.0001	1	1	1	1
0.85	1.0161	1.0045	1.0017	1.0005	1.0001	1.0001	1	1	1	1	1
0.8	1.0019	1.0006	1.0002	1.0001	1	1	1	1	1	1	1
0.75	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2

1	198.6582	25.2508	2.5833	1.2201	1.0578	1.0183	1.0065	1.0019	1.0008	1.0002	1
0.95	3.5696	1.3343	1.0785	1.0243	1.0079	1.0028	1.0010	1.0002	1.0001	1	1
0.9	1.1079	1.0316	1.0107	1.0033	1.0010	1.0003	1.0001	1	1	1	1
0.85	1.0138	1.0043	1.0011	1.0003	1.0001	1	1	1	1	1	1
0.8	1.0016	1.0004	1.0001	1	1	1	1	1	1	1	1
0.75	1.0001	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.8997	36.4464	3.8059	1.2098	1.0556	1.0176	1.0056	1.0020	1.0006	1.0002	1.0001
0.95	7.2009	1.3290	1.0732	1.0217	1.0068	1.0024	1.0009	1.0002	1.0001	1	1
0.9	1.1020	1.0306	1.0094	1.0030	1.0008	1.0002	1.0001	1	1	1	1
0.85	1.0123	1.0037	1.0011	1.0003	1.0001	1	1	1	1	1	1
0.8	1.0015	1.0004	1.0001	1	1	1	1	1	1	1	1
0.75	1.0001	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.7894	41.7594	5.1901	1.2085	1.0533	1.0164	1.0055	1.0020	1.0004	1.0001	1
0.95	10.1663	1.3384	1.0721	1.0225	1.0073	1.0023	1.0007	1.0002	1.0001	1	1
0.9	1.0985	1.0281	1.0090	1.0027	1.0009	1.0003	1.0001	1	1	1	1
0.85	1.0121	1.0038	1.0010	1.0003	1.0001	1	1	1	1	1	1
0.8	1.0013	1.0005	1.0001	1	1	1	1	1	1	1	1
0.75	1.0001	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S33 ARL of the EWMA TBFA schemes based on the statistic Z_3 when variable A following the normal distribution and T following the gamma distribution assuming a Clayton copula when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.5612	21.6423	8.2591	4.8513	3.3895	2.5903	2.0893	1.7303	1.4593	1.2604	1.1333
0.95	9.9707	5.8479	4.0744	3.0869	2.4809	2.0712	1.7582	1.4960	1.2973	1.1579	1.0741

0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.1534	49.0563	14.4036	5.4753	2.7832	1.7969	1.3690	1.1674	1.0757	1.0319	1.0127
0.95	26.9627	9.7073	4.3511	2.4463	1.6656	1.3129	1.1446	1.0638	1.0263	1.0105	1.0040
0.9	5.8275	3.1398	2.0056	1.4901	1.2360	1.1085	1.0483	1.0195	1.0072	1.0030	1.0009
0.85	2.2121	1.6156	1.3077	1.1496	1.0673	1.0293	1.0112	1.0045	1.0016	1.0006	1.0002
0.8	1.3222	1.1625	1.0773	1.0339	1.0135	1.0051	1.0019	1.0007	1.0002	1.0001	1
0.75	1.0664	1.0295	1.0121	1.0052	1.0018	1.0006	1.0002	1.0001	1	1	1
0.7	1.0082	1.0032	1.0012	1.0005	1.0002	1	1	1	1	1	1
0.65	1.0006	1.0002	1.0001	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.9036	52.1317	15.7391	5.9189	2.8987	1.8208	1.3683	1.1669	1.0744	1.0301	1.0119
0.95	29.8519	10.9779	4.7634	2.5577	1.6910	1.3183	1.1427	1.0616	1.0257	1.0101	1.0034
0.9	6.6831	3.4336	2.0910	1.5002	1.2336	1.1052	1.0443	1.0187	1.0069	1.0027	1.0011
0.85	2.3398	1.6480	1.3091	1.1464	1.0643	1.0270	1.0110	1.0035	1.0013	1.0004	1.0002
0.8	1.3263	1.1611	1.0719	1.0315	1.0126	1.0046	1.0018	1.0006	1.0002	1.0001	1
0.75	1.0615	1.0265	1.0122	1.0047	1.0015	1.0006	1.0002	1.0001	1.0001	1	1
0.7	1.0071	1.0030	1.0010	1.0004	1.0001	1	1	1	1	1	1
0.65	1.0004	1.0001	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.3118	12.8343	5.1440	3.1857	2.3386	1.8406	1.4722	1.2424	1.1113	1.0491	1.0201
0.95	6.0619	3.7570	2.7100	2.1618	1.7815	1.4480	1.2252	1.1051	1.0441	1.0180	1.0070
0.9	2.8891	2.3094	1.9953	1.6899	1.3811	1.1863	1.0813	1.0348	1.0134	1.0049	1.0016
0.85	2.0363	1.8524	1.5504	1.2802	1.1279	1.0539	1.0223	1.0087	1.0033	1.0010	1.0004
0.8	1.6689	1.3595	1.1631	1.0702	1.0268	1.0102	1.0036	1.0014	1.0003	1.0002	1.0001
0.75	1.1633	1.0688	1.0265	1.0103	1.0036	1.0013	1.0003	1.0001	1	1	1
0.7	1.0184	1.0063	1.0023	1.0008	1.0002	1.0001	1	1	1	1	1
0.65	1.0008	1.0003	1.0001	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.8008	17.2112	4.6885	2.6004	1.8003	1.3882	1.1818	1.0799	1.0336	1.0130	1.0050

0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.0436	51.9796	11.5340	3.1069	1.6233	1.2311	1.0921	1.0351	1.0139	1.0053	1.0019
0.95	20.7842	6.3006	2.2722	1.4010	1.1549	1.0587	1.0230	1.0083	1.0028	1.0010	1.0003
0.9	3.0854	1.5877	1.2020	1.0783	1.0302	1.0113	1.0041	1.0013	1.0004	1.0001	1.0001
0.85	1.2081	1.0765	1.0294	1.0104	1.0037	1.0013	1.0003	1.0001	1.0001	1	1
0.8	1.0189	1.0070	1.0021	1.0007	1.0003	1	1	1	1	1	1
0.75	1.0008	1.0002	1.0001	1	1	1	1	1	1	1	1
0.7	1.0001	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.8$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.4370	5.8564	2.6854	1.8792	1.3223	1.1000	1.0330	1.0120	1.0042	1.0015	1.0005
0.95	3.0588	2.0838	1.6119	1.1621	1.0478	1.0165	1.0056	1.0019	1.0006	1.0002	1.0001
0.9	1.8428	1.2176	1.0505	1.0160	1.0053	1.0017	1.0005	1.0001	1.0001	1	1
0.85	1.0353	1.0103	1.0032	1.0008	1.0002	1.0001	1.0001	1	1	1	1
0.8	1.0009	1.0003	1.0001	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.0756	7.1075	2.3647	1.4624	1.1263	1.0408	1.0137	1.0049	1.0018	1.0006	1.0002
0.95	2.7229	1.7833	1.1829	1.0505	1.0152	1.0053	1.0018	1.0006	1.0003	1.0001	1
0.9	1.2015	1.0416	1.0127	1.0045	1.0013	1.0004	1.0001	1	1	1	1
0.85	1.0054	1.0018	1.0006	1.0001	1	1	1	1	1	1	1
0.8	1.0001	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2

0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S34 ARL of the EWMA TBEA schemes based on the statistic Z_1 when variable A following the normal distribution and T following the gamma distribution assuming a Frank copula when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.2938	31.0243	9.9643	5.3257	3.5888	2.7534	2.2625	1.9438	1.7049	1.4915	1.3192
0.95	10.0029	5.3105	3.6028	2.7555	2.2619	1.9441	1.7024	1.4974	1.3197	1.1849	1.0985
0.9	3.5969	2.7506	2.2624	1.9454	1.7063	1.4991	1.3212	1.1864	1.0995	1.0479	1.0224
0.85	2.2589	1.9432	1.7047	1.4958	1.3211	1.1876	1.0979	1.0475	1.0218	1.0094	1.0038
0.8	1.7068	1.5000	1.3234	1.1844	1.0983	1.0467	1.0217	1.0092	1.0035	1.0015	1.0005
0.75	1.3229	1.1871	1.0993	1.0468	1.0215	1.0088	1.0033	1.0012	1.0007	1.0001	1.0001
0.7	1.0999	1.0466	1.0210	1.0094	1.0037	1.0012	1.0004	1.0002	1.0000	1.0001	1
0.65	1.0207	1.0088	1.0034	1.0012	1.0004	1.0002	1.0001	1	1	1	1
0.6	1.0032	1.0011	1.0004	1.0001	1	1	1	1	1	1	1
0.55	1.0004	1.0001	1	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.2560	41.6817	12.6448	5.7893	3.5569	2.5603	2.0155	1.6790	1.4343	1.2594	1.1435
0.95	12.5812	5.8025	3.5529	2.5589	2.0223	1.6783	1.4328	1.2597	1.1409	1.0723	1.0350
0.9	3.5516	2.5577	2.0198	1.6790	1.4337	1.2597	1.1403	1.0701	1.0333	1.0155	1.0063
0.85	2.0252	1.6783	1.4341	1.2605	1.1434	1.0711	1.0325	1.0145	1.0058	1.0028	1.0011
0.8	1.4354	1.2606	1.1424	1.0704	1.0326	1.0142	1.0063	1.0025	1.0009	1.0004	1.0001
0.75	1.1407	1.0723	1.0333	1.0142	1.0055	1.0022	1.0008	1.0003	1.0001	1	1
0.7	1.0323	1.0146	1.0057	1.0022	1.0007	1.0003	1.0001	1	1	1	1
0.65	1.0057	1.0020	1.0008	1.0003	1.0001	1	1	1	1	1	1
0.6	1.0008	1.0002	1.0001	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.0014	51.7317	16.6213	6.9767	3.8828	2.5908	1.9470	1.5710	1.3351	1.1839	1.0956
0.95	16.5088	7.0044	3.8776	2.5957	1.9481	1.5710	1.3360	1.1877	1.0944	1.0466	1.0210
0.9	3.8816	2.6021	1.9503	1.5749	1.3348	1.1842	1.0941	1.0451	1.0205	1.0087	1.0034
0.85	1.9506	1.5748	1.3375	1.1845	1.0937	1.0458	1.0208	1.0082	1.0035	1.0012	1.0006
0.8	1.3348	1.1856	1.0933	1.0445	1.0198	1.0078	1.0035	1.0011	1.0005	1.0002	1.0001
0.75	1.0948	1.0445	1.0199	1.0081	1.0036	1.0011	1.0005	1.0002	1.0001	1	1
0.7	1.0194	1.0075	1.0033	1.0012	1.0004	1.0001	1	1	1	1	1
0.65	1.0029	1.0010	1.0004	1.0001	1.0001	1	1	1	1	1	1
0.6	1.0002	1.0001	1.0001	1	1	1	1	1	1	1	1

0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.4994	62.0402	21.5603	9.0193	4.5700	2.8076	1.9852	1.5524	1.2997	1.1569	1.0773
0.95	21.5563	8.9961	4.5865	2.8274	1.9879	1.5500	1.3004	1.1573	1.0788	1.0362	1.0163
0.9	4.6106	2.8284	1.9942	1.5487	1.3017	1.1565	1.0760	1.0363	1.0159	1.0067	1.0026
0.85	1.9946	1.5509	1.3031	1.1578	1.0765	1.0354	1.0154	1.0061	1.0025	1.0008	1.0003
0.8	1.3016	1.1551	1.0772	1.0359	1.0153	1.0060	1.0025	1.0010	1.0004	1.0002	1
0.75	1.0773	1.0358	1.0155	1.0060	1.0023	1.0007	1.0003	1.0001	1	1	1
0.7	1.0152	1.0057	1.0023	1.0008	1.0003	1.0001	1	1	1	1	1
0.65	1.0022	1.0008	1.0003	1.0001	1.0001	1	1	1	1	1	1
0.6	1.0003	1.0001	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.4751	69.4811	26.6837	11.4293	5.6831	3.2374	2.1200	1.5761	1.2922	1.1447	1.0713
0.95	26.6720	11.4677	5.6426	3.2464	2.1203	1.5710	1.2943	1.1470	1.0705	1.0327	1.0137
0.9	5.6755	3.2365	2.1213	1.5725	1.2966	1.1459	1.0699	1.0310	1.0143	1.0061	1.0024
0.85	2.1309	1.5781	1.2972	1.1449	1.0691	1.0308	1.0138	1.0061	1.0022	1.0007	1.0004
0.8	1.2940	1.1468	1.0686	1.0309	1.0136	1.0056	1.0021	1.0007	1.0003	1.0001	1
0.75	1.0691	1.0311	1.0131	1.0054	1.0019	1.0007	1.0002	1.0001	1	1	1
0.7	1.0127	1.0050	1.0018	1.0007	1.0002	1.0001	1	1	1	1	1
0.65	1.0021	1.0005	1.0002	1.0001	1	1	1	1	1	1	1
0.6	1.0002	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.2247	71.3402	28.6270	12.7134	6.2564	3.4904	2.2140	1.6045	1.2980	1.1423	1.0669
0.95	28.2080	12.6372	6.2235	3.4966	2.2211	1.6045	1.2975	1.1452	1.0699	1.0306	1.0136
0.9	6.2547	3.4912	2.2235	1.6038	1.3004	1.1419	1.0680	1.0302	1.0124	1.0057	1.0020
0.85	2.2180	1.6057	1.3016	1.1468	1.0671	1.0300	1.0133	1.0052	1.0021	1.0007	1.0003
0.8	1.2979	1.1449	1.0669	1.0293	1.0126	1.0049	1.0020	1.0007	1.0003	1.0001	1
0.75	1.0657	1.0288	1.0128	1.0052	1.0017	1.0005	1.0002	1.0001	1.0001	1	1
0.7	1.0124	1.0053	1.0020	1.0005	1.0003	1.0001	1	1	1	1	1
0.65	1.0015	1.0007	1.0002	1	1	1	1	1	1	1	1
0.6	1.0002	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.1063	18.1192	5.7457	3.3094	2.3861	1.9320	1.6072	1.3252	1.1399	1.0549	1.0203

0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.6495	56.7535	17.3045	6.0367	2.7002	1.6046	1.2238	1.0807	1.0291	1.0104	1.0037
0.95	17.1988	6.0757	2.7088	1.6058	1.2225	1.0790	1.0288	1.0100	1.0033	1.0014	1.0004
0.9	2.7090	1.6130	1.2222	1.0797	1.0289	1.0101	1.0034	1.0011	1.0004	1.0001	1.0001
0.85	1.2228	1.0803	1.0290	1.0097	1.0030	1.0010	1.0004	1.0001	1	1	1
0.8	1.0275	1.0097	1.0033	1.0009	1.0003	1.0001	1	1	1	1	1
0.75	1.0031	1.0012	1.0003	1.0001	1.0001	1	1	1	1	1	1
0.7	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.8582	60.2825	19.5160	6.9973	2.9626	1.6558	1.2245	1.0789	1.0281	1.0096	1.0035
0.95	19.4547	6.9966	2.9790	1.6547	1.2245	1.0802	1.0269	1.0096	1.0034	1.0011	1.0005
0.9	2.9729	1.6611	1.2261	1.0793	1.0273	1.0100	1.0030	1.0012	1.0004	1.0002	1.0001
0.85	1.2270	1.0792	1.0279	1.0094	1.0030	1.0011	1.0003	1.0001	1.0001	1	1
0.8	1.0270	1.0093	1.0028	1.0009	1.0003	1.0001	1	1	1	1	1
0.75	1.0030	1.0009	1.0003	1.0001	1	1	1	1	1	1	1
0.7	1.0003	1.0001	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.8$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.3875	5.8282	2.4316	1.7051	1.1456	1.0306	1.0082	1.0019	1.0006	1.0002	1
0.95	2.4356	1.7032	1.1475	1.0296	1.0079	1.0020	1.0005	1.0001	1	1	1
0.9	1.1458	1.0293	1.0074	1.0019	1.0005	1.0002	1	1	1	1	1
0.85	1.0073	1.0017	1.0004	1.0001	1	1	1	1	1	1	1
0.8	1.0004	1.0001	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2

0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.8300	37.9094	5.4878	1.3288	1.0494	1.0112	1.0037	1.0009	1.0002	1.0001	1
0.95	5.5058	1.3315	1.0479	1.0115	1.0031	1.0007	1.0002	1	1	1	1
0.9	1.0469	1.0109	1.0027	1.0007	1.0001	1	1	1	1	1	1
0.85	1.0024	1.0007	1.0001	1	1	1	1	1	1	1	1
0.8	1.0002	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S35 ARL of the EWMA TBEA schemes based on the statistic Z_2 when variable A following the normal distribution and T following the gamma distribution assuming a Frank copula when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.3245	24.1438	9.1636	5.3133	3.6839	2.7954	2.2380	1.8553	1.5637	1.3353	1.1757
0.95	10.7966	6.1136	4.1707	3.1451	2.5052	2.0805	1.7673	1.5093	1.2998	1.1540	1.0755
0.9	4.7018	3.5137	2.7848	2.3060	1.9697	1.6986	1.4634	1.2683	1.1358	1.0630	1.0268
0.85	3.0774	2.5371	2.1620	1.8867	1.6451	1.4232	1.2402	1.1197	1.0556	1.0229	1.0093
0.8	2.3518	2.0499	1.8223	1.5998	1.3830	1.2128	1.1044	1.0452	1.0189	1.0076	1.0030
0.75	1.9738	1.7707	1.5577	1.3479	1.1890	1.0909	1.0387	1.0156	1.0061	1.0024	1.0007
0.7	1.7248	1.5184	1.3166	1.1682	1.0780	1.0328	1.0123	1.0046	1.0018	1.0006	1.0001
0.65	1.4785	1.2860	1.1475	1.0676	1.0286	1.0108	1.0038	1.0013	1.0005	1.0002	1
0.6	1.2563	1.1294	1.0578	1.0231	1.0088	1.0026	1.0009	1.0003	1.0001	1.0001	1
0.55	1.1143	1.0490	1.0190	1.0075	1.0022	1.0006	1.0003	1.0001	1	1	1
0.5	1.0411	1.0159	1.0054	1.0017	1.0004	1.0001	1.0001	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.0397	33.1498	10.0585	4.9220	3.1204	2.2652	1.7707	1.4481	1.2442	1.1207	1.0554
0.95	13.6061	6.0531	3.6593	2.5871	2.0007	1.6211	1.3616	1.1931	1.0936	1.0420	1.0178
0.9	4.2891	2.9552	2.2545	1.8161	1.5087	1.2956	1.1486	1.0696	1.0314	1.0124	1.0049
0.85	2.5249	2.0262	1.6853	1.4236	1.2374	1.1180	1.0531	1.0220	1.0082	1.0030	1.0012
0.8	1.8579	1.5697	1.3483	1.1885	1.0892	1.0385	1.0154	1.0061	1.0022	1.0009	1.0003
0.75	1.4855	1.2816	1.1464	1.0676	1.0287	1.0111	1.0043	1.0014	1.0006	1.0001	1.0001
0.7	1.2285	1.1145	1.0503	1.0204	1.0075	1.0025	1.0008	1.0002	1.0001	1.0001	1
0.65	1.0853	1.0378	1.0146	1.0055	1.0019	1.0006	1.0002	1	1	1	1
0.6	1.0266	1.0105	1.0037	1.0012	1.0003	1.0001	1.0001	1	1	1	1

0.55	1.0069	1.0023	1.0007	1.0002	1	1	1	1	1	1	1
0.5	1.0015	1.0005	1.0002	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.4760	43.0928	12.8635	5.5203	3.1680	2.1511	1.6360	1.3389	1.1686	1.0790	1.0347
0.95	19.7112	7.4322	3.9089	2.5332	1.8664	1.4849	1.2563	1.1254	1.0569	1.0243	1.0098
0.9	4.9186	3.0092	2.1460	1.6700	1.3706	1.1933	1.0907	1.0401	1.0165	1.0065	1.0026
0.85	2.4756	1.8827	1.5181	1.2864	1.1432	1.0661	1.0278	1.0110	1.0044	1.0016	1.0007
0.8	1.6957	1.4081	1.2205	1.1067	1.0464	1.0189	1.0074	1.0024	1.0009	1.0003	1.0001
0.75	1.3179	1.1626	1.0762	1.0330	1.0118	1.0047	1.0017	1.0006	1.0002	1	1
0.7	1.1235	1.0541	1.0227	1.0086	1.0029	1.0012	1.0004	1.0002	1	1	1
0.65	1.0371	1.0157	1.0049	1.0017	1.0006	1.0002	1.0001	1	1	1	1
0.6	1.0102	1.0034	1.0009	1.0002	1.0001	1	1	1	1	1	1
0.55	1.0021	1.0007	1.0001	1	1	1	1	1	1	1	1
0.5	1.0004	1.0001	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.2596	51.6944	16.4138	6.6681	3.4472	2.1920	1.5988	1.2986	1.1424	1.0641	1.0278
0.95	26.9424	9.7454	4.5744	2.6796	1.8520	1.4370	1.2185	1.1000	1.0445	1.0190	1.0079
0.9	6.3210	3.3875	2.2012	1.6276	1.3203	1.1552	1.0702	1.0303	1.0124	1.0049	1.0020
0.85	2.6646	1.8826	1.4644	1.2325	1.1104	1.0486	1.0193	1.0082	1.0027	1.0010	1.0005
0.8	1.6550	1.3423	1.1700	1.0778	1.0334	1.0139	1.0052	1.0017	1.0006	1.0002	1.0001
0.75	1.2536	1.1209	1.0535	1.0217	1.0084	1.0030	1.0011	1.0004	1.0001	1	1
0.7	1.0886	1.0362	1.0148	1.0055	1.0018	1.0007	1.0002	1.0001	1	1	1
0.65	1.0251	1.0093	1.0032	1.0010	1.0002	1.0001	1	1	1	1	1
0.6	1.0062	1.0022	1.0006	1.0002	1.0001	1	1	1	1	1	1
0.55	1.0011	1.0004	1.0001	1	1	1	1	1	1	1	1
0.5	1.0002	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.6763	58.2087	19.9320	8.0772	3.9321	2.3116	1.6088	1.2841	1.1307	1.0583	1.0244
0.95	33.9317	12.5743	5.5785	2.9985	1.9201	1.4318	1.2010	1.0900	1.0397	1.0165	1.0067
0.9	8.3140	4.0553	2.3878	1.6511	1.3089	1.1389	1.0606	1.0258	1.0098	1.0044	1.0016
0.85	3.0845	1.9795	1.4574	1.2164	1.0987	1.0423	1.0170	1.0062	1.0024	1.0010	1.0003
0.8	1.6937	1.3292	1.1516	1.0668	1.0280	1.0114	1.0040	1.0012	1.0004	1.0001	1.0001
0.75	1.2334	1.1045	1.0451	1.0181	1.0065	1.0022	1.0008	1.0002	1.0001	1	1
0.7	1.0724	1.0298	1.0117	1.0039	1.0017	1.0005	1.0002	1	1	1	1
0.65	1.0191	1.0076	1.0025	1.0009	1.0002	1.0001	1	1	1	1	1
0.6	1.0043	1.0013	1.0004	1.0001	1	1	1	1	1	1	1
0.55	1.0008	1.0004	1.0001	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.5115	61.3609	21.6269	8.7728	4.2243	2.4073	1.6368	1.2892	1.1290	1.0559	1.0242
0.95	37.1358	14.0792	6.2011	3.1954	1.9750	1.4458	1.1979	1.0892	1.0373	1.0152	1.0059

0.9	9.4885	4.5044	2.5349	1.6904	1.3119	1.1361	1.0590	1.0233	1.0099	1.0038	1.0017
0.85	3.4012	2.0716	1.4817	1.2173	1.0956	1.0399	1.0161	1.0061	1.0023	1.0008	1.0003
0.8	1.7469	1.3372	1.1470	1.0641	1.0266	1.0101	1.0035	1.0012	1.0004	1.0001	1.0001
0.75	1.2338	1.1042	1.0422	1.0170	1.0064	1.0019	1.0007	1.0002	1.0001	1	1
0.7	1.0694	1.0279	1.0109	1.0037	1.0014	1.0004	1.0001	1.0001	1	1	1
0.65	1.0173	1.0068	1.0022	1.0006	1.0002	1.0001	1	1	1	1	1
0.6	1.0040	1.0013	1.0004	1.0001	1	1	1	1	1	1	1
0.55	1.0008	1.0002	1	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.8053	15.0609	5.8530	3.5651	2.5560	2.0127	1.6467	1.3255	1.1325	1.0502	1.0190
0.95	6.7211	4.0071	2.8425	2.2129	1.8447	1.5161	1.2416	1.0941	1.0352	1.0128	1.0054
0.9	3.1441	2.4285	2.0115	1.7139	1.4025	1.1736	1.0652	1.0249	1.0083	1.0030	1.0010
0.85	2.1640	1.8804	1.5991	1.3072	1.1253	1.0452	1.0164	1.0057	1.0019	1.0008	1.0002
0.8	1.7806	1.4846	1.2255	1.0863	1.0314	1.0108	1.0037	1.0012	1.0003	1.0001	1
0.75	1.3815	1.1617	1.0588	1.0211	1.0074	1.0024	1.0007	1.0002	1.0001	1	1
0.7	1.1138	1.0421	1.0134	1.0047	1.0015	1.0005	1.0001	1.0001	1	1	1
0.65	1.0278	1.0086	1.0030	1.0010	1.0003	1.0001	1	1	1	1	1
0.6	1.0058	1.0020	1.0006	1.0001	1	1	1	1	1	1	1
0.55	1.0011	1.0003	1.0001	1	1	1	1	1	1	1	1
0.5	1.0003	1.0001	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.2090	21.3619	5.8793	3.0629	2.0770	1.5607	1.2485	1.0951	1.0342	1.0124	1.0047
0.95	7.3371	3.5467	2.3382	1.7626	1.3814	1.1555	1.0571	1.0200	1.0075	1.0026	1.0009
0.9	2.6296	1.9590	1.5455	1.2498	1.0949	1.0347	1.0119	1.0041	1.0013	1.0006	1.0002
0.85	1.7179	1.3780	1.1557	1.0580	1.0200	1.0068	1.0021	1.0007	1.0002	1.0001	1
0.8	1.2505	1.0977	1.0345	1.0109	1.0039	1.0011	1.0004	1.0001	1	1	1
0.75	1.0574	1.0198	1.0066	1.0022	1.0006	1.0002	1.0001	1	1	1	1
0.7	1.0114	1.0035	1.0012	1.0003	1.0001	1	1	1	1	1	1
0.65	1.0023	1.0006	1.0001	1.0001	1	1	1	1	1	1	1
0.6	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.4929	31.2839	7.4557	3.1953	1.9687	1.4253	1.1649	1.0602	1.0216	1.0085	1.0027
0.95	10.5548	3.9532	2.2760	1.6067	1.2540	1.0963	1.0337	1.0120	1.0048	1.0014	1.0006
0.9	2.6578	1.8230	1.3805	1.1505	1.0535	1.0187	1.0068	1.0022	1.0006	1.0003	1.0001
0.85	1.5433	1.2349	1.0893	1.0313	1.0101	1.0037	1.0010	1.0002	1.0001	1	1
0.8	1.1363	1.0499	1.0175	1.0060	1.0018	1.0006	1.0002	1	1	1	1
0.75	1.0264	1.0096	1.0027	1.0008	1.0003	1.0001	1	1	1	1	1
0.7	1.0045	1.0014	1.0004	1.0001	1	1	1	1	1	1	1
0.65	1.0009	1.0002	1.0001	1	1	1	1	1	1	1	1

0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.3642	40.8063	10.1464	3.6908	1.9736	1.3706	1.1346	1.0492	1.0170	1.0058	1.0022
0.95	15.8211	4.9833	2.4100	1.5535	1.2100	1.0745	1.0264	1.0094	1.0029	1.0011	1.0004
0.9	3.0192	1.8044	1.3193	1.1198	1.0418	1.0147	1.0045	1.0017	1.0005	1.0002	1.0001
0.85	1.4731	1.1813	1.0631	1.0223	1.0073	1.0023	1.0009	1.0003	1.0000	1.0001	1
0.8	1.1029	1.0357	1.0112	1.0038	1.0013	1.0003	1.0002	1	1	1	1
0.75	1.0186	1.0058	1.0020	1.0005	1.0002	1	1	1	1	1	1
0.7	1.0032	1.0008	1.0003	1.0001	1	1	1	1	1	1	1
0.65	1.0005	1.0001	1	1	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.6161	48.9987	13.3986	4.5339	2.1030	1.3692	1.1230	1.0439	1.0150	1.0052	1.0020
0.95	21.8451	6.7665	2.7533	1.5801	1.1945	1.0668	1.0230	1.0075	1.0030	1.0009	1.0003
0.9	3.7992	1.9088	1.3078	1.1050	1.0367	1.0130	1.0045	1.0015	1.0005	1.0001	1
0.85	1.4812	1.1652	1.0569	1.0182	1.0061	1.0018	1.0006	1.0002	1.0001	1	1
0.8	1.0874	1.0304	1.0102	1.0030	1.0010	1.0002	1.0001	1	1	1	1
0.75	1.0149	1.0053	1.0015	1.0004	1.0001	1	1	1	1	1	1
0.7	1.0025	1.0008	1.0003	1	1	1	1	1	1	1	1
0.65	1.0004	1.0002	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.0011	51.8505	14.9995	5.0548	2.2116	1.3738	1.1217	1.0411	1.0142	1.0052	1.0019
0.95	24.7289	7.8055	3.0132	1.6150	1.1946	1.0645	1.0220	1.0074	1.0026	1.0009	1.0003
0.9	4.3374	2.0137	1.3172	1.1008	1.0344	1.0114	1.0040	1.0014	1.0004	1.0001	1
0.85	1.5016	1.1618	1.0545	1.0182	1.0059	1.0018	1.0006	1.0002	1.0001	1	1
0.8	1.0850	1.0279	1.0092	1.0027	1.0009	1.0003	1.0001	1.0001	1	1	1
0.75	1.0149	1.0042	1.0015	1.0004	1.0001	1	1	1	1	1	1
0.7	1.0026	1.0007	1.0002	1.0001	1	1	1	1	1	1	1
0.65	1.0003	1.0001	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.8$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2

0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.2139	29.8064	3.6825	1.2446	1.0433	1.0109	1.0032	1.0009	1.0003	1.0001	1
0.95	5.8824	1.4235	1.0595	1.0139	1.0041	1.0012	1.0004	1.0001	1	1	1
0.9	1.0912	1.0194	1.0053	1.0014	1.0003	1.0001	1.0001	1	1	1	1
0.85	1.0075	1.0017	1.0005	1.0002	1.0001	1	1	1	1	1	1
0.8	1.0005	1.0002	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.9473	33.7708	4.4351	1.2432	1.0410	1.0106	1.0031	1.0010	1.0002	1.0001	1
0.95	7.5128	1.4496	1.0600	1.0141	1.0039	1.0009	1.0003	1.0001	1.0001	1	1
0.9	1.0902	1.0191	1.0049	1.0013	1.0003	1.0001	1	1	1	1	1
0.85	1.0068	1.0018	1.0004	1.0002	1.0001	1	1	1	1	1	1
0.8	1.0005	1.0002	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S36 ARL of the EWMA TBEA schemes based on the statistic Z_3 when variable A following the normal distribution and T following the gamma distribution assuming a Frank copula when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.9035	23.1844	8.7560	5.1056	3.5487	2.6966	2.1682	1.7967	1.5146	1.2942	1.1510
0.95	10.5852	6.1753	4.2659	3.2269	2.5818	2.1466	1.8304	1.5629	1.3412	1.1778	1.0837
0.9	4.5810	3.5533	2.8868	2.4162	2.0854	1.8258	1.5936	1.3691	1.1985	1.0956	1.0431
0.85	2.9980	2.5634	2.2392	2.0010	1.7977	1.5854	1.3702	1.2036	1.0976	1.0431	1.0181
0.8	2.2849	2.0774	1.9182	1.7525	1.5511	1.3491	1.1883	1.0908	1.0398	1.0167	1.0063
0.75	1.9610	1.8379	1.6767	1.4805	1.2946	1.1567	1.0746	1.0311	1.0128	1.0052	1.0019
0.7	1.7355	1.5621	1.3779	1.2188	1.1121	1.0524	1.0216	1.0087	1.0033	1.0013	1.0005
0.65	1.4076	1.2485	1.1377	1.0660	1.0299	1.0122	1.0051	1.0016	1.0006	1.0002	1.0001
0.6	1.1328	1.0689	1.0314	1.0134	1.0057	1.0021	1.0008	1.0002	1.0001	1	1

0.55	1.0268	1.0122	1.0044	1.0018	1.0008	1.0003	1.0001	1	1	1	1
0.5	1.0032	1.0014	1.0005	1.0001	1.0001	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.1683	30.9896	9.2789	4.5565	2.9200	2.1314	1.6787	1.3814	1.1954	1.0917	1.0411
0.95	12.9948	5.9624	3.6654	2.6160	2.0293	1.6451	1.3795	1.1963	1.0920	1.0419	1.0164
0.9	4.0422	2.9340	2.3022	1.8797	1.5793	1.3418	1.1806	1.0847	1.0371	1.0150	1.0060
0.85	2.3957	2.0144	1.7229	1.4829	1.2815	1.1455	1.0681	1.0295	1.0119	1.0048	1.0016
0.8	1.7777	1.5601	1.3602	1.2049	1.1026	1.0477	1.0195	1.0078	1.0031	1.0010	1.0004
0.75	1.3849	1.2350	1.1230	1.0604	1.0266	1.0108	1.0046	1.0015	1.0005	1.0002	1.0001
0.7	1.1211	1.0603	1.0277	1.0120	1.0048	1.0021	1.0007	1.0002	1	1	1
0.65	1.0234	1.0105	1.0041	1.0016	1.0005	1.0002	1.0001	1	1	1	1
0.6	1.0027	1.0013	1.0004	1.0001	1	1	1	1	1	1	1
0.55	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.2714	40.0144	11.4974	4.9640	2.8702	1.9852	1.5211	1.2668	1.1280	1.0562	1.0243
0.95	18.6821	7.2774	3.8780	2.5386	1.8609	1.4831	1.2522	1.1199	1.0536	1.0220	1.0090
0.9	4.5153	2.9378	2.1527	1.7019	1.3996	1.2121	1.1008	1.0448	1.0185	1.0073	1.0025
0.85	2.2778	1.8319	1.5213	1.2982	1.1541	1.0729	1.0320	1.0129	1.0048	1.0018	1.0007
0.8	1.5575	1.3453	1.1949	1.0983	1.0453	1.0196	1.0077	1.0031	1.0009	1.0004	1.0001
0.75	1.1946	1.1048	1.0502	1.0228	1.0095	1.0036	1.0013	1.0004	1.0002	1.0001	1
0.7	1.0448	1.0208	1.0087	1.0037	1.0012	1.0005	1.0001	1	1	1	1
0.65	1.0060	1.0026	1.0009	1.0004	1.0001	1.0001	1	1	1	1	1
0.6	1.0005	1.0002	1.0001	1	1	1	1	1	1	1	1
0.55	1.0001	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.9467	48.3257	14.6217	5.8286	3.0561	1.9713	1.4776	1.2227	1.1005	1.0438	1.0176
0.95	26.5073	9.6848	4.5488	2.6522	1.8348	1.4206	1.2019	1.0913	1.0400	1.0165	1.0063
0.9	5.7638	3.2991	2.2018	1.6411	1.3359	1.1633	1.0735	1.0322	1.0125	1.0047	1.0019
0.85	2.3813	1.7881	1.4442	1.2326	1.1125	1.0501	1.0214	1.0082	1.0030	1.0010	1.0005
0.8	1.4700	1.2637	1.1394	1.0649	1.0290	1.0118	1.0045	1.0016	1.0007	1.0002	1.0001
0.75	1.1333	1.0669	1.0311	1.0138	1.0052	1.0020	1.0007	1.0003	1.0001	1	1
0.7	1.0254	1.0109	1.0050	1.0018	1.0006	1.0003	1.0001	1	1	1	1
0.65	1.0032	1.0014	1.0005	1.0002	1	1	1	1	1	1	1
0.6	1.0003	1.0001	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.1702	54.8931	17.8945	6.9855	3.3917	2.0409	1.4652	1.2072	1.0902	1.0383	1.0152
0.95	34.5584	12.7284	5.6009	2.9722	1.8887	1.4083	1.1860	1.0800	1.0348	1.0135	1.0052

0.9	7.7904	4.0085	2.4020	1.6702	1.3179	1.1446	1.0622	1.0258	1.0099	1.0040	1.0015
0.85	2.7176	1.8572	1.4281	1.2078	1.0949	1.0413	1.0171	1.0068	1.0025	1.0008	1.0004
0.8	1.4611	1.2393	1.1168	1.0533	1.0232	1.0092	1.0033	1.0014	1.0004	1.0002	1.0001
0.75	1.1085	1.0526	1.0239	1.0103	1.0040	1.0014	1.0006	1.0001	1.0001	1	1
0.7	1.0185	1.0088	1.0034	1.0010	1.0005	1.0002	1.0001	1	1	1	1
0.65	1.0019	1.0009	1.0003	1.0001	1	1	1	1	1	1	1
0.6	1.0002	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.9501	57.2532	19.0484	7.6267	3.6018	2.0978	1.4731	1.2014	1.0870	1.0353	1.0148
0.95	37.8342	14.2855	6.2131	3.1801	1.9438	1.4155	1.1840	1.0756	1.0322	1.0133	1.0049
0.9	8.9828	4.4468	2.5463	1.6963	1.3132	1.1418	1.0591	1.0243	1.0098	1.0035	1.0012
0.85	2.9450	1.9263	1.4376	1.2021	1.0909	1.0390	1.0159	1.0059	1.0023	1.0010	1.0003
0.8	1.4770	1.2348	1.1104	1.0488	1.0206	1.0080	1.0033	1.0010	1.0005	1.0001	1
0.75	1.1034	1.0485	1.0217	1.0085	1.0036	1.0013	1.0005	1.0001	1	1	1
0.7	1.0177	1.0072	1.0029	1.0012	1.0005	1.0001	1	1	1	1	1
0.65	1.0021	1.0007	1.0002	1.0001	1	1	1	1	1	1	1
0.6	1.0002	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.2919	14.7354	5.7570	3.5033	2.5099	1.9861	1.6264	1.3272	1.1403	1.0561	1.0215
0.95	6.7452	4.1149	2.9549	2.2963	1.9261	1.6187	1.3198	1.1345	1.0521	1.0199	1.0075
0.9	3.1382	2.4961	2.0895	1.8437	1.5672	1.2811	1.1128	1.0426	1.0159	1.0057	1.0021
0.85	2.1493	1.9483	1.7491	1.4598	1.2081	1.0785	1.0295	1.0100	1.0034	1.0014	1.0003
0.8	1.8225	1.5858	1.3043	1.1194	1.0433	1.0143	1.0049	1.0018	1.0006	1.0001	1.0001
0.75	1.3429	1.1468	1.0539	1.0185	1.0064	1.0022	1.0007	1.0003	1.0001	1	1
0.7	1.0521	1.0180	1.0059	1.0016	1.0006	1.0003	1	1	1	1	1
0.65	1.0048	1.0014	1.0004	1.0002	1	1	1	1	1	1	1
0.6	1.0003	1.0002	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.2407	20.7160	5.6854	2.9722	2.0268	1.5342	1.2371	1.0906	1.0349	1.0136	1.0048
0.95	7.3556	3.6578	2.4249	1.8351	1.4449	1.1889	1.0717	1.0260	1.0094	1.0034	1.0013
0.9	2.5936	2.0142	1.6327	1.3174	1.1241	1.0461	1.0154	1.0060	1.0020	1.0005	1.0003
0.85	1.7110	1.4073	1.1765	1.0650	1.0211	1.0077	1.0028	1.0008	1.0003	1.0001	1
0.8	1.1895	1.0721	1.0249	1.0078	1.0027	1.0008	1.0003	1.0001	1	1	1
0.75	1.0233	1.0078	1.0025	1.0009	1.0003	1.0001	1.0001	1	1	1	1
0.7	1.0018	1.0006	1.0002	1.0001	1	1	1	1	1	1	1
0.65	1.0002	1	1	1	1	1	1	1	1	1	1

0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.9679	30.0258	7.1384	3.0708	1.9053	1.3925	1.1539	1.0576	1.0214	1.0082	1.0027
0.95	10.9326	4.1636	2.3797	1.6733	1.2941	1.1127	1.0411	1.0145	1.0049	1.0017	1.0006
0.9	2.6130	1.8702	1.4388	1.1812	1.0640	1.0221	1.0083	1.0027	1.0009	1.0003	1.0001
0.85	1.4976	1.2257	1.0842	1.0295	1.0097	1.0034	1.0012	1.0003	1.0001	1	1
0.8	1.0857	1.0295	1.0105	1.0032	1.0010	1.0004	1.0002	1	1	1	1
0.75	1.0084	1.0029	1.0010	1.0003	1.0001	1	1	1	1	1	1
0.7	1.0006	1.0001	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.4873	38.7885	9.5626	3.5142	1.9090	1.3419	1.1230	1.0454	1.0164	1.0059	1.0023
0.95	16.7985	5.4287	2.5563	1.6264	1.2379	1.0859	1.0302	1.0111	1.0038	1.0012	1.0005
0.9	2.9803	1.8614	1.3603	1.1344	1.0468	1.0161	1.0055	1.0020	1.0005	1.0002	1.0001
0.85	1.4052	1.1617	1.0585	1.0192	1.0060	1.0023	1.0008	1.0002	1.0001	1	1
0.8	1.0562	1.0199	1.0064	1.0020	1.0007	1.0002	1.0001	1	1	1	1
0.75	1.0053	1.0017	1.0005	1.0001	1.0001	1	1	1	1	1	1
0.7	1.0003	1.0001	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.0211	46.4972	12.5583	4.2491	2.0012	1.3253	1.1101	1.0391	1.0150	1.0049	1.0018
0.95	24.0140	7.5925	3.0425	1.6563	1.2198	1.0737	1.0252	1.0092	1.0033	1.0011	1.0004
0.9	3.8460	1.9908	1.3445	1.1168	1.0384	1.0128	1.0046	1.0014	1.0003	1.0002	1.0001
0.85	1.3900	1.1398	1.0483	1.0158	1.0050	1.0016	1.0006	1.0002	1.0002	1	1
0.8	1.0444	1.0150	1.0047	1.0016	1.0006	1.0002	1.0001	1	1	1	1
0.75	1.0038	1.0011	1.0004	1.0001	1	1	1	1	1	1	1
0.7	1.0003	1.0001	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.8193	49.7543	14.0829	4.7464	2.1044	1.3359	1.1117	1.0390	1.0137	1.0049	1.0019

0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.1154	28.2694	4.1635	1.5440	1.0916	1.0219	1.0061	1.0024	1.0006	1.0002	1.0001
0.95	7.0749	1.9731	1.1585	1.0258	1.0070	1.0020	1.0005	1.0002	1	1	1
0.9	1.1999	1.0256	1.0051	1.0012	1.0003	1.0001	1	1	1	1	1
0.85	1.0043	1.0007	1.0001	1	1	1	1	1	1	1	1
0.8	1.0001	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.6724	36.9373	6.4505	1.6042	1.0838	1.0212	1.0059	1.0019	1.0007	1.0003	1
0.95	13.1551	2.4309	1.1409	1.0233	1.0059	1.0016	1.0005	1.0002	1	1	1
0.9	1.1785	1.0212	1.0042	1.0012	1.0003	1	1	1	1	1	1
0.85	1.0037	1.0005	1.0001	1	1	1	1	1	1	1	1
0.8	1.0001	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.4313	40.3383	7.7696	1.6728	1.0820	1.0189	1.0059	1.0017	1.0005	1.0002	1.0001
0.95	16.5025	2.9468	1.1397	1.0218	1.0057	1.0019	1.0004	1.0001	1.0001	1	1
0.9	1.1761	1.0209	1.0041	1.0010	1.0003	1.0001	1	1	1	1	1
0.85	1.0035	1.0007	1.0001	1	1	1	1	1	1	1	1
0.8	1.0001	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S37 ARL of the EWMA TBFA schemes based on the statistic Z_1 when variable A following the normal distribution and T following the weibull distribution assuming a Clayton copula when

$\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.0121	32.5724	10.2778	5.4016	3.6348	2.7688	2.2705	1.9534	1.7204	1.5179	1.3395
0.95	10.2651	5.4067	3.6308	2.7694	2.2711	1.9475	1.7192	1.5166	1.3401	1.1921	1.0946
0.9	3.6339	2.7708	2.2670	1.9514	1.7187	1.5218	1.3416	1.1932	1.0934	1.0393	1.0143
0.85	2.2705	1.9482	1.7204	1.5219	1.3395	1.1928	1.0939	1.0372	1.0129	1.0034	1.0009
0.8	1.7199	1.5217	1.3394	1.1907	1.0932	1.0370	1.0121	1.0034	1.0008	1.0002	1
0.75	1.3401	1.1924	1.0922	1.0369	1.0118	1.0031	1.0007	1.0001	1	1	1
0.7	1.0908	1.0356	1.0118	1.0028	1.0008	1.0001	1	1	1	1	1
0.65	1.0113	1.0027	1.0008	1.0001	1	1	1	1	1	1	1
0.6	1.0007	1.0001	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.6134	44.8020	13.4540	6.0845	3.6788	2.6187	2.0603	1.7087	1.4608	1.2804	1.1481
0.95	13.5403	6.1083	3.6761	2.6216	2.0626	1.7121	1.4660	1.2777	1.1459	1.0672	1.0259
0.9	3.6695	2.6215	2.0616	1.7078	1.4640	1.2789	1.1470	1.0651	1.0253	1.0082	1.0025
0.85	2.0609	1.7128	1.4630	1.2792	1.1444	1.0669	1.0243	1.0081	1.0022	1.0004	1.0001
0.8	1.4658	1.2785	1.1463	1.0645	1.0236	1.0074	1.0021	1.0004	1.0001	1	1
0.75	1.1442	1.0651	1.0238	1.0069	1.0015	1.0003	1.0001	1	1	1	1
0.7	1.0239	1.0069	1.0016	1.0004	1	1	1	1	1	1	1
0.65	1.0013	1.0004	1.0001	1	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.1366	56.5281	18.3237	7.6218	4.1505	2.7202	2.0270	1.6242	1.3648	1.1987	1.0984
0.95	18.3931	7.6507	4.1564	2.7335	2.0253	1.6253	1.3669	1.1985	1.0946	1.0387	1.0137
0.9	4.1259	2.7281	2.0266	1.6282	1.3639	1.1980	1.0932	1.0375	1.0127	1.0035	1.0010
0.85	2.0283	1.6221	1.3682	1.1990	1.0922	1.0368	1.0120	1.0035	1.0010	1.0002	1.0001
0.8	1.3687	1.1955	1.0915	1.0368	1.0125	1.0032	1.0006	1.0002	1	1	1
0.75	1.0907	1.0360	1.0116	1.0037	1.0006	1.0001	1	1	1	1	1
0.7	1.0111	1.0030	1.0006	1.0001	1	1	1	1	1	1	1
0.65	1.0007	1.0001	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.0571	66.3375	23.9342	10.0230	5.0438	3.0304	2.0939	1.6086	1.3325	1.1646	1.0752
0.95	23.9376	10.0169	5.0490	3.0493	2.1107	1.6140	1.3258	1.1641	1.0720	1.0285	1.0095

0.9	5.0349	3.0452	2.1048	1.6166	1.3318	1.1637	1.0715	1.0283	1.0089	1.0026	1.0005
0.85	2.1028	1.6176	1.3296	1.1630	1.0727	1.0272	1.0083	1.0025	1.0005	1.0001	1
0.8	1.3322	1.1640	1.0717	1.0270	1.0082	1.0020	1.0005	1.0001	1	1	1
0.75	1.0718	1.0258	1.0083	1.0017	1.0003	1.0001	1	1	1	1	1
0.7	1.0082	1.0020	1.0004	1.0001	1	1	1	1	1	1	1
0.65	1.0003	1.0001	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.1716	74.4788	29.5167	12.9799	6.3824	3.5800	2.3109	1.6622	1.3300	1.1539	1.0657
0.95	29.5750	13.0158	6.4025	3.5999	2.3045	1.6738	1.3329	1.1557	1.0668	1.0252	1.0077
0.9	6.3734	3.5961	2.3144	1.6698	1.3321	1.1560	1.0653	1.0229	1.0072	1.0017	1.0005
0.85	2.3141	1.6734	1.3299	1.1537	1.0628	1.0227	1.0065	1.0017	1.0004	1.0001	1
0.8	1.3329	1.1516	1.0629	1.0221	1.0065	1.0014	1.0003	1	1	1	1
0.75	1.0620	1.0222	1.0064	1.0016	1.0002	1.0001	1	1	1	1	1
0.7	1.0058	1.0013	1.0003	1.0001	1	1	1	1	1	1	1
0.65	1.0003	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.7340	76.8220	31.8369	14.3440	7.1394	3.9535	2.4547	1.7119	1.3437	1.1529	1.0640
0.95	31.6532	14.4316	7.1244	3.9438	2.4603	1.7174	1.3423	1.1548	1.0628	1.0227	1.0078
0.9	7.1438	3.9792	2.4553	1.7157	1.3452	1.1524	1.0635	1.0217	1.0071	1.0016	1.0004
0.85	2.4605	1.7179	1.3417	1.1525	1.0605	1.0207	1.0067	1.0012	1.0004	1	1
0.8	1.3400	1.1495	1.0600	1.0204	1.0061	1.0015	1.0003	1.0001	1	1	1
0.75	1.0589	1.0209	1.0056	1.0013	1.0004	1.0001	1	1	1	1	1
0.7	1.0054	1.0012	1.0002	1	1	1	1	1	1	1	1
0.65	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.4951	17.3440	5.4252	3.1491	2.2829	1.8623	1.5365	1.2580	1.0984	1.0306	1.0084
0.95	5.4098	3.1447	2.2842	1.8594	1.5409	1.2618	1.0964	1.0299	1.0069	1.0013	1.0002
0.9	2.2858	1.8598	1.5405	1.2587	1.0940	1.0289	1.0065	1.0014	1.0002	1	1
0.85	1.5410	1.2638	1.0964	1.0273	1.0067	1.0012	1.0001	1	1	1	1
0.8	1.0941	1.0273	1.0061	1.0011	1.0001	1	1	1	1	1	1
0.75	1.0062	1.0009	1.0001	1	1	1	1	1	1	1	1
0.7	1.0001	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1

0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.9351	26.9887	6.5114	3.1450	2.1286	1.6285	1.2981	1.1134	1.0351	1.0099	1.0024
0.95	6.5132	3.1679	2.1277	1.6291	1.2992	1.1133	1.0358	1.0089	1.0018	1.0003	1
0.9	2.1275	1.6290	1.2978	1.1124	1.0339	1.0080	1.0016	1.0003	1	1	1
0.85	1.3034	1.1119	1.0334	1.0075	1.0013	1.0001	1.0001	1	1	1	1
0.8	1.0309	1.0077	1.0015	1.0002	1	1	1	1	1	1	1
0.75	1.0013	1.0001	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.0028	38.6910	9.1531	3.5929	2.1293	1.5382	1.2223	1.0770	1.0240	1.0054	1.0011
0.95	9.1502	3.6064	2.1385	1.5421	1.2210	1.0761	1.0223	1.0050	1.0010	1.0001	1
0.9	2.1390	1.5361	1.2245	1.0752	1.0209	1.0047	1.0009	1.0001	1	1	1
0.85	1.2224	1.0756	1.0206	1.0046	1.0009	1.0001	1	1	1	1	1
0.8	1.0192	1.0039	1.0006	1.0001	1	1	1	1	1	1	1
0.75	1.0007	1.0001	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.4765	49.1708	13.2536	4.5770	2.2888	1.5174	1.1957	1.0627	1.0172	1.0041	1.0008
0.95	13.2701	4.5867	2.3012	1.5221	1.1943	1.0629	1.0170	1.0037	1.0007	1.0001	1
0.9	2.3047	1.5214	1.1935	1.0611	1.0164	1.0039	1.0007	1.0001	1	1	1
0.85	1.1927	1.0592	1.0159	1.0032	1.0005	1	1	1	1	1	1
0.8	1.0146	1.0031	1.0005	1	1	1	1	1	1	1	1
0.75	1.0004	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.5715	57.3211	18.0265	6.2255	2.6684	1.5534	1.1849	1.0565	1.0151	1.0038	1.0006

0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.6241	24.1260	3.4736	1.7134	1.1037	1.0128	1.0017	1.0002	1	1	1
0.95	3.4918	1.7167	1.1001	1.0118	1.0013	1.0001	1	1	1	1	1
0.9	1.1027	1.0116	1.0016	1.0001	1	1	1	1	1	1	1
0.85	1.0011	1.0001	1	1	1	1	1	1	1	1	1
0.8	1	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.0325	36.4601	5.6850	1.7284	1.0847	1.0109	1.0013	1.0001	1	1	1
0.95	5.6794	1.7338	1.0847	1.0101	1.0011	1.0001	1	1	1	1	1
0.9	1.0836	1.0092	1.0010	1.0001	1	1	1	1	1	1	1
0.85	1.0009	1.0001	1	1	1	1	1	1	1	1	1
0.8	1	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.2156	46.5381	10.0077	1.9438	1.0805	1.0092	1.0012	1.0001	1	1	1
0.95	9.8290	1.9708	1.0790	1.0086	1.0010	1.0001	1	1	1	1	1
0.9	1.0780	1.0082	1.0009	1.0001	1	1	1	1	1	1	1
0.85	1.0009	1	1	1	1	1	1	1	1	1	1
0.8	1	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2

1	200.8633	50.6629	12.5330	2.2555	1.0786	1.0091	1.0012	1.0001	1	1	1
0.95	12.1450	2.2643	1.0781	1.0088	1.0011	1.0001	1	1	1	1	1
0.9	1.0784	1.0081	1.0009	1.0001	1	1	1	1	1	1	1
0.85	1.0006	1.0001	1	1	1	1	1	1	1	1	1
0.8	1	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S38 ARL of the EWMA TBEA schemes based on the statistic Z_2 when variable A following the normal distribution and T following the weibull distribution assuming a Clayton copula when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.7376	25.6599	9.4625	5.4665	3.7550	2.8382	2.2694	1.8805	1.5946	1.3685	1.1917
0.95	11.2565	6.2833	4.2653	3.1913	2.5404	2.1094	1.7913	1.5433	1.3344	1.1671	1.0662
0.9	4.7960	3.5732	2.8268	2.3313	1.9871	1.7268	1.5026	1.2965	1.1433	1.0538	1.0159
0.85	3.1186	2.5687	2.1789	1.8980	1.6804	1.4681	1.2684	1.1229	1.0417	1.0116	1.0025
0.8	2.3769	2.0617	1.8374	1.6393	1.4313	1.2386	1.1022	1.0340	1.0081	1.0015	1.0002
0.75	1.9818	1.7923	1.6035	1.3946	1.2082	1.0849	1.0258	1.0059	1.0008	1.0001	1
0.7	1.7549	1.5651	1.3588	1.1793	1.0690	1.0190	1.0041	1.0006	1.0001	1	1
0.65	1.5331	1.3259	1.1553	1.0569	1.0148	1.0029	1.0003	1	1	1	1
0.6	1.2932	1.1336	1.0448	1.0105	1.0017	1.0001	1	1	1	1	1
0.55	1.1111	1.0348	1.0076	1.0012	1.0001	1	1	1	1	1	1
0.5	1.0268	1.0054	1.0008	1.0001	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.1866	37.7817	11.5008	5.3895	3.3422	2.3901	1.8548	1.5128	1.2810	1.1328	1.0503
0.95	15.7579	6.7175	3.9423	2.7426	2.0926	1.7002	1.4203	1.2206	1.0970	1.0350	1.0096
0.9	4.6554	3.1446	2.3710	1.8997	1.5827	1.3485	1.1745	1.0694	1.0215	1.0053	1.0010
0.85	2.6637	2.1188	1.7608	1.4902	1.2822	1.1314	1.0481	1.0130	1.0027	1.0004	1.0001
0.8	1.9409	1.6475	1.4164	1.2259	1.0964	1.0318	1.0080	1.0014	1.0003	1	1
0.75	1.5631	1.3479	1.1743	1.0692	1.0206	1.0045	1.0008	1	1	1	1
0.7	1.2868	1.1335	1.0475	1.0124	1.0023	1.0002	1	1	1	1	1
0.65	1.0993	1.0320	1.0069	1.0013	1.0002	1	1	1	1	1	1
0.6	1.0199	1.0038	1.0007	1	1	1	1	1	1	1	1
0.55	1.0023	1.0003	1	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.6685	50.7576	15.4696	6.4673	3.5684	2.3637	1.7615	1.4065	1.1987	1.0850	1.0295
0.95	24.3056	8.9805	4.4957	2.8125	2.0270	1.5859	1.3118	1.1440	1.0564	1.0174	1.0045

0.55	1.0001	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.2185	13.4777	5.3213	3.2637	2.3656	1.8730	1.5127	1.2412	1.0901	1.0256	1.0057
0.95	6.0535	3.6490	2.6184	2.0635	1.7058	1.3947	1.1677	1.0529	1.0133	1.0027	1.0004
0.9	2.8837	2.2496	1.8826	1.5716	1.2888	1.1090	1.0306	1.0066	1.0009	1.0001	1
0.85	2.0277	1.7493	1.4524	1.2022	1.0664	1.0160	1.0026	1.0004	1	1	1
0.8	1.6365	1.3452	1.1344	1.0378	1.0082	1.0013	1.0001	1	1	1	1
0.75	1.2476	1.0848	1.0201	1.0036	1.0003	1	1	1	1	1	1
0.7	1.0499	1.0105	1.0016	1.0002	1	1	1	1	1	1	1
0.65	1.0051	1.0005	1.0001	1	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.4370	18.3362	5.1297	2.7507	1.8979	1.4429	1.1818	1.0592	1.0160	1.0033	1.0005
0.95	6.2928	3.1604	2.1293	1.6101	1.2816	1.1037	1.0292	1.0060	1.0011	1.0002	1
0.9	2.3818	1.7895	1.4147	1.1708	1.0538	1.0127	1.0023	1.0003	1	1	1
0.85	1.5666	1.2674	1.0959	1.0255	1.0044	1.0006	1.0001	1	1	1	1
0.8	1.1601	1.0475	1.0103	1.0017	1.0002	1	1	1	1	1	1
0.75	1.0222	1.0039	1.0006	1	1	1	1	1	1	1	1
0.7	1.0012	1.0001	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.8104	26.4518	6.2092	2.7772	1.7657	1.3207	1.1135	1.0330	1.0075	1.0015	1.0003
0.95	8.4752	3.3582	2.0224	1.4596	1.1814	1.0571	1.0142	1.0027	1.0003	1	1
0.9	2.3251	1.6402	1.2715	1.0936	1.0251	1.0050	1.0008	1.0001	1	1	1
0.85	1.3972	1.1534	1.0462	1.0100	1.0015	1.0002	1	1	1	1	1
0.8	1.0774	1.0193	1.0033	1.0004	1	1	1	1	1	1	1
0.75	1.0072	1.0009	1.0001	1	1	1	1	1	1	1	1
0.7	1.0003	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.5301	35.1185	8.1201	3.0686	1.7468	1.2769	1.0911	1.0249	1.0056	1.0010	1.0001

0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.8557	8.2827	2.4903	1.5623	1.0936	1.0135	1.0019	1.0002	1	1	1
0.95	2.8173	1.7684	1.1706	1.0231	1.0032	1.0003	1	1	1	1	1
0.9	1.3053	1.0412	1.0054	1.0005	1	1	1	1	1	1	1
0.85	1.0094	1.0010	1.0001	1	1	1	1	1	1	1	1
0.8	1.0002	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.0886	14.9242	2.6572	1.4200	1.0585	1.0082	1.0009	1	1	1	1
0.95	3.2295	1.6365	1.1028	1.0138	1.0014	1.0001	1	1	1	1	1
0.9	1.1798	1.0239	1.0028	1.0003	1	1	1	1	1	1	1
0.85	1.0047	1.0005	1	1	1	1	1	1	1	1	1
0.8	1.0001	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.5217	24.2998	3.3862	1.3702	1.0486	1.0073	1.0006	1.0001	1	1	1
0.95	4.8019	1.6104	1.0826	1.0105	1.0011	1.0001	1	1	1	1	1
0.9	1.1413	1.0177	1.0023	1.0001	1	1	1	1	1	1	1
0.85	1.0035	1.0004	1	1	1	1	1	1	1	1	1
0.8	1	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2

1	200.8471	32.5140	5.0041	1.3681	1.0417	1.0059	1.0006	1.0001	1	1	1
0.95	7.9347	1.6877	1.0722	1.0094	1.0010	1.0001	1	1	1	1	1
0.9	1.1253	1.0147	1.0016	1.0002	1	1	1	1	1	1	1
0.85	1.0032	1.0002	1	1	1	1	1	1	1	1	1
0.8	1	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.6726	35.9798	6.1182	1.3944	1.0414	1.0055	1.0007	1	1	1	1
0.95	9.8842	1.8000	1.0697	1.0090	1.0010	1.0001	1	1	1	1	1
0.9	1.1237	1.0143	1.0017	1.0001	1	1	1	1	1	1	1
0.85	1.0027	1.0002	1	1	1	1	1	1	1	1	1
0.8	1	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S39 ARL of the EWMA TBEA schemes based on the statistic Z_3 when variable A following the normal distribution and T following the weibull distribution assuming a Clayton copula when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.6702	25.9903	9.5131	5.4649	3.7582	2.8362	2.2682	1.8851	1.6073	1.3714	1.1904
0.95	11.6149	6.6089	4.5375	3.3944	2.7065	2.2335	1.9056	1.6583	1.4312	1.2356	1.0987
0.9	4.8577	3.7495	3.0265	2.5239	2.1614	1.8963	1.6797	1.4671	1.2640	1.1166	1.0397
0.85	3.1519	2.6895	2.3238	2.0610	1.8644	1.6796	1.4781	1.2760	1.1250	1.0429	1.0113
0.8	2.3843	2.1412	1.9661	1.8230	1.6560	1.4589	1.2654	1.1209	1.0416	1.0111	1.0022
0.75	2.0001	1.8932	1.7683	1.6000	1.4059	1.2253	1.0985	1.0330	1.0088	1.0017	1.0002
0.7	1.8204	1.6814	1.5059	1.3210	1.1652	1.0685	1.0224	1.0050	1.0010	1.0001	1
0.65	1.5438	1.3696	1.2124	1.0991	1.0382	1.0114	1.0024	1.0004	1	1	1
0.6	1.2151	1.1109	1.0450	1.0152	1.0041	1.0009	1.0002	1	1	1	1
0.55	1.0395	1.0139	1.0039	1.0011	1.0002	1	1	1	1	1	1
0.5	1.0028	1.0008	1.0002	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.0505	38.7385	11.5153	5.4003	3.3290	2.3704	1.8466	1.5087	1.2744	1.1189	1.0406
0.95	17.0471	7.2530	4.2553	2.9320	2.2419	1.8075	1.5060	1.2843	1.1293	1.0445	1.0117

0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.7564	70.4504	26.8941	11.3714	5.4219	2.9414	1.8533	1.3551	1.1314	1.0402	1.0106
0.95	58.3819	23.1526	10.1060	4.9193	2.7713	1.7810	1.3332	1.1241	1.0392	1.0097	1.0018
0.9	16.3422	7.6018	3.9784	2.3867	1.6260	1.2669	1.0991	1.0298	1.0077	1.0014	1.0002
0.85	5.0295	2.9429	1.9247	1.4243	1.1796	1.0648	1.0191	1.0043	1.0007	1	1
0.8	2.0542	1.5182	1.2391	1.0992	1.0336	1.0095	1.0023	1.0003	1	1	1
0.75	1.2405	1.1052	1.0397	1.0128	1.0032	1.0006	1.0001	1	1	1	1
0.7	1.0339	1.0116	1.0032	1.0008	1.0002	1	1	1	1	1	1
0.65	1.0018	1.0003	1.0001	1	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.5340	15.2252	5.8529	3.5356	2.5407	2.0017	1.6474	1.3601	1.1539	1.0505	1.0118
0.95	6.8533	4.1658	2.9806	2.3183	1.9294	1.6390	1.3575	1.1511	1.0467	1.0108	1.0021
0.9	3.1599	2.5106	2.0997	1.8460	1.5892	1.3143	1.1263	1.0372	1.0083	1.0011	1.0001
0.85	2.1555	1.9429	1.7574	1.4900	1.2359	1.0851	1.0234	1.0047	1.0007	1.0001	1
0.8	1.8303	1.6098	1.3338	1.1341	1.0410	1.0099	1.0016	1.0003	1	1	1
0.75	1.3731	1.1594	1.0519	1.0135	1.0025	1.0005	1	1	1	1	1
0.7	1.0433	1.0110	1.0026	1.0003	1	1	1	1	1	1	1
0.65	1.0011	1.0002	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.6175	22.8158	6.0333	3.1060	2.1001	1.5939	1.2798	1.1066	1.0299	1.0061	1.0011
0.95	7.9935	3.8288	2.5075	1.8858	1.4971	1.2275	1.0813	1.0216	1.0040	1.0005	1.0001
0.9	2.6783	2.0621	1.6800	1.3674	1.1524	1.0484	1.0113	1.0020	1.0002	1	1
0.85	1.7589	1.4618	1.2110	1.0757	1.0195	1.0042	1.0007	1.0001	1	1	1
0.8	1.2271	1.0804	1.0230	1.0055	1.0010	1.0002	1	1	1	1	1
0.75	1.0166	1.0040	1.0006	1.0001	1	1	1	1	1	1	1
0.7	1.0003	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.7611	33.6860	7.8564	3.2893	2.0079	1.4638	1.1875	1.0600	1.0150	1.0026	1.0004

0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.8$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.3879	9.7074	3.8578	2.4553	1.8777	1.5013	1.1352	1.0263	1.0039	1.0005	1
0.95	4.3873	2.8219	2.0650	1.8050	1.3751	1.0742	1.0124	1.0016	1.0001	1	1
0.9	2.1375	1.9152	1.6589	1.1635	1.0236	1.0033	1.0003	1	1	1	1
0.85	1.7790	1.2609	1.0309	1.0045	1.0004	1	1	1	1	1	1
0.8	1.0206	1.0022	1.0002	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.2058	19.2987	4.1156	2.2179	1.6228	1.1730	1.0311	1.0046	1.0006	1.0001	1
0.95	5.1802	2.6093	1.8772	1.4103	1.0745	1.0108	1.0013	1.0001	1	1	1
0.9	1.9449	1.6344	1.1262	1.0166	1.0022	1.0003	1	1	1	1	1
0.85	1.1364	1.0133	1.0014	1.0002	1	1	1	1	1	1	1
0.8	1.0005	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.0440	34.5368	6.2720	2.3393	1.5133	1.1127	1.0184	1.0028	1.0003	1	1
0.95	10.3330	3.0156	1.8345	1.2627	1.0409	1.0066	1.0005	1.0001	1	1	1
0.9	1.9376	1.4325	1.0589	1.0078	1.0008	1.0001	1	1	1	1	1
0.85	1.0484	1.0052	1.0006	1.0001	1	1	1	1	1	1	1
0.8	1.0001	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2

1	199.1132	46.9912	10.6205	2.8863	1.4828	1.0909	1.0155	1.0022	1.0002	1	1
0.95	20.6322	4.8781	1.9150	1.2092	1.0304	1.0041	1.0005	1.0001	1	1	1
0.9	2.1322	1.3402	1.0405	1.0051	1.0005	1.0001	1	1	1	1	1
0.85	1.0305	1.0035	1.0004	1	1	1	1	1	1	1	1
0.8	1	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.1604	56.1340	15.5692	4.2569	1.5454	1.0842	1.0128	1.0016	1.0002	1	1
0.95	33.2787	9.0272	2.4299	1.1963	1.0269	1.0037	1.0003	1	1	1	1
0.9	3.3662	1.3276	1.0349	1.0046	1.0005	1	1	1	1	1	1
0.85	1.0238	1.0026	1.0002	1	1	1	1	1	1	1	1
0.8	1.0001	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.2075	59.3972	17.8502	5.0907	1.6169	1.0825	1.0132	1.0016	1.0001	1	1
0.95	39.2085	11.4834	2.9947	1.2013	1.0250	1.0037	1.0003	1	1	1	1
0.9	4.6492	1.3627	1.0331	1.0038	1.0005	1	1	1	1	1	1
0.85	1.0219	1.0024	1.0003	1	1	1	1	1	1	1	1
0.8	1	1	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1

Table S40 ARL of the EWMA TBFA schemes based on the statistic Z_1 when variable A following the normal distribution and T following the weibull distribution assuming a Frank copula when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.3112	37.4085	11.5771	5.9428	3.9598	2.9877	2.4246	2.0837	1.8452	1.6454	1.4560
0.95	11.5917	5.9490	3.9481	2.9886	2.4302	2.0785	1.8443	1.6452	1.4573	1.2909	1.1602

0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.0552	88.6771	39.8757	18.5572	9.1978	4.9793	3.0040	2.0237	1.5264	1.2625	1.1252
0.95	39.3007	18.5340	9.2145	4.9917	3.0049	2.0284	1.5276	1.2617	1.1217	1.0542	1.0217
0.9	9.2100	4.9818	3.0202	2.0408	1.5307	1.2637	1.1233	1.0512	1.0212	1.0072	1.0027
0.85	3.0223	2.0332	1.5281	1.2611	1.1193	1.0524	1.0198	1.0075	1.0021	1.0008	1.0002
0.8	1.5289	1.2586	1.1205	1.0510	1.0196	1.0068	1.0024	1.0005	1.0002	1	1
0.75	1.1201	1.0502	1.0195	1.0069	1.0021	1.0005	1.0001	1.0001	1	1	1
0.7	1.0195	1.0066	1.0020	1.0007	1.0001	1	1	1	1	1	1
0.65	1.0018	1.0005	1.0001	1	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.4498	91.7717	43.0114	20.8889	10.6059	5.6996	3.3384	2.1606	1.5702	1.2710	1.1237
0.95	42.2196	20.7157	10.5502	5.7197	3.3470	2.1616	1.5742	1.2696	1.1220	1.0517	1.0203
0.9	10.6092	5.7549	3.3440	2.1634	1.5718	1.2694	1.1226	1.0502	1.0203	1.0069	1.0024
0.85	3.3643	2.1733	1.5712	1.2664	1.1219	1.0494	1.0193	1.0066	1.0022	1.0008	1.0002
0.8	1.5665	1.2686	1.1191	1.0499	1.0190	1.0070	1.0022	1.0006	1.0002	1	1
0.75	1.1208	1.0497	1.0184	1.0068	1.0019	1.0005	1.0002	1	1	1	1
0.7	1.0184	1.0061	1.0019	1.0004	1.0001	1	1	1	1	1	1
0.65	1.0019	1.0005	1.0001	1	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.1866	22.7368	6.6645	3.7153	2.6313	2.0989	1.7993	1.5259	1.2584	1.0996	1.0342
0.95	6.6495	3.7235	2.6324	2.0991	1.8012	1.5232	1.2624	1.1024	1.0333	1.0108	1.0027
0.9	2.6316	2.0979	1.7970	1.5282	1.2647	1.1011	1.0328	1.0104	1.0027	1.0007	1.0002
0.85	1.8009	1.5303	1.2646	1.1001	1.0326	1.0094	1.0028	1.0007	1.0002	1.0001	1
0.8	1.2630	1.0998	1.0319	1.0097	1.0028	1.0008	1.0002	1	1	1	1
0.75	1.0316	1.0102	1.0028	1.0008	1.0002	1	1	1	1	1	1
0.7	1.0028	1.0008	1.0002	1	1	1	1	1	1	1	1
0.65	1.0001	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.7293	38.6727	9.2176	4.0499	2.5649	1.9438	1.5707	1.2829	1.1086	1.0361	1.0109

0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.0284	80.5613	32.2997	13.0447	5.5094	2.5889	1.5217	1.1653	1.0520	1.0153	1.0048
0.95	31.8923	13.0712	5.5037	2.5906	1.5256	1.1680	1.0516	1.0145	1.0043	1.0014	1.0003
0.9	5.4985	2.6043	1.5276	1.1663	1.0505	1.0153	1.0045	1.0011	1.0003	1.0001	1
0.85	1.5304	1.1668	1.0509	1.0157	1.0043	1.0013	1.0003	1	1	1	1
0.8	1.0503	1.0152	1.0043	1.0010	1.0003	1	1	1	1	1	1
0.75	1.0043	1.0011	1.0002	1.0001	1	1	1	1	1	1	1
0.7	1.0004	1.0001	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.8$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.5117	9.2744	3.2761	2.1026	1.7613	1.1899	1.0216	1.0039	1.0006	1.0001	1
0.95	3.2814	2.1052	1.7606	1.1912	1.0210	1.0035	1.0007	1.0001	1	1	1
0.9	1.7620	1.1901	1.0215	1.0035	1.0007	1.0001	1.0001	1	1	1	1
0.85	1.0203	1.0033	1.0005	1.0001	1	1	1	1	1	1	1
0.8	1.0007	1.0001	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.4572	20.3191	3.7171	2.0633	1.5299	1.0701	1.0099	1.0019	1.0003	1.0001	1
0.95	3.7264	2.0651	1.5354	1.0695	1.0092	1.0014	1.0003	1	1	1	1
0.9	1.5367	1.0717	1.0088	1.0017	1.0003	1	1	1	1	1	1
0.85	1.0093	1.0017	1.0004	1.0001	1	1	1	1	1	1	1
0.8	1.0004	1.0001	1	1	1	1	1	1	1	1	1
0.75	1	1	1	1	1	1	1	1	1	1	1
0.7	1	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2

0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1

Table S41 ARL of the EWMA TBEA schemes based on the statistic Z_2 when variable A following the normal distribution and T following the weibull distribution assuming a Frank copula when $\sigma_{0,A} = 1, \sigma_{0,T} = 1$ and $\tau \in \{0.2, 0.5, 0.8\}$

$\tau = 0.2$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.4825	30.8986	10.9911	6.1733	4.1921	3.1260	2.4801	2.0440	1.7333	1.4880	1.2719
0.95	13.2692	7.1614	4.7654	3.5341	2.7942	2.2900	1.9473	1.6882	1.4557	1.2500	1.1120
0.9	5.3984	3.9709	3.1158	2.5550	2.1547	1.8732	1.6471	1.4269	1.2260	1.0952	1.0329
0.85	3.4641	2.8255	2.3779	2.0528	1.8216	1.6182	1.4012	1.2076	1.0827	1.0269	1.0075
0.8	2.6129	2.2379	1.9750	1.7868	1.5915	1.3739	1.1871	1.0720	1.0224	1.0056	1.0011
0.75	2.1311	1.9206	1.7580	1.5706	1.3490	1.1703	1.0613	1.0181	1.0046	1.0009	1.0001
0.7	1.8802	1.7305	1.5412	1.3253	1.1516	1.0516	1.0149	1.0029	1.0005	1.0001	1
0.65	1.7109	1.5164	1.3011	1.1319	1.0448	1.0114	1.0025	1.0004	1.0001	1	1
0.6	1.4874	1.2744	1.1185	1.0381	1.0097	1.0018	1.0002	1	1	1	1
0.55	1.2524	1.1027	1.0320	1.0078	1.0014	1.0002	1	1	1	1	1
0.5	1.0909	1.0273	1.0059	1.0010	1.0002	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.3164	52.2404	16.0020	7.0315	4.0787	2.8127	2.1273	1.7164	1.4263	1.2196	1.0957
0.95	23.3320	9.1310	4.9534	3.2848	2.4407	1.9488	1.6158	1.3600	1.1777	1.0730	1.0242
0.9	6.0530	3.8598	2.7959	2.1897	1.8141	1.5306	1.3083	1.1433	1.0539	1.0173	1.0044
0.85	3.1952	2.4655	2.0205	1.7142	1.4650	1.2542	1.1131	1.0396	1.0114	1.0027	1.0006
0.8	2.2411	1.8945	1.6347	1.4034	1.2101	1.0837	1.0275	1.0071	1.0016	1.0003	1.0001
0.75	1.8010	1.5704	1.3486	1.1682	1.0640	1.0196	1.0053	1.0012	1.0002	1	1
0.7	1.5087	1.2939	1.1363	1.0477	1.0137	1.0033	1.0006	1.0001	1	1	1
0.65	1.2485	1.1060	1.0357	1.0091	1.0019	1.0003	1.0001	1	1	1	1
0.6	1.0793	1.0252	1.0063	1.0012	1.0002	1	1	1	1	1	1
0.55	1.0174	1.0038	1.0008	1.0001	1	1	1	1	1	1	1
0.5	1.0023	1.0004	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.8181	69.0584	24.2336	9.8493	4.9349	3.0288	2.1252	1.6390	1.3351	1.1570	1.0625
0.95	38.7201	14.5733	6.6451	3.7773	2.5337	1.8968	1.5193	1.2674	1.1187	1.0427	1.0133
0.9	9.2577	4.8542	3.0618	2.2108	1.7339	1.4191	1.2098	1.0867	1.0298	1.0084	1.0019
0.85	3.7577	2.5931	1.9833	1.6123	1.3442	1.1667	1.0626	1.0196	1.0052	1.0011	1.0003
0.8	2.2814	1.8265	1.5100	1.2771	1.1243	1.0446	1.0134	1.0029	1.0005	1.0002	1
0.75	1.6984	1.4281	1.2198	1.0914	1.0306	1.0079	1.0018	1.0003	1.0001	1	1
0.7	1.3554	1.1723	1.0677	1.0204	1.0053	1.0013	1.0001	1	1	1	1
0.65	1.1316	1.0468	1.0131	1.0030	1.0005	1.0001	1	1	1	1	1
0.6	1.0320	1.0086	1.0016	1.0003	1	1	1	1	1	1	1
0.55	1.0056	1.0010	1.0001	1	1	1	1	1	1	1	1

0.5	1.0006	1.0001	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.5059	79.7042	32.0762	13.6282	6.4954	3.5503	2.2688	1.6405	1.3094	1.1334	1.0499
0.95	52.5148	21.4655	9.6468	4.8747	2.9052	1.9733	1.4989	1.2354	1.0980	1.0345	1.0101
0.9	14.7426	6.9548	3.8285	2.4418	1.7611	1.3860	1.1745	1.0682	1.0226	1.0061	1.0013
0.85	5.2008	3.1086	2.1146	1.5970	1.3012	1.1281	1.0456	1.0143	1.0032	1.0009	1.0001
0.8	2.6074	1.8891	1.4749	1.2294	1.0938	1.0323	1.0086	1.0019	1.0004	1.0001	1
0.75	1.7024	1.3739	1.1727	1.0651	1.0213	1.0048	1.0011	1.0002	1	1	1
0.7	1.2935	1.1290	1.0465	1.0132	1.0034	1.0005	1.0001	1	1	1	1
0.65	1.0922	1.0302	1.0085	1.0017	1.0004	1	1	1	1	1	1
0.6	1.0198	1.0047	1.0008	1.0002	1	1	1	1	1	1	1
0.55	1.0030	1.0005	1.0001	1	1	1	1	1	1	1	1
0.5	1.0003	1.0001	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.2931	86.5069	38.1009	17.4065	8.4057	4.3637	2.5546	1.7066	1.3127	1.1219	1.0437
0.95	62.6820	28.0102	13.1808	6.4781	3.5470	2.1816	1.5349	1.2261	1.0881	1.0303	1.0089
0.9	20.8688	10.0078	5.1337	2.9294	1.9005	1.3992	1.1682	1.0590	1.0178	1.0048	1.0013
0.85	7.7341	4.1071	2.4610	1.6772	1.2983	1.1175	1.0388	1.0122	1.0028	1.0006	1.0001
0.8	3.3513	2.1164	1.5165	1.2206	1.0813	1.0269	1.0068	1.0016	1.0003	1.0001	1
0.75	1.8480	1.3848	1.1573	1.0554	1.0164	1.0042	1.0007	1.0002	1	1	1
0.7	1.2850	1.1105	1.0369	1.0103	1.0026	1.0004	1.0001	1	1	1	1
0.65	1.0780	1.0238	1.0059	1.0014	1.0003	1	1	1	1	1	1
0.6	1.0144	1.0037	1.0006	1.0001	1	1	1	1	1	1	1
0.55	1.0020	1.0002	1.0001	1	1	1	1	1	1	1	1
0.5	1.0002	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.3422	88.7372	40.7204	19.0161	9.3197	4.7993	2.7303	1.7601	1.3172	1.1237	1.0435
0.95	66.2426	30.8624	14.7403	7.3964	3.9243	2.3332	1.5743	1.2317	1.0849	1.0283	1.0082
0.9	23.6349	11.6246	5.9203	3.2479	1.9991	1.4268	1.1650	1.0580	1.0178	1.0049	1.0012
0.85	9.1327	4.8046	2.7352	1.7587	1.3120	1.1148	1.0394	1.0103	1.0027	1.0007	1.0001
0.8	3.9054	2.3152	1.5648	1.2201	1.0795	1.0246	1.0064	1.0015	1.0003	1.0001	1
0.75	1.9862	1.4145	1.1604	1.0528	1.0146	1.0039	1.0006	1.0001	1	1	1
0.7	1.3035	1.1095	1.0345	1.0095	1.0019	1.0003	1	1	1	1	1
0.65	1.0741	1.0226	1.0058	1.0012	1.0002	1	1	1	1	1	1
0.6	1.0139	1.0032	1.0006	1.0001	1	1	1	1	1	1	1
0.55	1.0018	1.0002	1.0001	1	1	1	1	1	1	1	1
0.5	1.0001	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.4964	20.5226	7.2423	4.2337	2.9719	2.2782	1.8836	1.5853	1.2780	1.0909	1.0233
0.95	8.4267	4.7911	3.3203	2.5399	2.0605	1.7803	1.4965	1.2062	1.0607	1.0149	1.0031

0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.8266	82.1070	32.9732	13.2235	5.3191	2.4567	1.4607	1.1331	1.0341	1.0079	1.0015
0.95	51.2363	20.8348	8.3514	3.5692	1.8528	1.2616	1.0721	1.0165	1.0034	1.0006	1.0001
0.9	13.2537	5.4420	2.5290	1.4963	1.1437	1.0357	1.0078	1.0015	1.0003	1.0001	1
0.85	3.6336	1.9024	1.2825	1.0766	1.0174	1.0035	1.0008	1.0001	1	1	1
0.8	1.5309	1.1542	1.0368	1.0078	1.0015	1.0003	1.0001	1	1	1	1
0.75	1.0852	1.0183	1.0038	1.0006	1.0001	1	1	1	1	1	1
0.7	1.0088	1.0019	1.0004	1.0001	1	1	1	1	1	1	1
0.65	1.0005	1.0002	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.4928	84.8453	35.6985	15.0540	6.2215	2.7386	1.5008	1.1321	1.0336	1.0078	1.0016
0.95	55.9442	23.7966	9.9445	4.1952	1.9974	1.2747	1.0691	1.0157	1.0033	1.0007	1.0001
0.9	15.8711	6.6282	2.8956	1.5598	1.1444	1.0349	1.0076	1.0014	1.0003	1.0001	1
0.85	4.4901	2.1002	1.3014	1.0744	1.0167	1.0033	1.0005	1.0001	1	1	1
0.8	1.6061	1.1584	1.0362	1.0082	1.0014	1.0003	1	1	1	1	1
0.75	1.0812	1.0176	1.0035	1.0006	1.0001	1	1	1	1	1	1
0.7	1.0082	1.0014	1.0003	1.0001	1	1	1	1	1	1	1
0.65	1.0007	1.0001	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.8$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.0885	10.2655	3.9824	2.5311	1.9194	1.5625	1.0752	1.0070	1.0007	1.0001	1
0.95	4.4628	2.8019	2.0160	1.7943	1.2457	1.0211	1.0021	1.0002	1	1	1
0.9	2.1819	1.8983	1.5700	1.0685	1.0059	1.0006	1.0001	1	1	1	1
0.85	1.7945	1.2423	1.0183	1.0015	1.0002	1	1	1	1	1	1
0.8	1.0657	1.0054	1.0004	1	1	1	1	1	1	1	1
0.75	1.0014	1.0001	1	1	1	1	1	1	1	1	1
0.7	1.0001	1	1	1	1	1	1	1	1	1	1
0.65	1	1	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.0314	27.3679	4.6671	2.3560	1.7551	1.1652	1.0148	1.0015	1.0001	1	1

$\lambda = 0.5$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	201.6092	67.6858	23.7977	9.5143	4.7549	2.9010	2.0503	1.5811	1.2920	1.1221	1.0397
0.95	44.6116	16.3759	7.3296	4.0719	2.6877	1.9941	1.5802	1.3095	1.1328	1.0455	1.0121
0.9	9.5274	5.1391	3.3014	2.3934	1.8760	1.5373	1.2915	1.1281	1.0446	1.0123	1.0028
0.85	3.5800	2.6431	2.0955	1.7323	1.4642	1.2496	1.1085	1.0373	1.0099	1.0021	1.0004
0.8	2.1662	1.8413	1.5777	1.3615	1.1859	1.0795	1.0255	1.0075	1.0015	1.0003	1
0.75	1.6123	1.4117	1.2395	1.1169	1.0465	1.0150	1.0043	1.0009	1.0001	1	1
0.7	1.2424	1.1287	1.0573	1.0225	1.0080	1.0018	1.0004	1	1	1	1
0.65	1.0554	1.0232	1.0087	1.0029	1.0006	1.0002	1	1	1	1	1
0.6	1.0074	1.0026	1.0007	1.0002	1	1	1	1	1	1	1
0.55	1.0004	1.0002	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.9246	78.6996	31.2500	13.2348	6.2458	3.4065	2.1732	1.5737	1.2590	1.0974	1.0303
0.95	63.3140	25.5791	11.1142	5.5049	3.1587	2.1070	1.5662	1.2665	1.1049	1.0327	1.0087
0.9	17.0638	8.0468	4.3756	2.7484	1.9489	1.5098	1.2405	1.0952	1.0308	1.0081	1.0017
0.85	5.1716	3.2470	2.2823	1.7419	1.4099	1.1964	1.0792	1.0254	1.0067	1.0014	1.0001
0.8	2.4099	1.8763	1.5301	1.2895	1.1364	1.0520	1.0168	1.0045	1.0007	1.0001	1
0.75	1.5603	1.3353	1.1779	1.0793	1.0306	1.0094	1.0023	1.0006	1.0001	1	1
0.7	1.1758	1.0855	1.0380	1.0131	1.0041	1.0010	1.0002	1.0001	1	1	1
0.65	1.0337	1.0132	1.0049	1.0014	1.0002	1	1	1	1	1	1
0.6	1.0038	1.0012	1.0003	1	1	1	1	1	1	1	1
0.55	1.0002	1.0001	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.1089	85.0032	37.0986	16.7575	8.0309	4.1655	2.4278	1.6218	1.2527	1.0882	1.0262
0.95	76.9735	34.1677	15.6583	7.5864	4.0460	2.3929	1.6168	1.2581	1.0925	1.0280	1.0061
0.9	26.5390	12.4980	6.3598	3.5183	2.1799	1.5483	1.2302	1.0849	1.0254	1.0061	1.0012
0.85	8.5840	4.6730	2.8011	1.8923	1.4154	1.1807	1.0653	1.0195	1.0045	1.0010	1.0002
0.8	3.1518	2.1280	1.5763	1.2752	1.1160	1.0427	1.0131	1.0029	1.0009	1.0001	1
0.75	1.6245	1.3237	1.1515	1.0648	1.0222	1.0069	1.0016	1.0003	1.0001	1	1
0.7	1.1525	1.0679	1.0278	1.0101	1.0028	1.0007	1.0001	1.0001	1	1	1
0.65	1.0254	1.0097	1.0033	1.0008	1.0002	1	1	1	1	1	1
0.6	1.0026	1.0009	1.0002	1.0001	1	1	1	1	1	1	1
0.55	1.0002	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.1384	86.8863	39.0638	18.2194	8.8406	4.5970	2.5933	1.6666	1.2572	1.0851	1.0246
0.95	82.1748	37.5101	17.7447	8.7417	4.5228	2.5679	1.6652	1.2647	1.0876	1.0252	1.0059
0.9	30.8453	15.0293	7.5459	4.0279	2.3685	1.5920	1.2316	1.0800	1.0237	1.0058	1.0011
0.85	10.7341	5.6512	3.2190	2.0241	1.4446	1.1774	1.0619	1.0178	1.0044	1.0009	1.0002

0.8	3.7842	2.3559	1.6391	1.2855	1.1130	1.0391	1.0117	1.0029	1.0006	1.0001	1
0.75	1.6966	1.3377	1.1494	1.0590	1.0210	1.0061	1.0017	1.0003	1	1	1
0.7	1.1445	1.0655	1.0248	1.0088	1.0027	1.0006	1.0001	1	1	1	1
0.65	1.0221	1.0086	1.0031	1.0008	1.0003	1.0001	1	1	1	1	1
0.6	1.0021	1.0005	1.0002	1	1	1	1	1	1	1	1
0.55	1.0001	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.5$											
$\lambda = 0.1$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.3841	21.8542	7.6110	4.3975	3.0650	2.3490	1.9171	1.6437	1.3562	1.1333	1.0362
0.95	9.0160	5.2177	3.6462	2.7798	2.2338	1.9031	1.6793	1.3926	1.1474	1.0402	1.0085
0.9	3.8737	3.0439	2.4865	2.0888	1.8787	1.6810	1.3954	1.1456	1.0379	1.0075	1.0013
0.85	2.5867	2.1971	1.9771	1.8445	1.6447	1.3483	1.1195	1.0303	1.0057	1.0010	1.0001
0.8	2.0114	1.9159	1.7854	1.5517	1.2580	1.0771	1.0178	1.0040	1.0006	1.0001	1
0.75	1.8389	1.6556	1.3769	1.1456	1.0397	1.0090	1.0017	1.0004	1	1	1
0.7	1.4118	1.1820	1.0602	1.0163	1.0042	1.0008	1.0002	1	1	1	1
0.65	1.0630	1.0194	1.0061	1.0013	1.0003	1	1	1	1	1	1
0.6	1.0055	1.0017	1.0004	1.0001	1	1	1	1	1	1	1
0.55	1.0004	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.3$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.2121	44.5041	11.3133	4.7242	2.8562	2.0635	1.6451	1.3220	1.1067	1.0270	1.0047
0.95	17.0195	6.3282	3.5930	2.5030	1.9572	1.6219	1.3081	1.1001	1.0234	1.0044	1.0007
0.9	3.9252	2.7981	2.1815	1.8390	1.5544	1.2556	1.0742	1.0179	1.0033	1.0005	1.0001
0.85	2.2654	1.9435	1.7168	1.4285	1.1703	1.0451	1.0099	1.0019	1.0003	1.0001	1
0.8	1.7736	1.5277	1.2536	1.0842	1.0216	1.0048	1.0010	1.0002	1	1	1
0.75	1.2701	1.1042	1.0329	1.0088	1.0019	1.0003	1.0001	1	1	1	1
0.7	1.0350	1.0106	1.0029	1.0007	1.0001	1	1	1	1	1	1
0.65	1.0028	1.0007	1.0002	1	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.5$											
$\rho_A \backslash \rho_T$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.5927	64.4344	19.8878	6.9797	3.3134	2.1021	1.5620	1.2303	1.0664	1.0146	1.0024
0.95	37.6442	11.8429	4.8632	2.7628	1.9477	1.5154	1.2051	1.0564	1.0121	1.0022	1.0003
0.9	5.8968	3.2848	2.2611	1.7686	1.4141	1.1514	1.0384	1.0086	1.0015	1.0002	1
0.85	2.3850	1.9114	1.5757	1.2731	1.0855	1.0215	1.0045	1.0008	1.0001	1	1
0.8	1.6325	1.3461	1.1328	1.0378	1.0098	1.0019	1.0004	1.0001	1	1	1
0.75	1.1418	1.0474	1.0144	1.0036	1.0008	1.0002	1	1	1	1	1
0.7	1.0147	1.0048	1.0010	1.0003	1	1	1	1	1	1	1
0.65	1.0009	1.0003	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1

0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.7$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	198.9187	75.6779	28.2057	10.7521	4.5074	2.3556	1.5538	1.1918	1.0510	1.0106	1.0018
0.95	58.9261	21.2582	8.1318	3.6592	2.1098	1.4785	1.1626	1.0425	1.0083	1.0012	1.0002
0.9	11.9354	5.0443	2.7021	1.8109	1.3537	1.1161	1.0280	1.0054	1.0011	1.0002	1
0.85	2.9883	2.0219	1.5249	1.2115	1.0607	1.0144	1.0026	1.0005	1.0001	1	1
0.8	1.5758	1.2689	1.0937	1.0253	1.0059	1.0013	1.0002	1.0001	1	1	1
0.75	1.0972	1.0325	1.0090	1.0022	1.0005	1.0001	1	1	1	1	1
0.7	1.0091	1.0030	1.0007	1.0002	1	1	1	1	1	1	1
0.65	1.0007	1.0002	1	1	1	1	1	1	1	1	1
0.6	1.0001	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\lambda = 0.9$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	199.6428	82.8148	34.7889	14.6492	6.3225	2.9151	1.6292	1.1851	1.0437	1.0092	1.0016
0.95	74.9005	30.8436	12.8885	5.4898	2.6081	1.5251	1.1516	1.0350	1.0074	1.0011	1.0002
0.9	21.7373	9.0106	3.9944	2.0860	1.3575	1.0996	1.0228	1.0045	1.0007	1.0001	1
0.85	5.0308	2.5695	1.5867	1.1900	1.0517	1.0114	1.0021	1.0005	1.0001	1	1
0.8	1.6635	1.2499	1.0773	1.0213	1.0049	1.0010	1.0002	1	1	1	1
0.75	1.0791	1.0265	1.0075	1.0017	1.0004	1.0001	1	1	1	1	1
0.7	1.0078	1.0022	1.0006	1.0001	1	1	1	1	1	1	1
0.65	1.0005	1.0001	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
Shewhart ($\lambda = 1$)											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.3333	84.9843	37.0956	16.3914	7.2119	3.2779	1.7101	1.1848	1.0432	1.0088	1.0013
0.95	80.2966	34.9094	15.1555	6.6103	2.9711	1.5836	1.1520	1.0336	1.0067	1.0009	1.0002
0.9	26.7705	11.5649	4.9959	2.3515	1.3880	1.0973	1.0211	1.0042	1.0007	1.0001	1
0.85	6.7593	3.1338	1.6873	1.1912	1.0484	1.0111	1.0022	1.0004	1.0001	1	1
0.8	1.8004	1.2562	1.0737	1.0193	1.0041	1.0009	1.0002	1	1	1	1
0.75	1.0778	1.0237	1.0064	1.0015	1.0003	1.0001	1	1	1	1	1
0.7	1.0068	1.0021	1.0005	1.0001	1	1	1	1	1	1	1
0.65	1.0004	1.0001	1	1	1	1	1	1	1	1	1
0.6	1	1	1	1	1	1	1	1	1	1	1
0.55	1	1	1	1	1	1	1	1	1	1	1
0.5	1	1	1	1	1	1	1	1	1	1	1
$\tau = 0.8$											
$\lambda = 0.1$											
$\rho_T \backslash \rho_A$	1	1.1	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2
1	200.2616	14.0762	4.8847	2.9840	2.1482	1.8062	1.4794	1.0937	1.0130	1.0015	1.0001
0.95	5.6312	3.4726	2.5627	1.9580	1.8031	1.4179	1.0549	1.0061	1.0006	1.0001	1

[illegible]